

Exact mixed-locality representation selection for resource-constrained QAOA

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A quantum circuit can keep higher-order terms or reduce them with auxiliary qubits. Both choices can exceed resource limits. We study exact representations that reduce some terms and keep others. For direct pair substitutions, we derive feasibility boundaries and a capacity formula for pair-star forests. We construct infinite families where only mixed representations fit the budgets, including gate and depth budgets under a fixed synthesis. Compiled circuits show mixed-only regions on sparse devices. A five-seed test retains 52.9% of baseline mixed-only budget points across all tested seeds within a fixed candidate library. Separate QAOA tests favour native encoding on mean solution quality. A joint resource-and-quality test has no mixed-only success at its primary budget, but gives conditional examples on a fixed secondary grid. These results support exact representation selection under resource limits. They do not establish a general QAOA quality advantage or quantum advantage.

1 Introduction

Some binary optimisation problems contain products of three or more bits. Examples include clause penalties, hypergraph objectives and many-body spin models [1, 2]. We write such a problem as

$$\min_{x \in \{0,1\}^n} f(x), \quad f(x) = \sum_{A \subseteq [n]} c_A \prod_{i \in A} x_i. \quad (1)$$

Here x_i is a bit and c_A is the coefficient of the product on support A . QAOA uses f as a diagonal cost Hamiltonian. It alternates evolution under this Hamiltonian with mixing operations [3, 4].

The same classical objective can have several exact representations. They can use different numbers of qubits, different interaction graphs and different penalties. Their compiled circuits can also have different gate counts and depths. We therefore treat representation choice as part of the algorithm. This choice matters when the device has limited qubits, connectivity or circuit depth.

The first common choice is to keep the original objective. We call this the *native* representation. It uses no auxiliary qubits or added penalties. Previous comparisons have found benefits for this choice on some problem families [5–8].

A native higher-order phase can still be implemented with one- and two-qubit gates. Circuit optimisation can reduce its cost [9]. However, using fewer logical qubits does not always mean using

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Table 1: Scope of the cited methods. “Not studied” means the cited formulation does not address that question; it does not mean the method cannot do so. The last column asks whether a theorem separates native, full and mixed feasibility under joint budgets. Exactness here means recovering the original objective by minimising over auxiliaries. It does not mean equal QAOA dynamics.

Method	Pointwise exact	Mixed output	Joint reduced-set / pair choice	Resource objective	Endpoint phase theorem
Rosenberg substitution [12, 13]	Yes	Partial use possible	Local construction	Not a compiler objective	Not studied
Compact quadratization [19]	Yes	Quadratic output	Construction dependent	Auxiliary bounds	Not studied
Automatic full reduction [24, 25]	Yes	Full-Q target	Substitution choice	Transformation and circuit costs	Not studied
Hardware-aware selection [26]	Yes	Full-Q target	Auxiliary selection	Graph structure and mapped depth	Not studied
Approximate reduction [27]	Not in general	Ansatz dependent	Different ansatz space	Qubit and circuit cost	Not studied
This work	Yes	Explicit output	Both $\mathcal{D}$ and ρ	Fixed-compiler joint budgets	Within direct pair grammar

fewer gates or less depth. The circuit also depends on which qubits interact and how those interactions are routed. Routing can be expensive on sparse devices [10, 11].

The second common choice is *full quadratization*. We replace all higher-order terms with a quadratic function $F(x, y)$. The extra bits y are auxiliaries. Exactness means that $f(x) = \min_y F(x, y)$ for every original assignment x [12–14].

A Rosenberg substitution replaces a product $x_i x_j$ with an auxiliary bit and adds a penalty for an incorrect auxiliary value. It lowers the interaction order. It also adds qubits, couplings and coefficients to the Hamiltonian. Full quadratization can therefore simplify individual phase gadgets while increasing other resources.

The exactness identity preserves the classical optimum and its original-bit solutions. It does not fix the energies of states with incorrect auxiliaries. QAOA can visit these states. Two classically exact representations can therefore have different QAOA dynamics at the same depth.

Earlier work provides many exact reduction methods. Graph-cut constructions reduce higher-order binary energies [15, 16]. General bounds and compact constructions control the number of auxiliaries [17–19]. Optimised and heuristic substitutions can reduce that number further [20, 21]. Recursive reformulations connect product decomposition, penalties and convexification [22]. These methods show that the choice of products to replace matters.

Recent work also studies quantum circuit costs. Herrman et al. consider depth in reductions for constrained problems [23]. Schmidbauer et al. study how reductions change QAOA circuits and develop automatic quadratization methods [24, 25]. Rovara et al. choose auxiliaries to control interaction-graph degree and improve device mapping [26]. Approximate quadratization can reduce circuit cost by accepting a change in the objective [27].

These studies use different representation classes and resource objectives. Table 1 compares their stated scope with ours. A method aimed at a quadratic output may also allow partial substitution. We do not claim otherwise.

There is also a third choice. We can reduce some higher-order terms and keep the rest. The resulting Hamiltonian has mixed locality. A digital quantum processor can implement it with phase gadgets.

Partial Rosenberg substitution is already possible, so the substitution itself is not our novelty claim. We ask a more specific question: can a mixed representation meet resource budgets that exclude both native and fully reduced representations? We answer this question for a stated class of direct pair substitutions.

Three issues need care. First, several terms may use the same auxiliary. Its penalty must cover

their combined effect. A bound on each coefficient separately may be too small. Second, a later substitution may use an earlier auxiliary. We must check that every step remains exact. Third, selection must account for qubits, gates, depth and penalty scale together.

Our aim is to choose which terms to reduce and which pairs to share, while preserving the original objective for every original-bit assignment.

Warm-start QAOA uses a classical relaxation to prepare an initial state and a matching mixer [28–30]. Adding an auxiliary also requires choosing its initial probability. For an auxiliary representing $x_i x_j$, we can use a pair moment from the relaxation or the product of two single-bit probabilities. These values can differ. Neither choice makes a product state satisfy $y_p = x_i x_j$ in every measurement outcome. We study initialisation separately from exactness and resource feasibility.

Our main theoretical result concerns resource feasibility. We define K_Q as the largest number of cubic terms that can be reduced within auxiliary, penalty and auxiliary-degree budgets. We compute K_Q for pair-star forests. This gives explicit boundaries between the feasible representation classes. It also gives an infinite family for which only mixed representations are feasible. A separate result adds gate and depth budgets under a fixed phase-gadget synthesis. The static result needs no compiler. The gate and depth result depends on the specified synthesis.

We also give the penalty needed for one shared Rosenberg substitution and show how exact substitutions can be applied in sequence. These results apply to the stated construction, not to every possible reformulation. For higher-degree terms, finding the smallest valid penalty can be difficult. The sum of absolute residual coefficients gives a simple sufficient bound. It can be larger than necessary. Our small exact search checks do not imply that global selection scales to large instances.

The first two fresh experiments ask separate questions. The compilation study asks which exact representations meet joint resource budgets. Full and selective search use the same resource score and comparable candidate budgets. The QAOA study asks how preparation and optimisation affect solution quality. It includes warm $p = 0$ controls and larger optimisation budgets.

A third study checks resources and solution quality on the same circuit (Section 2.5.4). It uses 60 further random instances and 12 separate mechanism instances. No instance passes the primary test with only a mixed representation. Some instances pass at particular budgets in the fixed secondary grid. For some of these, exhaustive checks rule out every full representation in the stated class. The three guardrail rules always select the same design. The diagnostics help predict auxiliary inconsistency to a limited extent, but do not show a selection benefit. These studies use separate instance sets and are not pooled.

The earlier 14-instance QAOA study, 20-instance signal-enriched study and 18-instance noise study remain development evidence. Native has the best mean objective in Table S5. This is an important limit of the proposed approach. Smaller circuits and classical exactness do not by themselves imply better QAOA solutions. A general benefit from auxiliary pair moments is also not established.

2 Results

We first define the exact representations. We then state when a mixed representation is needed to meet the budgets. Finally, we test circuit resources and QAOA quality separately. The proofs do not rely on the experimental data. The experimental conclusions apply to the tested instances and protocols.

2.1 Representation framework

Let $\mathbb{B} = \{0, 1\}$. Write $x_A = \prod_{i \in A} x_i$ and $x_\emptyset = 1$. To encode a bit on a qubit, use $N_i = (I - Z_i)/2$. The cost Hamiltonian is then

$$H_f = f(N_1, \dots, N_n), \quad N_i = (I - Z_i)/2, \quad H_f|x\rangle = f(x)|x\rangle. \quad (2)$$

Its largest nonconstant interaction order is $\deg(f)$ [2].

Now add a_σ auxiliary bits and choose a representation F_σ . We call it *pointwise exact* if

$$f(x) = \min_{y \in \mathbb{B}^{a_\sigma}} F_\sigma(x, y) \quad \text{for every } x \in \mathbb{B}^n. \quad (3)$$

Table 2: Core notation. Subscripts σ indicate dependence on the chosen representation.

Symbol	Meaning
$\mathbb{B}, [n]$	$\{0, 1\}$ and $\{1, \dots, n\}$.
$x, y, u = (x, y)$	Original bits, auxiliary bits, and the combined register.
x_A, c_A	$\prod_{i \in A} x_i$ and its coefficient in f ; $x_\emptyset = 1$.
f, F_σ	Original objective and its pointwise-exact representation.
$\sigma, a_\sigma, N_\sigma$	Representation choice, auxiliary count, and total qubits $n + a_\sigma$.
ℓ, L_σ	Target locality of reduced terms and maximum retained locality.
$\mathcal{D}, \mathcal{R}$	Higher-order terms chosen for reduction and those retained.
p, y_p, P_ρ	A variable pair, its product auxiliary, and the active pair set.
$M_p, T_p, M_{\max, \sigma}$	Pair penalty, its exactness threshold, and largest active penalty.
$\mathcal{Y}_F(x)$	Auxiliary assignments minimising $F(x, \cdot)$.
H_F, Z_i	Diagonal cost Hamiltonian and the Pauli Z operator on qubit i .
$G_{2q, \sigma}, D_{2q, \sigma}, S_\sigma$	Compiled two-qubit gate count, two-qubit depth, and SWAP count.
κ_σ	Largest-to-smallest magnitude ratio of nonzero, nonidentity Pauli coefficients.
$\bar{\mu}_i, \bar{q}_{ij}$	Raw single and pair pseudo-moments from a common relaxation.
μ_i, q_p	Preparation probabilities from clipped moments or the stated independence fallback.

In the constructions, p denotes a variable pair; in experimental tables and figures it denotes QAOA depth (written r in the ansatz).

For each fixed x , minimising only over the auxiliary bits recovers $f(x)$. This preserves the classical optimum and its projected solutions. It does not constrain all energies with incorrect auxiliary values.

We use the Rosenberg penalty [12]

$$P_R(a, b, y) = ab - 2ay - 2by + 3y, \quad P_R = 0 \text{ if } y = ab, \quad P_R \geq 1 \text{ otherwise.} \quad (4)$$

It is zero when the auxiliary is correct. It is positive when the auxiliary is incorrect.

Table 2 lists the notation. We distinguish raw relaxation moments from the probabilities used to prepare a state. Raw moments need not come from a joint probability distribution. Preparation uses clipped probabilities or the stated independence rule. Appendix A gives the identities and resource definitions.

2.2 Exact selective locality reduction

Start with a multilinear polynomial $q(u) = \sum_S d_S u_S$ on $\mathbb{B}^N$. Combine equal monomials before making a substitution. Choose a pair $p = \{i, j\}$. Let $\mathcal{S}_p$ be the selected higher-order terms that contain this pair.

Factor $u_i u_j$ out of those terms. Their remaining factor is $A_p(v) = \sum_{S \in \mathcal{S}_p} d_S u_{S \setminus p}$, where v contains all coordinates except u_i, u_j . Let B_p contain all unselected terms. Then

$$q(u) = B_p(u) + u_i u_j A_p(v). \quad (5)$$

Introduce a new bit y_p . Replace the selected product $u_i u_j$ with y_p and add a penalty:

$$Q_{p, M_p}(u, y_p) = B_p(u) + y_p A_p(v) + M_p P_R(u_i, u_j, y_p). \quad (6)$$

This is a Rosenberg substitution [12, 13]. The penalty must control the whole residual A_p , because all selected terms use the same auxiliary.

Proposition 2.1 (Sharp aggregate-residual penalty).

For $M_p \geq 0$, with $\|A_p\|_\infty := \max_{v \in \mathbb{B}^{N-2}} |A_p(v)|$,

$$q(u) = \min_{y_p \in \mathbb{B}} Q_{p, M_p}(u, y_p) \quad \text{for every } u \quad \iff \quad M_p \geq \|A_p\|_\infty. \quad (7)$$

For every fixed u , the correct value $y_p = u_i u_j$ is the unique minimiser if and only if $M_p > \|A_p\|_\infty$.

Choosing the correct value $y_p = u_i u_j$ leaves the objective unchanged. Choosing the other value can lower the selected block by at most $\|A_p\|_\infty$. Its penalty is at least M_p . This explains why the stated threshold is sufficient. Appendix B.1 checks every case and proves that the threshold is also necessary.

We can repeat this operation. At each step, compute the residual from the *current* polynomial. Treat earlier auxiliaries as independent bits when computing the bound.

Theorem 2.2 (Sequential compositional exactness). *Let $g^{(0)} = f$, and form $g^{(r)}$ from $g^{(r-1)}$ by Eq. (6) using a fresh y_r and $M_r \geq \|A_r\|_\infty$. Then the recursively consistent lift $\Phi(x)$ satisfies*

$$f(x) = \min_{y \in \mathbb{B}^R} g^{(R)}(x, y) = g^{(R)}(x, \Phi(x)). \quad (8)$$

Here $\Phi(x)$ is obtained by setting each new auxiliary to its intended product, in substitution order. We call it the consistent lift of x . If every inequality is strict, this is the unique auxiliary minimiser.

Each exact step cannot lower the preceding polynomial. Choosing all auxiliaries consistently gives equality at every step. These two facts prove the sequential result. Appendix B.2 shows the intermediate inequalities one by one.

The penalty threshold is the smallest one for this Rosenberg substitution. It is not a minimum over all reduction methods. At higher degree, computing $\|A_p\|_\infty$ can require an exponential search. The simpler bound $M_p \geq \sum_{S \in \mathcal{S}_p} |d_S|$ is sufficient, although it can be too large.

A selected degree- k term can be reduced to order $\ell \geq 2$ using at most $\max\{k - \ell, 0\}$ new auxiliaries. Terms that are not selected remain native. Corollary B.1 gives the steps and the total auxiliary bound. Sharing products may save auxiliaries, but we must recompute the combined penalties when we share them.

2.3 The cubic design space

Our main experiments use cubic objectives. Write $f = f_{\leq 2} + \sum_{A \in \mathcal{C}_3(f)} c_A x_A$, where $\mathcal{C}_3(f) = \{A \subseteq [n] : |A| = 3, c_A \neq 0\}$. The function $f_{\leq 2}$ contains the terms of degree at most two.

A representation makes two choices. First, choose the cubic supports to reduce: $\mathcal{D} \subseteq \mathcal{C}_3(f)$. Second, choose a pair $\rho(A) \in \binom{A}{2}$ for each selected support A . We denote these choices by $\sigma = (\mathcal{D}, \rho)$.

The remaining cubic supports are $\mathcal{R} = \mathcal{C}_3(f) \setminus \mathcal{D}$. The used pairs are $P_\rho = \rho(\mathcal{D})$. Terms assigned to pair p form $\mathcal{D}_p = \{A \in \mathcal{D} : \rho(A) = p\}$. They share one auxiliary y_p . Their residual is $A_p(x) = \sum_{A \in \mathcal{D}_p} c_A x_{A \setminus p}$. The full representation is

$$F_{\rho, M}(x, y) = f_{\leq 2}(x) + \sum_{A \in \mathcal{R}} c_A x_A + \sum_{p \in P_\rho} y_p A_p(x) + \sum_{p = \{i, j\} \in P_\rho} M_p P_R(x_i, x_j, y_p). \quad (9)$$

Each residual is linear in distinct remaining bits. To maximise it, set the positive-coefficient bits to one and the negative-coefficient bits to zero. This gives C_p^+ . Reversing those choices gives the most negative value, $-C_p^-$. Hence

$$C_p^+ = \sum_{A \in \mathcal{D}_p, c_A > 0} c_A, \quad C_p^- = \sum_{A \in \mathcal{D}_p, c_A < 0} |c_A|, \quad T_p := \max\{C_p^+, C_p^-\} = \|A_p\|_\infty. \quad (10)$$

Proposition 2.3 (Exact cubic mixed-sign reduction).

For nonnegative penalties, $f(x) = \min_y F_{\rho, M}(x, y)$ for every x if and only if $M_p \geq T_p$ for every active pair. If every penalty inequality is strict, the unique correct auxiliary assignment is $\phi_\rho(x) = (x_i x_j)_{p = \{i, j\} \in P_\rho}$.

For fixed x , each auxiliary appears in its own separate block. We can therefore minimise one block at a time. The one-step threshold applies to each block. Appendix B.4 gives both directions of the argument, including the necessity of each penalty bound.

In selection, we fix positive margins and use

$$M_p = T_p + \varepsilon_p, \quad \varepsilon_p > 0 \text{ fixed}. \quad (11)$$

There are then finitely many representations. We denote this family by $\Sigma_{\text{ex}}^\varepsilon$. The native endpoint has $\mathcal{D} = \emptyset$. A fully reduced endpoint has $\mathcal{D} = \mathcal{C}_3(f)$. A strictly mixed representation has $0 < |\mathcal{D}| < |\mathcal{C}_3(f)|$.

One auxiliary can reduce several terms. The auxiliary count is therefore $a_\sigma = |P_\rho|$, not $|\mathcal{D}|$. Appendix B.5 describes this reuse as a pair-cover problem.

2.4 When resources force a mixed representation

We now ask when an exact representation meets all resource budgets. This question comes before QAOA optimisation. It does not ask which representation gives the best solution quality.

The results below use the direct cubic pair substitutions just defined. They do not cover every possible quadratization. In particular, they do not cover coefficient splitting or penalties that couple different auxiliaries.

2.4.1 Static capacity and exact phase boundaries

Let $E = \mathcal{C}_3$ be the cubic support set, and let $m = |E|$. A representation is $\sigma = (\mathcal{D}, \rho)$. For each used pair p , let $q_p = |\mathcal{D}_p|$ be its number of assigned terms. Its threshold is

$$T_p = \max \left\{ \sum_{A \in \mathcal{D}_p: c_A > 0} c_A, \sum_{A \in \mathcal{D}_p: c_A < 0} |c_A| \right\}.$$

Use the same margin $\varepsilon \geq 0$ for every pair, so $M_p = T_p + \varepsilon$. A zero margin preserves objective values but can allow an incorrect auxiliary to tie. A positive margin gives a unique correct auxiliary assignment. A penalty budget must bound the actual penalty $T_p + \varepsilon$.

We consider four static resources, in this order: auxiliary count, largest penalty, largest auxiliary degree, and number of retained cubic terms. Their values are

$$\left(|P_\rho|, \max_{p \in P_\rho} (T_p + \varepsilon), \max_{p \in P_\rho} (q_p + 2), |E \setminus \mathcal{D}| \right),$$

The empty-set maxima are zero. An auxiliary for pair $p = \{i, j\}$ couples to i, j and to the q_p distinct third bits. Its degree is therefore $q_p + 2$.

Let the four budgets be $(r_0, M_0, \Delta_0, h_0)$. All four budgets are nonnegative. Here r_0 and h_0 are integers. First ignore the retained-term budget h_0 . Let $\mathcal{Q}(r_0, M_0, \Delta_0)$ contain the reduced sets that can meet the other three budgets with some pair assignment. Define

$$K_Q(E, c; r_0, M_0, \Delta_0) = \max_{\mathcal{D} \in \mathcal{Q}(r_0, M_0, \Delta_0)} |\mathcal{D}|. \quad (12)$$

Thus K_Q is the most cubic terms we can reduce under those three budgets.

Removing terms from a valid assignment cannot add auxiliaries or increase either signed coefficient sum. It also cannot increase auxiliary degree. Every subset of a valid reduced set is therefore valid. This property is called downward closure.

Theorem 2.4 (Exact static feasibility boundary). *A feasible direct-pair representation exists if and only if $K_Q + h_0 \geq m$. The native endpoint is feasible if and only if $h_0 \geq m$; a fully reduced representation is feasible if and only if $K_Q = m$. Consequently, feasible representations exist and all are strictly mixed exactly when*

$$K_Q < m, \quad h_0 < m, \quad K_Q + h_0 \geq m. \quad (13)$$

We must reduce at least $m - h_0$ terms to satisfy the retained-term budget. We can reduce at most K_Q under the other three budgets. Feasibility therefore asks whether these two limits overlap. If they overlap but neither endpoint fits, every feasible representation must be strictly mixed. Appendix B.6 proves each direction.

A *pair-star* has a shared pair $\{a, b\}$ and distinct third bits $w_1, \dots, w_\ell$. Its supports are $\{\{a, b, w_t\} : 1 \leq t \leq \ell\}$. We call $\{a, b\}$ the core and the w_t bits the leaves (or petals). A pair-star forest is a union of such stars on disjoint vertex sets. A one-leaf star is just an isolated cubic term.

Theorem 2.5 (Closed form for a unit-weight pair-star forest). *Suppose all cubic coefficients are +1 and the star sizes are $\ell_1, \dots, \ell_s \geq 1$. Put $m = \sum_i \ell_i$ and*

$$C = \max\{0, \min\{\lfloor M_0 - \varepsilon \rfloor, \lfloor \Delta_0 \rfloor - 2\}\}.$$

If $r_0 = 0$ or $C = 0$, then $K_Q = 0$. Otherwise, let $g_i = \min\{\ell_i, C\} - 1$, sort these gains in nonincreasing order, and set $\rho_0 = \min\{r_0, s\}$. Then

$$K_Q = \min \left\{ m, r_0 + \sum_{j=1}^{\rho_0} g_{(j)} \right\}. \quad (14)$$

The formula has a simple interpretation. One used pair normally reduces one term. In star i , using the core can reduce up to $\min\{\ell_i, C\}$ terms. Its extra gain is $g_i = \min\{\ell_i, C\} - 1$. We spend the available auxiliaries on the largest gains first. Any remaining auxiliary can reduce one uncovered term through a private pair. Appendix B.7 proves the upper bound and shows how to attain it.

Write $R_Q = r_0 + \sum_{j=1}^{\rho_0} g_{(j)}$. Set $R_Q = 0$ if $r_0 = 0$ or $C = 0$. The following table gives the exact five feasibility classes for this construction. “Native only” and “full-Q only” refer to the endpoints. Mixed representations may also be feasible in those rows.

Feasibility class	Necessary and sufficient condition
Both endpoints	$R_Q \geq m, h_0 \geq m$
Native endpoint only	$R_Q < m, h_0 \geq m$
Full-Q endpoint only	$R_Q \geq m, h_0 < m$
Mixed only	$R_Q < m, h_0 < m, R_Q + h_0 \geq m$
None	$R_Q + h_0 < m$

Corollary 2.6 (An infinite static mixed-only family). *For every $L, H \geq 1$, take the disjoint union of an L -petal unit pair-star and H unit isolated triples. Under budgets $(r_0, M_0, \Delta_0, h_0) = (1, L + \varepsilon, L + 2, H)$, neither endpoint is feasible, whereas reducing the star through its core and retaining all isolated triples is feasible.*

The mixed design spends its one auxiliary on the reusable star core. Native fails because it keeps too many cubic terms. Full reduction fails because the isolated terms each need their own auxiliary. Appendix B.8 checks all four resources.

Neither endpoint is always more feasible than the other. For example, fix $h \geq 2$ and budgets $(r_0, M_0, \Delta_0, h_0) = (1, h + 1 + \varepsilon, h + 3, h)$. For h isolated unit cubic terms, native is feasible. Full reduction needs h auxiliaries and exceeds the auxiliary budget. For one unit pair-star with $h + 1$ leaves, full reduction through the core is feasible. Native retains $h + 1$ terms and exceeds the retained-term budget. The two outcomes use the same budgets. Interaction order alone cannot decide feasibility.

2.4.2 A compiled-resource consequence under a fixed synthesis

The static result does not count circuit gates or depth. For those resources, we must fix a synthesis. We use the reference synthesis Π in Appendix C.1. It follows these rules:

1. Expand $x_i = (1 - Z_i)/2$, combine equal Pauli strings and remove exact zeros.
2. Keep the cost angle symbolic. Order strings by weight, then by support.
3. Implement each string with a parity gadget whose target is its largest-index qubit.
4. Keep the emitted CNOT sequence. Place each CNOT in the first layer after both of its qubits are free.

This synthesis assumes all-to-all connectivity. It does not cancel gates between gadgets. G_{2q}^Π counts CNOTs in one cost layer. D_{2q}^Π is their scheduled depth. Single-qubit durations are excluded. The result below gives the costs of this specific circuit. It does not give lower bounds for every synthesis method.

Corollary 2.7 (An infinite compiled mixed-only family under Π). *Let $L \geq 2$, $H \geq 1$, and $\varepsilon > 0$. On $n = L + 2 + 3H$ original bits consider*

$$f_{L,H}(x) = x_a x_b \sum_{t=1}^L x_{w_t} + \sum_{j=1}^H x_{u_j} x_{v_j} x_{z_j}.$$

Order the original variables as $a, b, w_1, \dots, w_L$, followed by each isolated triple in order, and append auxiliaries in pair order. Use $M_p = T_p + \varepsilon$ and budgets

$$Q_{\max} = n + 1, \quad G_{\max} = 2L + 6 + 10H, \quad D_{\max} = 2L + 6, \quad M_{\max} = L + \varepsilon. \quad (15)$$

The native circuit violates both the gate and depth budgets; every fully reduced direct-pair representation violates the width budget. Reducing only the star through its core meets all four budgets and is pointwise exact with a unique consistent minimising lift.

The mixed circuit meets the stated gate and depth budgets with one auxiliary. Native exceeds both circuit budgets. Every full reduction needs at least $H+1$ auxiliaries, so it exceeds width. Appendix B.9 counts the Pauli supports, CNOTs and layers explicitly.

This result depends on the fixed synthesis Π . It excludes every full reduction in the stated direct-pair class. It makes no claim about QAOA solution quality. Gate cancellation, different gadget orderings and routing can change the counts. The experiments below test some of those changes. Adding affine terms to $f_{L,H}$ does not change the result: affine terms add only one-qubit phases.

2.5 Resource and QAOA evaluation

The three fresh studies use separate instance sets: 90 instances for cost-layer compilation, 60 for preparation and evolution, and 60 random plus 12 mechanism instances for joint complete-circuit feasibility. A later five-seed check reuses 60 instances from the first study. They use different candidate sets, penalty rules and parts of the circuit. We do not pool their results.

2.5.1 Fresh joint-budget compilation study

The compilation study asks whether a strictly mixed representation can be the only feasible choice under joint budgets. A small weighted score is a different criterion. QAOA solution quality is also a different question.

Before generating data, we froze the code, generators, candidate counts, resource weights, compiler settings and budget grid. We then generated 30 new Max3SAT and 30 new cubic-spin instances. Each has six variables and five distinct cubic supports. Max3SAT signs and spin coupling signs are sampled independently. After converting the spin objectives to bits, we divide them by eight. These instances come from a new stated distribution. They are not added to the old evaluation pool.

We also generated 30 constructed instances. Each has nine variables, a four-leaf pair-star and one isolated cubic term. Vertex labels are random. Positive weights are sampled independently from $\{0.75, 1, 1.25\}$. This family tests the mechanism behind the theorem.

The original search cap was 32 candidates. This family has only 27 full assignments that fit the 12-qubit device. Before obtaining any compiler result for this family, we recorded a technical amendment: both searches use 27 candidates. The 60 random instances follow the original protocol. The original code, amendment, times and hashes are included with the data. This was a local freeze before evaluation, not an external preregistration.

For each target device, full and mixed search use the same resource score:

$$J_{\text{new}} = \frac{1}{4} \left(\frac{a}{4} + \frac{G_{\text{CX}}}{160} + \frac{D_{\text{CX}}}{120} + \frac{M_{\max}}{8} \right). \quad (16)$$

Each search evaluates 32 distinct candidates per random instance and target. It evaluates 27 per constructed instance. Candidates must fit device capacity. Full search fixes $D = E$. Mixed search can keep a cubic term or reduce it through any of its three pairs. It includes both endpoints.

Family	Line-12	Ring-12	Grid-12
Max3SAT, $N = 30$	16/8	16/8	18/8
Cubic spin, $N = 30$	1/0	1/0	1/0
Constructed, $N = 30$	30/29	30/22	30/28

Table 3: Independent instances with at least one certified strict-mixed-only cell anywhere in the frozen grid. Each entry is canonical/reuse synthesis. The constructed cohort follows the disclosed technical amendment.

Both searches include minimum-auxiliary, maximum-reuse and distance-aware full assignments. Other proposals use a seeded permutation. Mixed search also includes native, shared-pair groups and reuse prefixes. This is a finite search. It does not prove a global minimum. Reusing a stored compiler result does not add candidates to either budget.

We enumerate all 3^5 full assignments to compute their logical width and penalty. For the constructed family, we also compile all 27 full assignments that fit device capacity.

Each baseline asks a specific question. Minimum-auxiliary full reduction tests the effect of minimising only auxiliary count. Maximum reuse tests a simple pair-sharing rule. Resource-optimised full reduction tests the value of allowing terms to remain native, with equal candidate budgets. The distance-aware rule chooses the shortest original-bit pair and uses reuse to break ties. It is our simple hardware-inspired comparator, not a reproduction of a named hardware-aware algorithm.

The random control matches the auxiliary count a and reduced-term count $|D|$ of the best strictly mixed candidate. When possible, it also matches $M_{\max}$ exactly. A distinct penalty-matched control exists in 495 of 540 instance–target–synthesis settings. In 15 settings, no distinct control matches even the required counts; we mark them as degenerate. Several settings can share an instance. They are not independent samples.

All representations use $M_p = 1.1 \max(C_p^+, C_p^-)$ and symbolic γ . We combine equal Pauli terms before building the cost circuit. Qiskit 2.5.2 compiles it to $\{CX, R_Z, SX, X\}$. We use optimisation level 3, SABRE routing, identity initial placement and seed 1729. We repeat selection on each target: line-12, ring-12 and 3×4 grid-12. All three devices have the same capacity.

We test two gadget strategies. The canonical strategy orders gadgets by size and then lexicographically. The reuse strategy groups commuting gadgets by a frequent root and greedily orders their parity supports. We insert no gadget barriers. The compiler can therefore optimise across gadget boundaries. These are two strategies within one fixed compiler. They do not test independent compilers. Section 2.5.2 adds a separate five-seed check on the Max-3SAT and constructed cohorts.

G is CX count. D is scheduled CX depth, not physical duration. Q includes occupied input sites and every site used during routing. We also save the logical width $n+a$. The reported circuit resources cover only the cost layer. They exclude preparation and mixers.

We use the following frozen budget grid:

$$\begin{aligned}
Q_{\max} &\in \{6, \dots, 12\}, & G_{\max} &\in \{0, 4, \dots, 240\}, \\
D_{\max} &\in \{24, 48, 72, 96, 120, 160, 240\}, \\
M_{\max} &\in \{0, 1.1, 2.2, 3.3, 4.4, 6.6, 8.8\}.
\end{aligned}$$

At each cell, we ask which candidate representations meet all four limits. The five classes are: both endpoints feasible, native endpoint only, full endpoint only, strictly mixed only, and none found. Endpoint-only labels can still include feasible mixed candidates. “None found” means no candidate in our library was feasible. It does not prove that no feasible representation exists.

We also identify a stronger, certified mixed-only subset. Three checks must pass. Native must fit width but fail gate or depth. A compiled mixed candidate must meet every budget. Finally, exhaustive logical enumeration must exclude every full assignment by width or penalty. The cells describe a fixed grid. They are correlated and are not independent trials.

At least one setting has a certified mixed-only region for 23/30 Max3SAT and 30/30 constructed instances. Such a region survives all six settings for 3/30 and 21/30 instances, respectively. The spin result is weaker: only 2/30 instances have a certified region in any setting. None has one under reuse

Joint-budget feasibility: first indexed pair-star + isolated instance

Fixed slice: CX depth ≤ 96 , penalty ≤ 4.4 . Library feasibility; endpoint-only labels may include mixed designs.

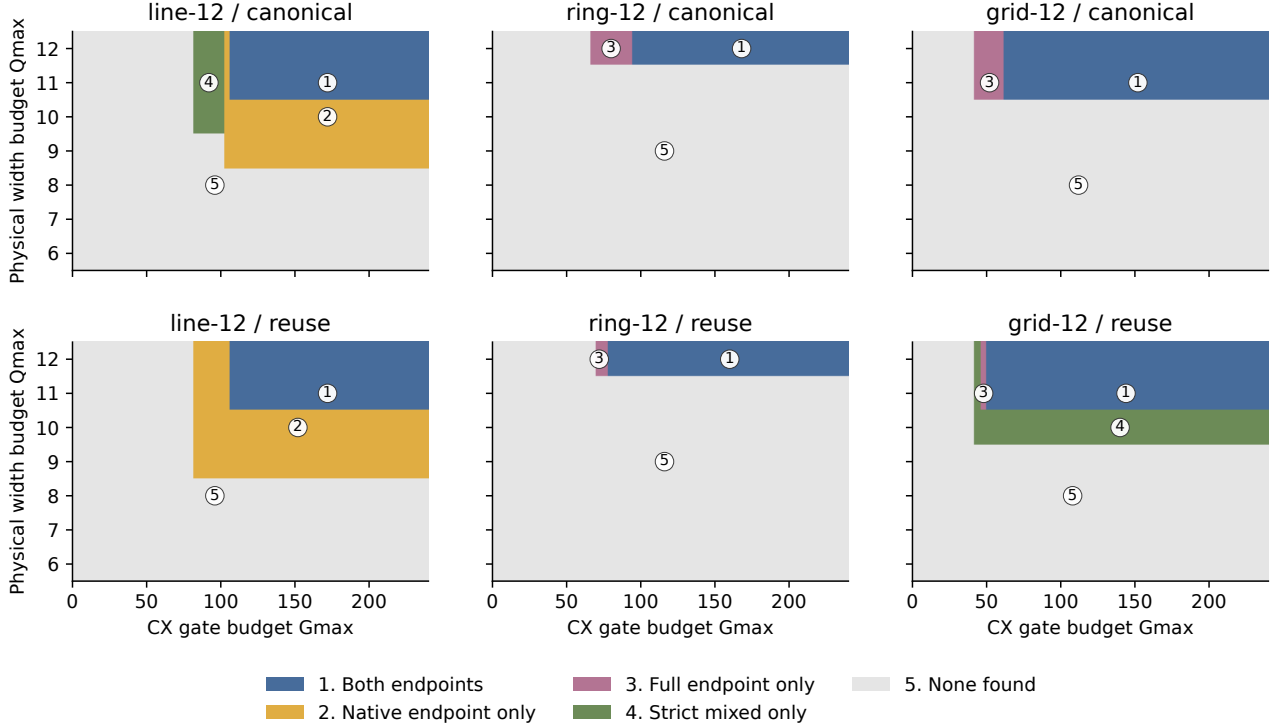


Figure 1: Real-compiler joint-budget phase maps for the first indexed constructed instance at $D_{\max} = 96$, $M_{\max} = 4.4$. Width is physical routing footprint. The same device capacity is used throughout. This is an illustrative slice; population counts use the complete frozen grid. The full endpoint is exhaustive over capacity-feasible assignments for this constructed family; the mixed candidate library remains finite.

synthesis. Mixed-only feasibility therefore occurs in the measured circuits, but it does not occur for every instance or every synthesis strategy.

The first indexed constructed instance gives a concrete example. Use ring-12 and canonical synthesis. The budgets are $(Q_{\max}, G_{\max}, D_{\max}, M_{\max}) = (12, 92, 48, 2.2)$. The measured resource vectors (Q, G, D, M) are:

- Native: $(12, 96, 55, 0)$. Width fits, but gates and depth exceed the limits.
- One full candidate: $(12, 80, 40, 3.85)$. Width, gates and depth fit, but penalty exceeds the limit.
- One strictly mixed candidate: $(12, 92, 43, 2.2)$. Every resource fits.

Checking one full candidate would not exclude the full endpoint. We therefore check all full assignments. Every one with logical width at most 12 has $M_{\max} \geq 2.475$. The others exceed width. Thus every full assignment fails, while the displayed mixed candidate fits. Recompile reproduces all three circuit records.

We chose this example after seeing the results, within the unchanged grid. It illustrates exact feasibility. It does not demonstrate better QAOA solution quality.

The best strictly mixed candidate has lower J_{new} than the resource-optimised full candidate on every instance in every setting. However, allowing native in the search changes the answer. The unrestricted search selects native in all random-instance settings and in 173/180 constructed settings.

The weighted score therefore usually favours native in this study. The reason to use mixed is the joint hard budgets: they can exclude native even when native has a lower weighted score. The matched random control sometimes equals or beats the selected mixed candidate. Our finite search does not guarantee that the chosen mixed candidate is the best in its matching class.

Changing synthesis can change representation rankings. We compare canonical and reuse scores on the same candidate identities. The median Spearman correlations on line, ring and grid are 0.913, 0.915, 0.949 for Max3SAT; 0.906, 0.905, 0.955 for cubic spin; and 0.771, 0.729, 0.812 for the constructed family. A correlation near one means that the score rankings mostly agree.

We also compare the Pareto sets in (Q, G, D, M) . A candidate is Pareto optimal within the library if no other candidate is no worse in every resource and better in at least one. Jaccard similarity measures the shared fraction of two Pareto sets. Its median values on line, ring and grid are 0.333, 0.250, 0.333 for Max3SAT; 1, 1, 1 for cubic spin; and 0.321, 0.351, 0.355 for the constructed family.

The winning full assignment changes on 6–14 of 30 Max3SAT instances, 10–13 of 30 spin instances and 0–4 of 30 constructed instances, depending on topology. Similar overall rankings therefore need not give the same frontier or winner. The study uses 31,577 compiler calls and 32,760 scheduled candidate evaluations before sharing repeated results. We independently check the logical phases and routed-circuit semantics.

2.5.2 What changes across five compiler seeds

We repeated cost-layer compilation for the same 30 Max-3SAT and 30 constructed instances from the preceding study. These are reused instances, not 60 new independent samples. We kept each target’s candidate library fixed and tested five routing seeds, three 12-qubit topologies and both synthesis strategies. All 101,460 planned compilations completed. The new run fixes eight SABRE trials. The earlier run used different effective routing settings, so we establish a new seed-1729 baseline. Methods in Section 4.5 give the full comparison.

Here a budget point is *mixed-only within the library* when an exact strictly mixed candidate meets all four limits, while every native or fully reduced candidate in that library fails at least one limit. This test does not exclude full reductions outside the library. It is therefore different from the exhaustive width/penalty certificate in the preceding subsection.

The new baseline has 149,439 such points across all instance, topology and synthesis settings. Of these, 79,069 remain mixed-only under every tested seed: a retained fraction of 52.91% (79,069/149,439). The same budget point must pass in all five seeds, but the feasible mixed candidate may change with the seed. This count does not prove that one candidate works for all five seeds. Budget points share instances and are not independent observations. Of the 360 instance–topology–synthesis settings, 49 retain every baseline TRUE point, 171 retain some but not all, 35 retain none of a nonempty baseline, and 105 have an empty baseline. Retaining every baseline point does not require identical regions: other seeds may gain new TRUE points. Supplementary Section S5 gives the results by family and topology, the changes in selected candidates and the corrected seed-by-seed display. Mixed-only budget points can survive these five seeds, but a substantial part of the baseline region does not.

2.5.3 Fresh evaluation: preparation and incremental evolution

The new QAOA study separates preparation from evolution. We freeze the code, seeds, hypotheses and budgets before generating data. Each family has 30 new independent instances with $n = 6$ and six clauses or spin interactions. Sampling is uniform under two conditions: all variables are covered, and at least two cubic monomials remain nonzero. The study is small, noiseless and limited to $p = 1$. Its freeze is local, not externally registered.

The earlier 14-, 20- and 18-instance sets remain exploratory. The signal-enriched set is a separate mechanism study.

We compare native, proper mixed ($0 < |D| < |E|$) and full reduction. Mixed and full selection use the same reference-compiler score $J_{\text{res}} = (a/4 + G/160 + D_2/120 + M/8)/4$. Each evaluates up to 128 distinct seeded candidates. If the domain is smaller, we evaluate it completely. Penalties equal the cubic threshold plus one. The fibre diagnostic is not used in selection.

The mixed comparison is conditional on using a proper mixed representation. If native is allowed, it has the smallest resource score among evaluated candidates on all 60 instances. This selection uses

Table 4: New $n = 6$ QAOA results, with 30 independent instances per family. Positive gain means a lower original objective. Intervals are paired 95% bootstrap intervals from 10,000 instance resamples. Evolution uses pair auxiliary warming and 3×120 equal-effort evaluations. Native evolution gain is the prespecified primary QAOA comparison. The other intervals are descriptive.

Family	Gain	Mean [95% interval]	Positive / 30
Weighted Max-3SAT	Preparation	0.484 [0.279, 0.703]	17
Weighted Max-3SAT	Native evolution	1.152 [0.959, 1.344]	30
Weighted Max-3SAT	Mixed evolution	1.086 [0.900, 1.277]	30
Weighted Max-3SAT	Full-Q evolution	0.778 [0.661, 0.891]	30
Cubic spin glass	Preparation	0.000 [0.000, 0.000]	0
Cubic spin glass	Native evolution	2.133 [2.004, 2.238]	30
Cubic spin glass	Mixed evolution	1.222 [1.063, 1.376]	30
Cubic spin glass	Full-Q evolution	0.648 [0.522, 0.780]	30

the all-to-all reference compiler. It is separate from the topology-specific compilation study. The two studies cannot establish a joint hardware-and-quality threshold on the same instances.

Every $p = 1$ equal-effort arm uses the same three parameter starts for an instance. Each restart receives exactly 120 objective evaluations. We run COBYLA first. If it stops early, we fill the remaining quota with uniform parameter samples from a fixed common seed. This is COBYLA plus quota completion, not 120 COBYLA iterations.

We minimise the exact original expectation $\mathbb{E}[f(x)]$. We sum over auxiliary outcomes. Encoded energy is a separate diagnostic. Starts are uniform on $[0, 2\pi) \times [0, \pi)$. We do not add $p = 0$ as an optimiser candidate. Positive gain over $p = 0$ is therefore not guaranteed by construction.

All representations use the same pair/triple McCormick–RLT relaxation. Original-bit probabilities are clipped to $[0.05, 0.95]$. For pair warming, we clip the auxiliary pair moments to this interval. For independence warming, we multiply the clipped original probabilities and clip the result to the same interval. Preparation and matched mixing otherwise use the same rules across arms.

Write $f_{\text{cold},0}$ for the cold initial expectation, $f_{\text{warm},0}$ for the warm initial expectation, and $f_{\text{warm},1}$ for the optimised warm $p = 1$ expectation. The total improvement is

$$\begin{aligned}
 f_{\text{cold},0} - f_{\text{warm},1} &= \underbrace{f_{\text{cold},0} - f_{\text{warm},0}}_{\text{classical preparation gain}} \\
 &\quad + \underbrace{f_{\text{warm},0} - f_{\text{warm},1}}_{\text{incremental evolution gain}} .
 \end{aligned}$$

Every representation has the same original-bit distribution at warm $p = 0$. Their original expectations are therefore equal. Independent sampling from these clipped marginals has the same expectation. We also report 2,048 actual samples per instance and deterministic rounding of the raw relaxation marginals, with ties rounded to zero. These controls measure preparation. They do not compare against a fully optimised classical algorithm.

For weighted Max-3SAT, the mean preparation gain is 0.484 [0.279, 0.703]. Native evolution adds 1.152 [0.959, 1.344] beyond warm $p = 0$. The mean original objectives for raw rounding, independent sampling and the prepared state are 2.267, 1.786 and 1.783.

Compared with independence warming, pair warming lowers the mean objective by 0.046 [-0.005, 0.112] for proper mixed and 0.119 [0.040, 0.202] for full-Q. The mixed interval includes zero. These are secondary results at a fixed optimisation budget. They do not establish a general pair-moment benefit.

For cubic spin glass, the mean preparation gain is 0.000 [0.000, 0.000]. Native evolution adds 2.133 [2.004, 2.238] beyond warm $p = 0$. The mean original objectives for raw rounding, independent sampling and the prepared state are -1.467, 0.010 and 0.000.

Compared with independence warming, pair warming changes the mean objective by gains of -0.051 [-0.175, 0.061] for proper mixed and 0.112 [-0.011, 0.238] for full-Q. Both intervals include zero. The spin data do not establish a pair-moment benefit.

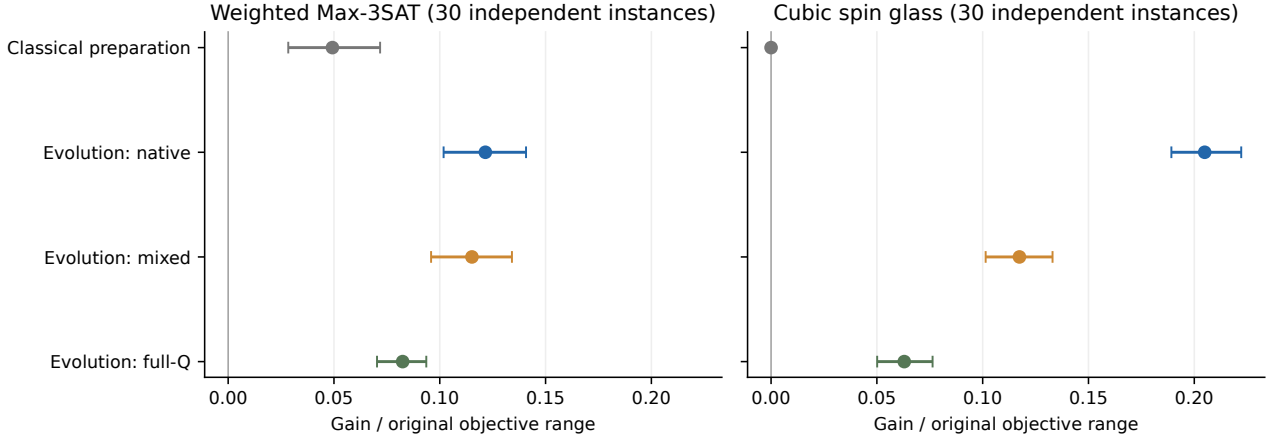


Figure 2: Classical preparation and subsequent $p = 1$ evolution on fresh instances. Each gain is divided by its instance’s exact original-objective range before averaging. Bars are paired instance-bootstrap 95% intervals; positive values reduce the original objective. The incremental bars use the same warm $p = 0$ comparator.

Table 5: Mean original objectives with pair warming. EE uses equal-effort COBYLA with quota completion. NM uses equal-effort Nelder–Mead. Scan uses a 65^2 grid and local refinement. Lower is better.

Family	Representation	EE	NM	Scan
Weighted Max-3SAT	Native	0.631	0.772	0.405
Weighted Max-3SAT	Proper mixed	0.697	0.799	0.533
Weighted Max-3SAT	Full-Q	1.005	1.057	0.891
Cubic spin glass	Native	-2.133	-1.979	-2.257
Cubic spin glass	Proper mixed	-1.222	-1.228	-1.495
Cubic spin glass	Full-Q	-0.648	-0.535	-0.888

Larger optimisation budgets. We also scan a 65×65 grid over the full (γ, β) domain. We then run four Nelder–Mead refinements, each with a 240-evaluation quota. This gives 5,185 evaluations per warm pair arm. If refinement stops early, the same uniform completion rule fills its quota. The encoded energies are integers, so the 2π phase period is valid. The zero-phase grid row includes the $p = 0$ expectation.

A second equal-effort comparison uses Nelder–Mead with the same three starts and 3×120 evaluations. These checks show how results depend on optimisation effort. The scan and local refinements do not prove a global optimum.

Of 172,800 equal-effort evaluations, 17,959 are uniform quota-completion samples. The second optimiser uses no completion samples. The scan-refinement stage uses 4,681. We retain all traces, early-stop messages and evaluation counts.

For weighted Max-3SAT, increasing effort from COBYLA to scan refinement changes the winning representation on 7/30 instances. The mean within-instance rank correlation is 0.717. Replacing COBYLA with equal-effort Nelder–Mead changes the winner on 9/30. One optimiser and one budget do not give a stable ranking for every instance.

For cubic spin glass, scan refinement changes the COBYLA winner on 1/30 instances. The mean within-instance rank correlation is 0.917. Equal-effort Nelder–Mead changes the winner on 2/30. The ranking is more stable here, but still depends on the protocol.

The new study uses independent instances and measures preparation and evolution separately. Its scope is small noiseless $p = 1$ simulation. It does not show a solution-quality advantage for mixed representations or a scalable quantum advantage. This study does not test diagnostic prediction. The separate joint study below addresses that question and distinguishes prediction from selection.

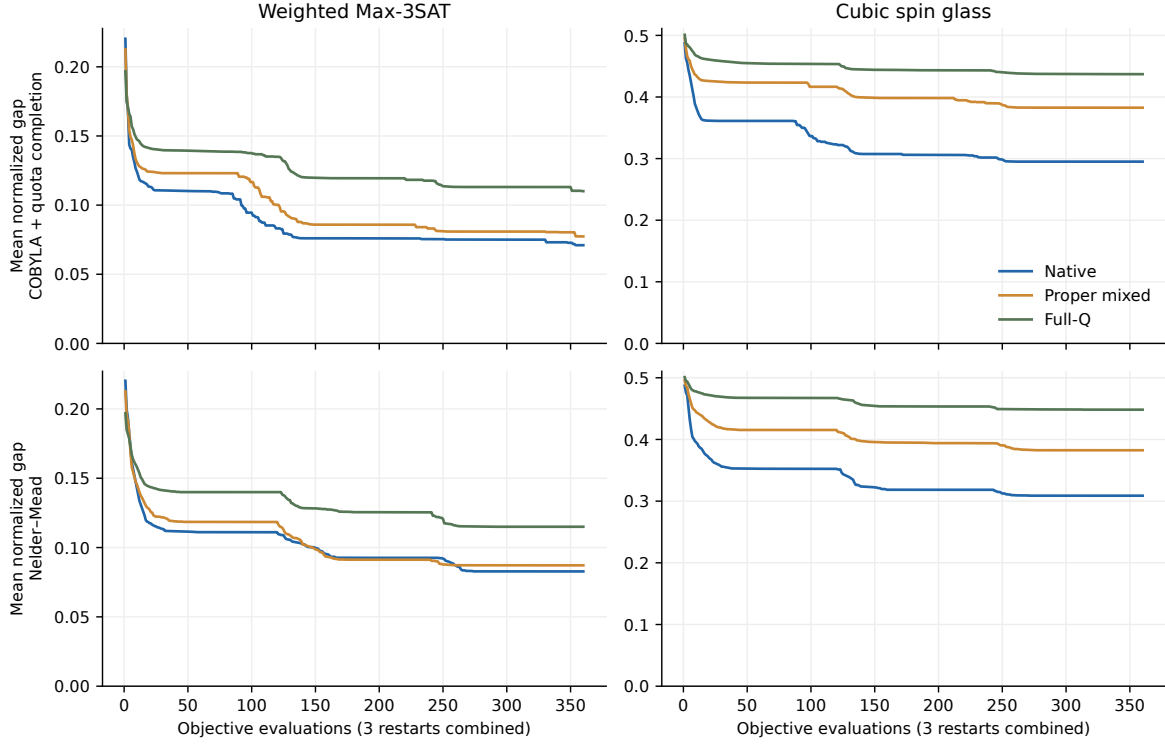


Figure 3: Incumbent original-objective gap, divided by each instance’s exact objective range, versus cumulative evaluations over three restarts. Each line averages 30 independent instances. All curves use pair warming. They distinguish equal-effort behaviour from the separately reported larger scan-and-refinement budget.

2.5.4 Joint resource and solution-quality test

The first two studies use different instances and circuit scopes. We therefore ran a third study that checks resources and quality on the same circuit. Its protocol and source hashes were saved before instance generation. This is a local freeze, not external preregistration. There are 30 new Max-3SAT and 30 new cubic-spin instances with six original bits. A separate set has 12 nine-bit pair-star mechanism instances. All 72 objectives are distinct, with no exact overlap with 462 supplied historical objectives. We analyse the three cohorts separately.

We compile the complete symbolic $p = 1$ circuit, including preparation, cost evolution and the matched mixer. The target is ring-12, with reuse synthesis and the fixed Qiskit level-3 settings. Full and proper-mixed selection use the same score in Eq. (16) and matched candidate quotas. Selection does not use QAOA outcomes. Each evaluated design uses three common starts and 120 Nelder–Mead evaluations per start, including the declared quota completion. Key designs, chosen before observing quality, also receive a 65^2 scan and four local refinements. Section 4.4 gives the protocol.

The primary test. The fixed resource limits are $(Q, G, D, M) \leq (12, 96, 64, 16)$. We measure quality on the original bits using the normalised gap

$$g = \frac{\mathbb{E}[f] - \min f}{\max f - \min f} \leq 0.2. \quad (17)$$

The extrema are found by exact enumeration for these small instances. Table 6 reports resource availability and joint success separately. Quality in the primary test uses the scan-refined results. No instance has mixed-only joint success at this primary budget. This is a negative result for the primary question.

Zero observed successes do not prove a zero population rate. The one-sided 95% binomial upper limits are 9.50% for 0/30 and 22.09% for 0/12. Resampling all-zero counts gives a degenerate bootstrap interval and does not resolve this uncertainty.

Table 6: Primary test. Each entry is the number that fits the resource budgets followed by the number that also meets $g \leq 0.2$. Mixed-only joint success is zero in every cohort.

Cohort	N	Native	Full	Proper mixed
Max-3SAT	30	30 $\rightarrow$ 30	8 $\rightarrow$ 7	30 $\rightarrow$ 30
Cubic spin	30	30 $\rightarrow$ 1	0 $\rightarrow$ 0	30 $\rightarrow$ 0
Mechanism	12	9 $\rightarrow$ 9	12 $\rightarrow$ 12	12 $\rightarrow$ 12

The secondary budget grid. The frozen secondary grid has 432 resource-budget combinations and quality targets 0.1, 0.2, 0.3. At $g \leq 0.2$, at least one cell has only the selected mixed design jointly successful for 7/30 Max-3SAT, 0/30 spin and 5/12 mechanism instances. These are instance counts, not independent cell trials. For 3/30, 0/30 and 4/12 instances, respectively, the evidence is stronger: native fails compiled resources, and exhaustive logical width/penalty checks exclude every full assignment in the stated grammar. Failure of one selected full design alone would not establish that result.

Table 7: Secondary grid at $g \leq 0.2$. Counts refer to independent instances with at least one qualifying cell. The last column also requires native resource failure and exhaustive resource exclusion of every full assignment.

Cohort	N	Selected mixed only	Full endpoint excluded
Max-3SAT	30	7	3
Cubic spin	30	0	0
Mechanism	12	5	4

One verified example. We chose `max3sat_07` after examining the results, within the unchanged secondary grid. Its budget is (8, 64, 64, 8). Native uses $(Q, G, D, M) = (6, 65, 50, 0)$, so it exceeds the CX limit. All 729 full pair assignments need at least nine logical qubits, so they exceed width. A selected mixed circuit uses (7, 58, 36, 7) and meets every resource limit. Its equal-effort gap is 0.1848065782, below 0.2; its warm $p = 0$ gap is 0.25. The saved-angle expectation was independently reproduced with the transpiled circuit statevector.

This example establishes conditional joint feasibility under this compiler, grammar and budget. It does not change the zero primary result. The best of 2,048 classical samples has gap zero on this instance. Across all cohorts, the best classical sample meets $g \leq 0.2$ on every instance. A best sample and a circuit expectation are different quantities. The comparison does not establish quantum advantage.

Optional fibre diagnostics. The no-guardrail, raw-moment and clipped prepared-state rules choose the same design on all 72 instances. The tested guardrails therefore give no observed selection benefit. We also test prediction on instances not used to fit the prediction model. The diagnostic helps predict final auxiliary inconsistency, but its benefit for final objective quality is uncertain. Better prediction does not show that a threshold improves selection. The guardrail remains optional. Supplementary Section S2 gives the results.

The study used 4,560 main compiler calls and 1,476,585 objective evaluations. Checks reproduce saved states and diagnostics; they do not rerun all optimisation. The Windows audit finds two total-gate-count differences, while all checked budget and score fields match. Appendix G.2 preserves the strict replay failure and the separate compatibility result. These are small noiseless simulations under one compiler configuration.

2.5.5 What the earlier studies establish

The earlier studies helped develop and check the method. They are not part of the fresh evaluation sets. Max-3SAT showed the clearest circuit savings; spin glass showed a gate-depth trade-off. Native had the best mean objective in the historical 14-instance QAOA comparison (Supplementary Table S5). The

20-instance signal-enriched warm-start study does not establish a general pair-moment advantage. The 18-instance noise test is exploratory and is not a calibrated hardware test. Supplementary Section S1 reports these studies.

3 Discussion

The main result is that mixed can be the only feasible exact representation. The static theorem needs no compiler. Corollary 2.7 also includes gates and depth, but only under its stated synthesis II. Its proof lists every contributing Pauli support and derives the CNOT layers. It is a count for this circuit, not a lower bound for every possible circuit. Both results concern the direct semantic pair-substitution grammar. They do not exclude other quadratization constructions.

The measured circuits show the same resource mechanism on sparse devices. Native can fail gates or depth while all full assignments fail width or penalty, leaving a feasible mixed design. This depends on the synthesis and target. Reuse synthesis removes the observed spin mixed-only regions. Similar score rankings can still give different Pareto sets and winners. Selection should therefore be repeated for the device and synthesis in use.

The five-seed follow-up gives a further limit on compiler robustness. Within its saved candidate libraries, 52.9% of the new seed-1729 mixed-only budget points remain mixed-only in all five seeds. The feasible candidate may differ between seeds. This supports the existence of shared feasible budget regions in the tested settings, but not a claim that every region or every selected representation is stable. It also does not exclude representations outside the libraries.

Hard budgets and weighted scores answer different questions. Native has the lowest unrestricted score in every random-instance setting of the compiler study. Mixed becomes useful when a hard budget excludes native. The separate QAOA study also favours native on mean original-objective quality under equal effort, a second optimiser and a larger parameter scan. Some instance winners change with effort. Exactness and smaller circuits do not determine QAOA quality. Earlier work on local structure and constraint density motivates this caution [31–33].

The third study tests resources and quality together. Its primary mixed-only counts are 0/30, 0/30 and 0/12. A fixed secondary grid contains conditional successes, including the verified `max3sat_07` example. This is evidence for existence at particular budgets, not a positive primary result or a general quality advantage. The best classical sample meets the quality target on every instance, so no quantum advantage is established.

Warm-start gains have two sources. The fresh study measures classical preparation and subsequent evolution separately. Native evolution improves on warm $p = 0$ on all 30 instances of each family under the equal-effort protocol. Pair warming has no benefit that holds across both families and all representations. The signal-enriched old study is a separate mechanism test. It cannot represent ordinary random instances. Results for standard QAOA started at a classical string [34] also do not directly apply to our matched product-state mixer.

The fibre study tests prediction and selection separately. After including the stated control variables, adding the prepared-state diagnostic reduces the mean squared prediction error for inconsistency by about 27.7% on held-out instances. The interval for better quality prediction includes zero. All three guardrail rules select the same designs. These results do not show that using the threshold improves the algorithm.

The historical 14-, 20- and 18-instance studies remain exploratory and are reported in the Supplementary Information. Different device capacities cannot isolate connectivity alone. A $p = 0$ row measures a budget limit, not completed positive-depth optimisation. A smaller noise-induced loss from a worse noiseless starting value need not give a better final value. The old coordinate correction and numerical environment changes also prevent attributing every change to the mixer convention alone.

The supported contribution is resource-constrained exact representation selection. The data are small and noiseless. Finite search returns the best design found; only a stated exhaustive certificate proves optimality or infeasibility. The primary and secondary tests, and the three fresh cohorts, must stay separate. Larger studies should test higher-degree search, additional compilers and positive-depth

quality under tight hardware budgets.

4 Methods

4.1 Resource-aware selection and certification

First fix a compilation protocol Π . Specify the device, synthesis, compiler seeds and aggregation rule. Then choose positive resource scales and nonnegative weights that sum to one. The score is

$$J_{\text{res}}(\sigma) = w_a \frac{a_\sigma}{s_a} + w_G \frac{G_{2q,\sigma}}{s_G} + w_D \frac{D_{2q,\sigma}}{s_D} + w_M \frac{M_{\text{max},\sigma}}{s_M}. \quad (18)$$

The compiler supplies gate count and depth. Polynomial degree alone does not determine them [9, 10, 24]. Primary selection uses resource constraints only. We compile candidates on the target being tested. Old all-to-all certificates still validate the old model. New finite-budget searches are heuristics unless we explicitly give an exhaustive certificate.

We keep the fibre excess only as an optional diagnostic (Appendix C.3). It measures the expected extra encoded energy from the auxiliary assignments:

$$E_{\text{fib}}(\sigma; \eta, \xi) = \mathbb{E}[F_\sigma(X, Y) - f(X)] \geq 0, \quad \hat{E}_{\text{fib}} = E_{\text{fib}}/s_{\text{fib}}. \quad (19)$$

Here all bits are independently Bernoulli with parameters (η, ξ) . The scale $s_{\text{fib}} > 0$ is the same for every design. Raw relaxation moments and clipped preparation probabilities give different product laws. This definition does not claim that raw moments come from one joint distribution.

The joint study tests whether this diagnostic adds predictive information beyond auxiliary count, compiled score, reduced-term count, prepared-state gap and family. It finds a limited benefit for predicting inconsistency on held-out instances. The benefit for predicting quality is uncertain, and the tested thresholds do not change the selected designs. Primary resource selection therefore does not use this diagnostic.

Keep only exact representations that meet all four budgets:

$$\mathcal{F}_{\text{sel}} = \left\{ \sigma \in \Sigma_{\text{ex}}^\varepsilon : \begin{array}{l} N_\sigma \leq Q_{\text{max}}, \quad G_{2q,\sigma} \leq B_G, \quad D_{2q,\sigma} \leq B_D, \\ M_{\text{max},\sigma} \leq B_M \end{array} \right\}, \quad (20)$$

Then minimise the score over that set:

$$J_{\text{res}}^\star = \min_{\sigma \in \mathcal{F}_{\text{sel}}} J_{\text{res}}(\sigma). \quad (21)$$

The set is finite, so a minimum exists if it is nonempty. To test the freedom to retain terms, compare full search with selective search:

$$\min_{\rho: (E, \rho) \in \mathcal{F}_{\text{sel}}} J_{\text{res}}(E, \rho) \quad \text{with} \quad \min_{\mathcal{D}, \rho: (\mathcal{D}, \rho) \in \mathcal{F}_{\text{sel}}} J_{\text{res}}(\mathcal{D}, \rho). \quad (22)$$

Every full representation is also selective. Exact search over the larger family therefore cannot have a worse best score. Finite searches may miss different candidates, so this ordering need not hold for their returned designs. We report best-found scores and compiler calls rather than calling them global optima.

Minimum-auxiliary full reduction tests a different resource objective. Maximum reuse tests a simple structural rule. A device-aware rule tests the value of device information. Random controls matched on a_σ and $|\mathcal{D}|$ compare choices with the same two counts. Those matches alone do not fix penalty or reuse.

Each cubic term can stay native or use one of three pairs. This gives a finite search tree. A Pareto beam keeps at most B prefixes at each step and records the best feasible design found so far. We call that design the *incumbent*. Discarding prefixes can miss better designs.

Optional branch and bound [35, 36] starts again from the root and reuses completed evaluations. It removes a subtree only after proving it infeasible or proving its score cannot improve the incumbent.

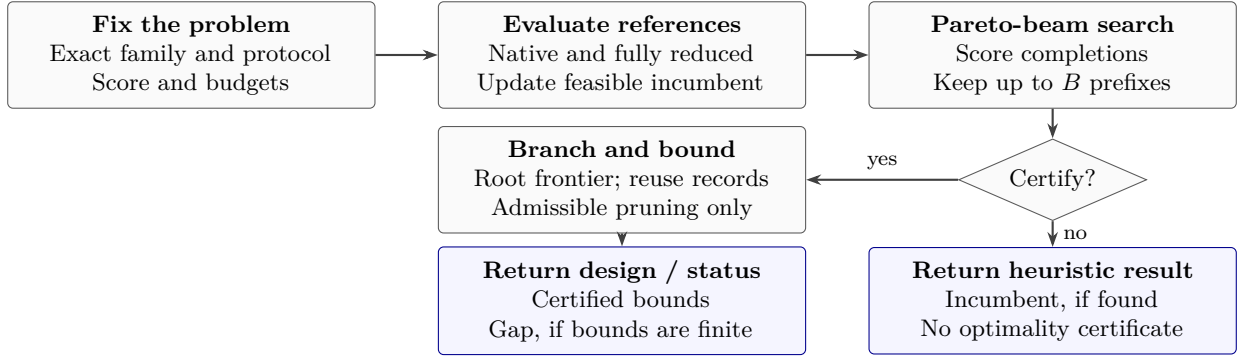


Figure 4: Representation selection separates a practical search from optional certification. The beam seeks a low-cost feasible design; its discarded prefixes remain part of the certification search. Branch and bound may return an optimal, infeasible or interrupted status, with bounds reported according to the stopping rules in Appendix C.5. The historical optional fibre constraint is not part of the primary selector.

A failed native completion alone is not such a proof: a descendant may reduce more terms and become feasible.

Let U_J be the score of the best feasible evaluated design. Each unresolved subtree has a valid lower score bound. Let L_J^{glob} be the minimum of these bounds and U_J . The frontier must still cover every unresolved design. If a feasible incumbent exists, then

$$L_J^{\text{glob}} \leq J_{\text{res}}^* \leq U_J, \quad U_J - J_{\text{res}}^* \leq U_J - L_J^{\text{glob}}. \quad (23)$$

The left inequality is a lower bound on the best possible score. The right inequality follows because the incumbent is feasible. Their difference bounds how far the incumbent can be from the optimum.

Equal bounds certify the optimal value only for the stated design space, compiler and constraints. Without an incumbent, we report that no feasible design has yet been found. We claim infeasibility only after exhausting the frontier. Figure 4 shows the procedure. The old optional fibre check in Algorithm 1 is disabled for primary selection. Appendix C gives the bounds and numerical rules. The proof of Proposition C.1 checks each part of the interval separately.

4.2 Warm-started QAOA and evaluation

For each representation, use the same relaxation to prepare a product warm start [28–30]. Clip original marginals by setting $\mu_i = \text{clip}_\delta(\bar{\mu}_i)$, where $\text{clip}_\delta(t) = \min\{1 - \delta, \max\{\delta, t\}\}$ and $0 < \delta < 1/2$.

The historical pair rule uses $q_p = \text{clip}_\delta(\bar{q}_{ij})$. Its independence fallback uses $q_p = \mu_i \mu_j$ without clipping again. The new QAOA study in Section 2.5.3 clips both auxiliary rules to the same interval. This is a stated difference between protocols.

Let $|\phi(t)\rangle = \sqrt{1-t}|0\rangle + \sqrt{t}|1\rangle$. The product state is

$$|\Phi_\sigma^{\text{ws}}\rangle = \bigotimes_{i=1}^n |\phi(\mu_i)\rangle \otimes \bigotimes_{p \in P_\rho} |\phi(q_p)\rangle. \quad (24)$$

For its ordered probabilities $\omega = (\mu, q)$, use the matched mixer

$$H_{M,\sigma}^{\text{ws}} = \sum_{j=1}^{N_\sigma} h_M^{(j)}(\omega_j), \quad h_M(t) = -2\sqrt{t(1-t)}X - (1-2t)Z. \quad (25)$$

The warm state is the unique ground state of this mixer Hamiltonian. Use $U_F(\gamma) = e^{-i\gamma H_{F\sigma}}$ and $U_M(\beta) = e^{+i\beta H_{M,\sigma}^{\text{ws}}}$. If every qubit has probability 1/2, then $H_{M,\sigma}^{\text{ws}} = -\sum_j X_j$. Thus $U_M(\beta) = e^{-i\beta \sum_j X_j}$, the cold mixer at the same angle.

After r layers, the state is

$$|\psi_{r,\sigma}\rangle = U_M(\beta_r)U_F(\gamma_r) \cdots U_M(\beta_1)U_F(\gamma_1)|\Phi_\sigma^{\text{ws}}\rangle, \quad (26)$$

At $r = 0$, it is the prepared state. All cost terms commute, so splitting the cost layer into phase gadgets is exact. The local mixer can leave the correct-auxiliary subspace. Penalties do not force every measurement to satisfy the auxiliary relations. Mixers that preserve a feasible subspace are another possible choice [37].

We score QAOA using the original objective: $\mathcal{E}_f(\psi) = \sum_{x,y} f(x)p_\psi(x,y)$, where $p_\psi(x,y) = |\langle x,y|\psi\rangle|^2$. Every auxiliary outcome contributes to this sum. Optimum probability also sums over all auxiliary outcomes.

We record auxiliary inconsistency separately as $R_{\text{inc}} = \mathbb{P}_\psi(Y \neq \phi_\rho(X))$. It is zero when there are no auxiliaries. Appendix D gives the circuit identities and diagnostic definitions. The protocol below specifies budgets and aggregation.

4.3 Benchmark construction and comparison protocol

Weighted Max-3SAT minimises violated-clause weight:

$$f_{\text{3SAT}}(\mathbf{x}) = \sum_C w_C \prod_{\ell_i \in C} u_{\ell_i}(x_i), \quad w_C > 0, \quad (27)$$

The factor is $u_{x_i} = 1 - x_i$ for a positive literal and $u_{\neg x_i} = x_i$ for a negative literal. A clause product is one exactly when every literal is false. The spin-glass benchmark uses

$$E(\mathbf{s}) = \sum_{\{i,j,k\} \in \mathcal{E}} K_{ijk} s_i s_j s_k, \quad s_i \in \{-1, 1\}, \quad K_{ijk} \in \{-1, 1\}, \quad (28)$$

Here $\mathcal{E}$ is the sampled set of triples. Set $s_i = 1 - 2x_i$, expand the products and combine equal monomials before reduction. Sampling changes sparsity and pair reuse. These two families provide different test problems. They do not isolate application type from hypergraph structure. Appendices G.3, E and F give the generators and expansions.

The old evaluation has three tiers: 60 oracle instances, 180 compilation instances and 14 test-labelled noiseless QAOA instances. Each tier is balanced between families. Oracle checks compare $f(\mathbf{x})$ with $\min_{\mathbf{y}} F_\sigma(\mathbf{x}, \mathbf{y})$ for every assignment. They also test threshold failures and consistency.

Separate studies use 18 noise instances and 20 strongest-screened warm-start instances. Warm-start screening uses only relaxation information. It does not use QAOA outcomes. The screened set is not a representative random sample and is not pooled with the original comparison. Candidate-pool generation targets differ from the numbers actually evaluated.

Each old instance has native, full, selective and matched-random designs. Full reduction uses a minimum-cardinality pair cover with deterministic tie-breaking. Five seeded random controls match only active auxiliary count. They may repeat and need not match circuit resources. Appendix H.3 gives their construction.

Within each family and tier, the split labels are 60%/20%/20% for training, validation and test. Settings use training and validation rows, with no direct selector retuning on test-labelled rows. However, the pool was available during protocol development. We therefore treat these labels as exploratory, not as an untouched test. Resource summaries pool all 180 compilation instances.

The old compiler study uses common synthesis settings and five transpiler seeds on all-to-all coupling and a 12-qubit 3×4 grid. It records width, two-qubit gates and depth, routing, penalties and coefficient range. Width failures are recorded separately. Paired device comparisons use only instances where all compared designs fit. Additional line, ring and heavy-hex tests compare selective and random designs. They do not rerun QAOA (Appendices C.2 and G).

Old QAOA runs use COBYLA [38], three restarts, at most 60 evaluations per restart and tolerance 10^{-3} . The loss is the original expectation $\mathcal{E}_f$. Encoded energy and consistency are diagnostics. Initial angles have $\gamma \in [0, 2\pi]$ and $\beta \in [0, \pi]$. These intervals choose starts; they do not bound the optimiser.

Equal-layer comparisons use $p = 1, 2$. For gate budget $B \in \{128, 256\}$, let $g_\sigma > 0$ be the median grid gate count of one cost layer. Assign

$$p_\sigma(B) = \lfloor B/g_\sigma \rfloor. \quad (29)$$

If no layer fits, keep the $p = 0$ initial-state baseline without optimisation. Report these counts separately, along with the subset where every design fits at least one layer. The targeted warm-start study uses only $p = 1, 2$.

Primary QAOA results use exact statevector expectations. The old E6 noise study instead uses 4096 shots per evaluation. It applies representative depolarising and readout errors on the grid, not a calibrated device model. Starting from Equation (29), it lowers depth until the complete transpiled measured circuit fits the budget. Comparisons use the resulting depth. Both noise levels reuse noiselessly optimised parameters without re-optimisation (Appendix G.7).

4.4 Joint-study protocol and statistical scope

The joint study saves its protocol, source hashes and six dependency versions before generating instances. The 30 Max-3SAT and 30 spin objectives use six variables and uniform generation, with variable coverage and at least two nonzero cubic supports. The 12 mechanism objectives use nine variables, four pair-star leaves and one disjoint triple. Labels are permuted uniformly. Cubic weights are sampled independently from $\{1, 2, 3\}$ and affine coefficients from $\{-2, -1, 0, 1, 2\}$. Their integer energies justify the 2π cost-angle period. Penalties are the exact construction-specific threshold plus one. Clipping uses $\delta = 0.05$.

We compile the complete symbolic $p = 1$ preparation, cost and matched-mixer circuit on ring-12. Settings are reuse synthesis, Qiskit 2.5.2 level 3, SABRE routing, identity initial layout, seed 1729 and basis $\{R_Z, SX, X, CX\}$. Width counts occupied input sites and all sites touched by routing. This differs from the cost-layer-only scope in Section 2.5.1.

The full and proper-mixed searches use the same candidate quota: the smaller of 32 and the sizes of their capacity-feasible design spaces. The mechanism quota is 27 in the initial freeze. Full candidates include assignments chosen for minimum auxiliary count, maximum pair reuse and short ring distances. Mixed candidates include groups that share a pair and prefixes of a reuse ordering. For each budget, full and mixed search separately choose the lowest score in Eq. (16). Ties use lexicographic actions. QAOA outcomes are not used to propose or select designs.

Every evaluated design uses three common parameter starts and 120 Nelder–Mead evaluations per start. If the optimiser stops early, seeded uniform samples fill the remaining quota. Before observing quality, we choose a set of designs for the larger scan. This set includes the native, full and three guardrail choices made under capacity limits alone, together with the full and mixed choices at the primary budget. Each distinct design receives a 65^2 grid and four refinements with 240 evaluations each. If several choices give the same design, they share one evaluation. The optimisation loss is the original expectation. Encoded energy and auxiliary consistency are separate diagnostics. Classical controls are the warm $p = 0$ expectation, deterministic rounding of raw marginals, and the mean and best of 2,048 independent samples.

The secondary grid is

$$\begin{aligned} Q_{\max} &\in \{8, 10, 12\}, & G_{\max} &\in \{64, 96, 128, 160, 192, 256\}, \\ D_{\max} &\in \{32, 64, 96, 128\}, & M_{\max} &\in \{2, 4, 8, 16, 32, 64\}. \end{aligned}$$

It has 432 cells, each checked at $g \leq 0.1, 0.2, 0.3$. Cells are correlated sensitivity checks, not independent trials. We distinguish a failed selected endpoint from exhaustive resource exclusion of all full assignments. The latter uses complete logical width/penalty enumeration under the same grammar and penalty rule.

The optional raw/prepared guardrails use the same proper-mixed bank with $Q \leq 12$ and threshold 0.02. The shared scale is $4 \sum_p (\sum_{e \supset p} |c_e| + 1)$, summing all cubic pairs. The raw rule uses relaxation moments; the prepared rule uses the clipped probabilities actually used to prepare the state. An empty eligible set is unavailable. We do not relax the threshold or insert a fallback.

For prediction, a fixed seed selects eight mixed candidates from each random instance before outcomes are observed. This gives 480 probes. We fit a linear model while leaving out all probes from one instance, then predict those probes. We repeat this for each instance. The control variables are auxiliary count, compiled score, reduced-term count, warm $p = 0$ gap and family. We add the raw

and prepared diagnostics separately and together. To form intervals, we resample the held-out errors by instance. We do not refit every model in each resample. These intervals are secondary, descriptive and unadjusted. Prediction does not test whether the threshold causes better selection. Instances are the statistical unit. The 12 mechanism instances are analysed separately from the random families. The two primary guardrail comparisons for the random families use 10,000 paired instance resamples and 97.5% intervals.

4.5 Five-seed compilation protocol

This follow-up reuses the Max-3SAT and constructed cohorts in Section 2.5.1. It contains 30 instances from each family and no cubic-spin instances. The smaller 10-per-family subset is nested in this sample; it is not a second sample. Instances are ordered by the saved SHA-256 selection rule.

For each instance and target, we use the saved full and mixed candidate libraries from the earlier study. We do not add candidates after examining the new seed results. There are 20,292 candidate–target–synthesis rows: 11,330 full, 8,602 strictly mixed and 360 native. Compiling each row at seeds {1729, 2718, 31415, 57721, 65537} gives 101,460 tasks. The full search is not exhaustive for the Max-3SAT family. All claims in this follow-up refer to the saved library.

The compiler is Qiskit 2.5.2 with basis $\{R_Z, SX, X, CX\}$ and optimisation level 3. The targets are line-12, ring-12 and grid-12. Each uses canonical and reuse synthesis. Initial placement is the identity. The explicit SABRE routing pass uses the decay heuristic and eight trials. Placement search is disabled. Thus the five seeds share the same routing settings. The earlier archive used implicit routing settings. Its seed-1729 output is not the control for this comparison: the new run supplies that control.

We use the same four-dimensional budget grid as in Section 2.5.1. It has $7 \times 61 \times 7 \times 7 = 20,923$ points per instance–target–synthesis–seed setting. Q counts occupied input sites and any extra sites used by routing. G counts CX gates, D is CX-only depth, and M is the largest Rosenberg penalty coefficient. These resources cover one cost layer; they exclude preparation and mixers. There is no quality threshold in this follow-up.

At each point we first check candidate exactness and strict mixed status. A point is TRUE when at least one such mixed candidate fits and no non-mixed candidate in the library fits. It is FALSE when this condition is ruled out by complete records. Missing evidence remains UNKNOWN; we do not count it as failure or as zero resource use. All saved tasks compiled successfully, and this run has no UNKNOWN points. For a fixed instance, topology and synthesis, we intersect the five TRUE sets. Retention is the size of this intersection divided by the size of the seed-1729 TRUE set. Pooled retention uses sums of these two counts. It is not the mean of the individual retention fractions.

For candidate rankings, we use J_{new} in Eq. (16) with the active auxiliary count in its width term. The archived selector is the lowest-score candidate within device capacity, without the other hard limits. Ties use the candidate identifier in lexicographic order. A changed winner and a changed feasible budget region answer different questions. Supplementary Section S5 reports both, and states the budgets used for the added check of the baseline winner. Appendix G.10 records the checks and their limits.

4.6 Use of language-model assistance

Language-model assistance was used to edit the text, develop and review code, and check proofs and saved numerical results. The archived protocols, source hashes and verification records distinguish scientific runs from later checks and editorial changes.

4.7 Statistics and reproducibility

The independent statistical unit is an instance. For old resource results, first take the median over transpiler seeds for each design. Then average the five random-design values within the instance. For old QAOA, first average three restarts per design. Then average the random-design means. Finally, form paired differences within each instance. We do not choose the best seed, restart or random draw.

Resource summaries use medians of paired differences. Old QAOA, warm-start and noise summaries use means and two-sided 95% Student- t intervals across instances [39]. The new studies use the aggregation and bootstrap rules stated in their own protocols.

Corrected warm-start intervals use $n - 1$ degrees of freedom. They are exploratory and unadjusted for multiple comparisons. An interval including zero does not establish equivalence. Seeds and restarts do not add independent instances. The descriptive regime labels use a separate endpoint-stability rule (Appendix H.7).

The old warm-start correction keeps the frozen instances, designs, probabilities, starts, seeds, budgets and loss. It does not resolve the LP or screen new instances. We rerun positive-depth cold controls because the optimiser environments are not known to be equivalent. We validate reuse at zero depth. Old and corrected outcomes are not pooled.

Original E4 uses the first HiGHS optimum. Selection and targeted screening use a deterministic representative of the LP optimum face. We preserve these different rules. Appendices H.4, G.5 and G.6 give the solver and execution records.

Data availability

The new evaluation data, frozen protocols, execution records and analysis scripts are supplied with this revision: the 17 September code-and-evidence archive, `URSS_Feedback_Iteration_v3.zip`, the Windows verification evidence, and the five-seed delivery described in Appendix G.10. The additional data corrections are supplied as a separate checked analysis. The review package preserves these inputs and their SHA-256 identities. The five-seed source and selected R12 acceptance evidence are available for review in the repository’s draft repair pull request; this does not imply publication of the complete scientific data or adoption of this manuscript. The historical processed data and source manifests are available in the [project repository](#). The complete raw data for the warm-start coordinate correction are publicly archived in the [versioned correction archive](#). The archive identity and SHA-256 are recorded in Appendix G.

Code availability

The five-seed source snapshot and the independent correction script accompany the additional results. The supplied snapshot has the coverage and environment limits stated in Appendix G.10.

The resource and QAOA evaluation code and reproduction instructions are included with this revision. The v3 scientific freeze is preserved; the separate portability verifier, effective audit and explicit audit amendment are supplied with the Windows evidence described in Appendix G.2. Historical code, together with the environment information and reproduction instructions, is available in the [project repository](#).

Acknowledgments

This work was supported in part by the National Hub for Edge AI, UK Research and Innovation (UKRI), through Research Grant EP/Y028813/1; by the UKRI Cross Council under Grant UKRI3734; by the Academic Centre of Excellence in Cyber Security Research, University of Warwick, under Grant EP/R007195/1; and by the University of Warwick Undergraduate Research Support Scheme (URSS).

Author contributions

J.L. developed the idea and theoretical framework, proved the analytical results and wrote the manuscript. Y.S. and R.Z. designed and implemented the numerical experiments, analysed the data and helped interpret the results. H.M.W., B.C. and C.M. provided research guidance and discussed the work. The final submission remains subject to all authors’ review and approval.

Competing interests

The authors declare no competing interests.

A Background identities and resource definitions

A.1 Multilinear and Pauli expansions

The same change between Boolean and spin variables is used in standard Ising formulations [40] and penalty-based QUBO models [41]. Here it gives the Pauli terms that we compile when we score each representation.

Every function $f : \mathbb{B}^n \rightarrow \mathbb{R}$ has the unique expansion

$$f(x) = \sum_{A \subseteq [n]} c_A x_A, \quad c_A = \sum_{B \subseteq A} (-1)^{|A|-|B|} f(\mathbf{1}_B), \quad (30)$$

where $\mathbf{1}_B$ is the characteristic vector of B [1, 2]. The monomial support, degree, and split at a target locality ℓ are

$$\text{msupp}(f) := \{A \subseteq [n] : c_A \neq 0\}, \quad \deg(f) := \max_{A \in \text{msupp}(f)} |A|, \quad (31)$$

$$\mathcal{M}_{>\ell}(f) := \{A \in \text{msupp}(f) : |A| > \ell\}, \quad f_{>\ell} := \sum_{A \in \mathcal{M}_{>\ell}(f)} c_A x_A, \quad (32)$$

$$f_{\leq \ell} := f - f_{>\ell}.$$

For the zero polynomial we use the harmless convention $\deg(0) = 0$. Since $N_i|x\rangle = x_i|x\rangle$, Eq. (2) implies $\lambda_{\min}(H_f) = \min_x f(x)$ and

$$U_f(\gamma) := e^{-i\gamma H_f}, \quad U_f(\gamma)|x\rangle = e^{-i\gamma f(x)}|x\rangle. \quad (33)$$

Expanding each $N_A = 2^{-|A|} \prod_{i \in A} (I - Z_i)$ and collecting terms gives [2]

$$H_f = \sum_{B \subseteq [n]} h_B Z_B, \quad h_B = (-1)^{|B|} \sum_{A \supseteq B} \frac{c_A}{2^{|A|}}, \quad Z_B := \prod_{i \in B} Z_i. \quad (34)$$

Suppose f is nonconstant, and choose a monomial of maximum degree d with support A . In the formula for h_A , no larger support can contribute: there are no monomials above degree d . Thus $h_A = (-1)^d c_A / 2^d \neq 0$. No Pauli term can have weight above d either. The largest nonzero Pauli weight therefore equals $\deg(f)$.

A.2 Consequences of pointwise exactness

For $F : \mathbb{B}^{n+m} \rightarrow \mathbb{R}$, define the optimal auxiliary fibre by

$$\mathcal{Y}_F(x) = \arg \min_{y \in \mathbb{B}^m} F(x, y). \quad (35)$$

Let $\mathcal{C}(x)$ contain the auxiliary assignments that satisfy the intended product relations. Constraint faithfulness means that every minimising auxiliary assignment is in this set: $\mathcal{Y}_F(x) \subseteq \mathcal{C}(x)$. If $\mathcal{C}(x) = \{\phi(x)\}$ has only one element, faithfulness means that $\phi(x)$ is the unique auxiliary minimiser.

Writing $\pi_x(x, y) = x$ and $f^* = \min_x f(x)$, Eq. (3) implies

$$F(x, y) \geq f(x), \quad \min_{x, y} F(x, y) = f^*, \quad \pi_x \left(\arg \min_{x, y} F(x, y) \right) = \arg \min_x f(x), \quad (36)$$

and

$$0 \leq f(x) - f^* \leq F(x, y) - f^*. \quad (37)$$

Here is each consequence separately.

Step 1: bound every encoded value. For fixed x , any value $F(x, y)$ is at least its minimum over auxiliaries. Exactness says that this minimum is $f(x)$. Thus $F(x, y) \geq f(x) \geq f^*$ for every (x, y) .

Step 2: attain the same global optimum. Choose an original minimiser x^* . Its finite auxiliary domain has a minimiser y^* . Exactness gives $F(x^*, y^*) = f(x^*) = f^*$. Step 1 rules out smaller values, so $\min_{x,y} F(x, y) = f^*$.

Step 3: check the projection in both directions. If (x, y) minimises F , then $F(x, y) = f^*$ by Step 2. Step 1 gives $f^* \leq f(x) \leq F(x, y) = f^*$. Hence x minimises f . Conversely, if x minimises f , choose an auxiliary minimiser at that x . Its encoded value is $f(x) = f^*$, so it is a global minimiser of F . This proves equality of the projected minimiser sets.

Step 4: compare the gaps. Subtract f^* from $f^* \leq f(x) \leq F(x, y)$. This gives Eq. (37). These statements concern classical values after minimising auxiliaries. They do not fix the energies of all other auxiliary assignments.

A.3 Representation and compiler resources

For $u = (x, y)$, a representation σ is written

$$F_\sigma(x, y) = \sum_{S \subseteq [N_\sigma]} c_S^{(\sigma)} u_S, \quad f(x) = \min_{y \in \mathbb{B}^{a_\sigma}} F_\sigma(x, y), \quad N_\sigma = n + a_\sigma, \quad L_\sigma = \max_{c_S^{(\sigma)} \neq 0} |S|. \quad (38)$$

Its quadratic interaction graph has vertex set $[N_\sigma]$ and edges $E_{2,\sigma} = \{\{i, j\} : c_{\{i,j\}}^{(\sigma)} \neq 0\}$; larger supports are hyperedges. For architecture $\mathcal{A}$ and compilation protocol $\mathcal{C}$, one compiled cost layer defines

$$\text{Compile}_{\mathcal{A},\mathcal{C}}(F_\sigma) = (G_{2q,\sigma}^{\mathcal{A},\mathcal{C}}, D_{2q,\sigma}^{\mathcal{A},\mathcal{C}}, S_\sigma^{\mathcal{A},\mathcal{C}}),$$

the two-qubit gate count, two-qubit depth, and inserted SWAP count [10, 24, 26]. We use the same architecture and protocol for every representation in a comparison and suppress their superscripts when fixed. The maximum active penalty is $M_{\max,\sigma} = \max_r M_{r,\sigma}$, with value zero if none is active.

For the collected expansion $H_{F_\sigma} = \sum_T h_T^{(\sigma)} Z_T$, define

$$\kappa_\sigma := \frac{\max_{T \neq \emptyset, h_T^{(\sigma)} \neq 0} |h_T^{(\sigma)}|}{\min_{T \neq \emptyset, h_T^{(\sigma)} \neq 0} |h_T^{(\sigma)}|}. \quad (39)$$

This definition applies when a nonidentity coefficient is present and is invariant under nonzero global rescaling. The identity coefficient is excluded because it contributes only a global phase to a cost layer. A wider range may demand finer relative control of the implemented Hamiltonian coefficients.

B Exact reduction: step-by-step proofs

B.1 Proof of the sharp one-step threshold

For the collected polynomial q and pair $p = \{i, j\}$ of Section 2.2, the selected support family obeys

$$\mathcal{S}_p \subseteq \{S \in \text{msupp}(q) : p \subseteq S, |S| \geq 3\} \quad (40)$$

The residual A_p does not contain u_i or u_j . The next inequality gives the energy gap for an incorrect auxiliary. Its indicator is one when $y_p \neq u_i u_j$ and zero otherwise. The bound is

$$Q_{p,M_p}(u, y_p) - q(u) \geq (M_p - \|A_p\|_\infty) \mathbf{1}_{\{y_p \neq u_i u_j\}}. \quad (41)$$

Proof of Proposition 2.1. Step 1: fix all original coordinates. Fix u and write $a = u_i$, $b = u_j$ and $A = A_p(v)$. Then $B_p(u)$ and A are fixed numbers. Only y_p can change. Subtracting the original polynomial gives

$$Q_{p,M_p}(u, y_p) - q(u) = (y_p - ab)A + M_p P_R(a, b, y_p).$$

Step 2: check the correct value. If $y_p = ab$, then the first term is zero and the penalty is zero. Thus $Q_{p,M_p}(u, ab) = q(u)$. This proves $\min_{y_p} Q_{p,M_p}(u, y_p) \leq q(u)$. To obtain equality, we must show that the incorrect value cannot be lower.

Step 3: check the incorrect value in every case. If $(a, b) = (0, 0)$, the incorrect value is $y_p = 1$. The objective change is A , and the penalty is $3M_p$. If $(a, b) = (0, 1)$ or $(1, 0)$, the incorrect value is also $y_p = 1$. The objective change is A , and the penalty is M_p . If $(a, b) = (1, 1)$, the incorrect value is $y_p = 0$. The objective change is $-A$, and the penalty is M_p . These calculations give

(a, b)	$Q_{p,M_p}(u, y_p) - q(u)$
$(0, 0)$	$A + 3M_p$
$(0, 1) \text{ or } (1, 0)$	$A + M_p$
$(1, 1)$	$M_p - A$

(42)

Step 4: prove the exact threshold. The second row is nonnegative for every v exactly when $M_p \geq -\min_v A_p(v)$. The third row is nonnegative for every v exactly when $M_p \geq \max_v A_p(v)$. The first row adds no further condition: $A + 3M_p = (A + M_p) + 2M_p$, and $M_p \geq 0$. We therefore need exactly

$$M_p \geq -\min_v A_p(v) \quad \text{and} \quad M_p \geq \max_v A_p(v), \quad (43)$$

Together these inequalities say $M_p \geq \max_v |A_p(v)| = \|A_p\|_\infty$. Under them, every incorrect value is no lower than $q(u)$. Together with Step 2, this proves pointwise exactness.

The same rows also prove necessity. If a positive maximum exceeds M_p , choose the maximising v and $(a, b) = (1, 1)$; the third row is negative. If a negative minimum has magnitude larger than M_p , choose that v and $(a, b) = (0, 1)$; the second row is negative. In either case the reduced minimum is below $q(u)$.

Step 5: bound the gap and check uniqueness. Each incorrect excess is at least $M_p - |A|$. It is therefore at least $M_p - \|A_p\|_\infty$. The correct excess is zero, so this proves Eq. (41) for both auxiliary choices.

If $M_p > \|A_p\|_\infty$, this gap is positive for every incorrect choice. The correct value is then the unique minimiser for every u . If $M_p = \|A_p\|_\infty$, the maximum absolute residual is attained because the Boolean domain is finite. A positive maximiser gives a zero third row. A negative maximiser in absolute value gives a zero second row. If the norm is zero, the second row is also zero. Thus equality allows a tie for some u . Below the threshold, Step 4 gives a strictly better incorrect value. This proves the stated if-and-only-if condition for uniqueness. $\square$

A small penalty example. Take the selected block $2x_1x_2x_3 - 3x_1x_2x_4$ and replace x_1x_2 . The residual is $A = 2x_3 - 3x_4$. For $(x_3, x_4) = (0, 0), (1, 0), (0, 1), (1, 1)$, its values are $0, 2, -3, -1$. Thus $\|A\|_\infty = 3$. A penalty $M = 3$ preserves the minimum value but allows a tie. For example, at $x_1 = 0, x_2 = 1, x_3 = 0, x_4 = 1$, choosing the incorrect $y = 1$ changes the value by $-3 + M = 0$. With $M > 3$, that change is positive and all incorrect choices are worse. With $M < 3$, the same assignment has a negative change, so exactness fails. If both coefficients were positive, the residual would be $2x_3 + 3x_4$. It reaches 5 when both bits are one. A penalty of 3 would then be too small, even though it bounds each coefficient separately. The penalty must therefore control the combined residual.

B.2 Proof of sequential compositional exactness

Start with $g^{(0)} = f$. Before step r , the available bits are $u^{(r-1)} = (x, y_{<r})$, where $y_{<r} = (y_1, \dots, y_{r-1})$. Choose two distinct current coordinates, indexed by i_r, j_r . Write their values as $a_r(u) = u_{i_r}$ and $b_r(u) = u_{j_r}$. Either coordinate can be an original bit or an earlier auxiliary. Factor their product out of the selected terms. Replace it with a new auxiliary y_r and add its penalty. The two polynomials are

$$\begin{aligned} g^{(r-1)}(u^{(r-1)}) &= B_r(u^{(r-1)}) + a_r b_r A_r(\hat{u}_r), \\ g^{(r)}(u^{(r-1)}, y_r) &:= B_r + y_r A_r + M_r P_R(a_r, b_r, y_r), \end{aligned} \quad (44)$$

Here $\hat{u}_r$ contains all current bits except the chosen pair. In the second line, B_r , A_r , a_r and b_r are evaluated at the same current assignment as in the first line. Compute A_r from this current polynomial. Do not restrict earlier auxiliaries to their intended products when taking its norm.

The margin δ_r is the penalty above the threshold. To define the correct auxiliary values, proceed in order. At step r , evaluate the chosen product using x and the previously defined correct auxiliaries. This gives $\phi_r(x)$. Collect them in $\Phi(x)$:

$$\begin{aligned}\delta_r &:= M_r - \|A_r\|_\infty, \\ \phi_r(x) &:= a_r(x, \phi_{<r}(x))b_r(x, \phi_{<r}(x)), \quad \Phi(x) := (\phi_1(x), \dots, \phi_R(x)).\end{aligned}\tag{45}$$

The notation $\phi_{<r}$ means $(\phi_1, \dots, \phi_{r-1})$. At the first step there are no earlier auxiliaries.

The next bound holds for any auxiliary vector y , including an incorrect one. Each indicator tests the product at that step using the actual earlier values $y_{<r}$:

$$g^{(R)}(x, y) - f(x) \geq \sum_{r=1}^R \delta_r \mathbf{1}_{\{y_r \neq a_r(x, y_{<r})b_r(x, y_{<r})\}} \geq 0.\tag{46}$$

If every margin is positive, only the recursively correct assignment minimises the final polynomial:

$$\arg \min_{y \in \mathbb{B}^R} g^{(R)}(x, y) = \{\Phi(x)\} \quad \text{for every } x \in \mathbb{B}^n.\tag{47}$$

Proof of Theorem 2.2. Step 1: apply the one-step bound to an arbitrary assignment. Fix x and an arbitrary full auxiliary vector y . At step r , use the prefix $(x, y_{<r})$ as the current assignment. We do not assume that its earlier auxiliaries are correct. The one-step bound applies to every current assignment, so it gives

$$g^{(r)}(x, y_{\leq r}) - g^{(r-1)}(x, y_{<r}) \geq \delta_r \mathbf{1}_{\{y_r \neq a_r(x, y_{<r})b_r(x, y_{<r})\}}.\tag{48}$$

Every margin $\delta_r = M_r - \|A_r\|_\infty$ is nonnegative.

Step 2: add the inequalities without skipping the cancellation. The left sides for the first two steps are

$$[g^{(1)}(x, y_1) - g^{(0)}(x)] + [g^{(2)}(x, y_1, y_2) - g^{(1)}(x, y_1)].$$

The two $g^{(1)}$ terms cancel. Adding the third step cancels $g^{(2)}$. Continuing through step R leaves only $g^{(R)}(x, y) - g^{(0)}(x) = g^{(R)}(x, y) - f(x)$. The right sides add to the sum in Eq. (46). That sum is nonnegative. Hence $g^{(R)}(x, y) \geq f(x)$ for every y , and therefore $\min_y g^{(R)}(x, y) \geq f(x)$.

Step 3: construct an assignment that gives equality. Set $y_1 = \phi_1(x)$ using the first chosen product. Then set $y_2 = \phi_2(x)$ using x and the correct first auxiliary. Continue recursively to obtain $y = \Phi(x)$. At every step the substituted product equals the auxiliary, and the penalty is zero. Thus each step leaves the preceding value unchanged:

$$g^{(R)}(x, \Phi(x)) = g^{(0)}(x) = f(x),\tag{49}$$

This gives $\min_y g^{(R)}(x, y) \leq f(x)$. Combining it with Step 2 proves Eq. (8).

Step 4: prove uniqueness when every margin is positive. Take any $y \neq \Phi(x)$. Let $r_\star$ be the first index at which they differ. All earlier auxiliaries agree, so $y_{<r_\star} = \phi_{<r_\star}(x)$. The intended product at step $r_\star$ is therefore exactly $\phi_{r_\star}(x)$. The chosen $y_{r_\star}$ differs from that product. Its indicator in Eq. (46) is one. All other terms in that sum are nonnegative. Hence

$$g^{(R)}(x, y) - f(x) \geq \delta_{r_\star} > 0.$$

Every other assignment has larger value than $\Phi(x)$. This proves Eq. (47). $\square$

B.3 Product grammar and target locality

Corollary B.1 (Selective target- ℓ reduction). *Fix $\ell \geq 2$ and $\mathcal{D} \subseteq \text{msupp}(f)$. There exists an acyclic Rosenberg reduction $F_{\mathcal{D},\ell}$ such that*

$$f(x) = \min_{y \in \mathbb{B}^m} F_{\mathcal{D},\ell}(x, y), \quad m \leq \sum_{A \in \mathcal{D}} \max\{|A| - \ell, 0\}, \quad (50)$$

and every selected monomial has terminal degree at most ℓ . A disjoint fresh-chain construction permits $M_{A,s} \geq |c_A|$ at each step; strict inequalities make the recursively consistent lift unique. If $\mathcal{M}_{>\ell}(f) \subseteq \mathcal{D}$, then $\deg(F_{\mathcal{D},\ell}) \leq \ell$.

A product grammar records which products can be replaced at each step. Its substitutions have the form $y_S = v_{B_S} v_{C_S}$, where $S = B_S \dot{\cup} C_S$ and $v_{\{i\}} = x_i$. If a block has more than one variable, its auxiliary must have been created at an earlier step. We can meet this requirement by processing blocks in order of size, so no circular dependency occurs. At the end, a support written as a product of t available blocks has locality t . When the auxiliary values are correct, that product equals the original monomial. This is the standard recursive Rosenberg construction [12, 13]. The proof below gives the auxiliary bound used here.

Proof of Corollary B.1. Step 1: handle one selected term. Let $A \in \mathcal{D}$ and write $k = |A|$. If $k \leq \ell$, leave the term unchanged; it needs no auxiliary. Suppose $k > \ell$. Order its variables as $A = \{i_1, \dots, i_k\}$. Introduce a fresh chain of $k - \ell$ auxiliaries:

$$y_{A,1} = x_{i_1} x_{i_2}, \quad y_{A,s} = y_{A,s-1} x_{i_{s+1}} \quad (2 \leq s \leq k - \ell). \quad (51)$$

The first step replaces $x_{i_1} x_{i_2}$ by $y_{A,1}$. The next step replaces $y_{A,1} x_{i_3}$ by $y_{A,2}$, and so on.

Step 2: check the term and penalty after each step. After the first step, the term is $c_A y_{A,1} x_{i_3} \cdots x_{i_k}$. After the second, it is $c_A y_{A,2} x_{i_4} \cdots x_{i_k}$. In general, after step s it is

$$c_A y_{A,s} \prod_{t=s+2}^k x_{i_t}. \quad (52)$$

The selected residual is $c_A \prod_{t=s+2}^k x_{i_t}$. Its absolute value is at most $|c_A|$ and reaches that value when all remaining bits are one. Its Boolean norm is therefore $|c_A|$. Proposition 2.1 allows $M_{A,s} \geq |c_A|$ at each step.

Step 3: check the final locality and count. After step s , the term has one auxiliary factor and $k - s - 1$ original factors. Its degree is $k - s$. At $s = k - \ell$, this degree is ℓ . The chain uses exactly $k - \ell$ auxiliaries.

Step 4: combine all selected terms. Use separate fresh chains for different selected supports. Do not apply later substitutions to the earlier quadratic penalty terms. Apply the chains in their dependency order. Each step is exact for the current polynomial, so Theorem 2.2 gives exactness of the whole construction. Summing the counts from Step 3 gives $\sum_{A \in \mathcal{D}} \max\{|A| - \ell, 0\}$ auxiliaries. The separate-chain construction already proves this existence bound. Allowing reuse adds further choices, so the minimum possible auxiliary count cannot increase. Strict penalties give a unique recursive lift by the same theorem.

Step 5: check the degree of the entire final polynomial. If every term above degree ℓ is selected, all terminal objective terms have degree at most ℓ . Every added Rosenberg penalty is quadratic. Since $\ell \geq 2$, those penalties also have degree at most ℓ . Thus $\deg(F_{\mathcal{D},\ell}) \leq \ell$. $\square$

B.4 Step-by-step proof of cubic mixed-sign exactness

Proof of Proposition 2.3. Step 1: separate the auxiliary choices. Fix x and let $z_p = x_i x_j$ for $p = \{i, j\}$. Subtract $f(x)$ from Eq. (9):

$$F_{\rho,M}(x, y) - f(x) = \sum_{p \in P_\rho} [(y_p - z_p) A_p(x) + M_p P_R(x_i, x_j, y_p)].$$

For fixed x , each summand depends on only one auxiliary y_p . If every auxiliary is correct, every summand is zero.

Step 2: compute the residual norm. Distinct cubic supports assigned to a fixed pair have distinct third bits. The residual is therefore a linear sum of independently selectable bits. Set positive-coefficient bits to one and negative-coefficient bits to zero to obtain its maximum C_p^+ . Reverse those choices to obtain its minimum $-C_p^-$. Thus $\|A_p\|_\infty = \max\{C_p^+, C_p^-\} = T_p$.

Step 3: prove sufficiency. If every $M_p \geq T_p$, the one-step proposition makes every summand nonnegative for every auxiliary choice. Their sum is therefore nonnegative. The all-correct choice gives zero, so $\min_y F_{\rho,M}(x, y) = f(x)$. This holds for every x . If every inequality is strict, every incorrect auxiliary gives a positive summand. Any incorrect auxiliary vector therefore has larger value than the all-correct vector. This proves uniqueness.

Step 4: prove necessity of every pair bound. Suppose some $M_p < T_p$. If $C_p^+ > M_p$, choose its third bits to reach $A_p = C_p^+$ and set $x_i = x_j = 1$. The incorrect value $y_p = 0$ has excess $M_p - C_p^+ < 0$. Otherwise $C_p^- > M_p$; choose $A_p = -C_p^-$ and set $x_i = 0, x_j = 1$. The incorrect value $y_p = 1$ has excess $M_p - C_p^- < 0$.

These choices are possible because A_p does not use x_i, x_j . After fixing x , set every other auxiliary to its correct product. Their summands are zero, even if their pairs share original bits with p . The total excess is then negative. Hence the auxiliary minimum is below $f(x)$ for this assignment. Pointwise exactness fails. This proves that every bound $M_p \geq T_p$ is necessary. $\square$

B.5 Auxiliary-reuse structure

For $\mathcal{D} \subseteq \mathcal{C}_3(f)$, define the pair shadow $\partial_2 \mathcal{D} := \bigcup_{A \in \mathcal{D}} \binom{A}{2}$. Within the one-pair-per-cubic-term construction, the minimum number of auxiliaries required to reduce every term in $\mathcal{D}$ is the set-cover number

$$\tau_2(\mathcal{D}) = \min \left\{ \sum_{p \in \partial_2 \mathcal{D}} z_p : \sum_{p \in \binom{A}{2}} z_p \geq 1 \ \forall A \in \mathcal{D}, \ z_p \in \{0, 1\} \right\}. \quad (53)$$

Related substitution-selection methods seek to minimise auxiliary variables [20, 21, 25]; Eq. (53) describes the minimum auxiliary count for this one-pair-per-cubic-term construction.

To see the first direction, take any pair assignment. Every selected cubic term contains its assigned pair. The active pairs therefore cover every selected support, so their number is at least the cover minimum. For the other direction, take any pair cover. For each cubic support, choose one covering pair that it contains. Use one auxiliary for each used pair and set its penalty above its residual threshold. This gives an exact reduction with no more auxiliaries than the cover size. An optimal cover therefore attains the minimum auxiliary count. This count does not minimise penalties or compiled resources.

If $\mathcal{D}$ is empty, no auxiliary is needed. For nonempty $\mathcal{D}$, define

$$\Delta_2(\mathcal{D}) := \max_{p \in \partial_2 \mathcal{D}} |\{A \in \mathcal{D} : p \subset A\}|,$$

one has

$$\left\lceil \frac{|\mathcal{D}|}{\Delta_2(\mathcal{D})} \right\rceil \leq \tau_2(\mathcal{D}) \leq |\mathcal{D}|.$$

For the lower bound, each pair belongs to at most $\Delta_2(\mathcal{D})$ selected cubic supports. Covering $|\mathcal{D}|$ supports therefore needs at least $\lceil |\mathcal{D}|/\Delta_2(\mathcal{D}) \rceil$ pairs. For the upper bound, choose one pair for each support; this uses at most $|\mathcal{D}|$ pairs.

We can also display pair reuse with an overlap graph G_{ov} , whose vertices are cubic supports and where A and B are adjacent when $|A \cap B| = 2$. All terms assigned to one auxiliary y_p belong to the clique

$$K(p) := \{A \in \mathcal{C}_3(f) : p \subset A\}.$$

B.6 Proof of the static feasibility boundary

Proof of Theorem 2.4. Recall that m is the total number of cubic terms. K_Q is the most terms we can reduce under the first three budgets. h_0 is the most terms we may leave native.

Step 1: prove necessity. Suppose a representation is feasible. It leaves at most h_0 native terms, so it reduces at least $m - h_0$ terms. Its reduced set meets the first three budgets, so it reduces at most K_Q terms. Thus

$$m - h_0 \leq |\mathcal{D}| \leq K_Q.$$

Rearranging gives $K_Q + h_0 \geq m$.

Step 2: prove sufficiency. Suppose $K_Q + h_0 \geq m$. The search space is finite, so some valid assignment reduces exactly K_Q terms. It meets the first three budgets by the definition of K_Q . It leaves $m - K_Q$ terms native. The assumed inequality gives $m - K_Q \leq h_0$, so it meets the fourth budget too. A feasible representation therefore exists.

Step 3: check native. Native uses no auxiliaries and no penalties. Its auxiliary degree is zero. It meets the first three budgets. It retains all m cubic terms. It is feasible exactly when $h_0 \geq m$.

Step 4: check full reduction. A full representation reduces m terms. Such a valid assignment exists exactly when $K_Q = m$, since K_Q cannot exceed m . It retains zero terms, so the fourth budget adds no restriction.

Step 5: identify the mixed-only region. Feasibility requires $K_Q + h_0 \geq m$. Excluding native requires $h_0 < m$. Excluding every full representation requires $K_Q < m$. If all three conditions hold, any feasible design must reduce at least one term and retain at least one term. It is strictly mixed. Conversely, if feasible designs exist and all are strictly mixed, neither endpoint can be feasible. The same three conditions must then hold. $\square$

B.7 Proof of the pair-star capacity formula

Proof of Theorem 2.5. We count how many terms each auxiliary can reduce, then show that the count can be attained.

Step 1: find the capacity of one pair. Suppose q unit-weight terms are assigned to a pair. Their penalty threshold is q , so the actual penalty is $q + \varepsilon$. The auxiliary has degree $q + 2$. It meets both budgets exactly when

$$q \leq \lfloor M_0 - \varepsilon \rfloor, \quad q \leq \lfloor \Delta_0 \rfloor - 2.$$

Thus a used pair can hold at most C terms. If $C = 0$, no pair can hold even one term. If $r_0 = 0$, no pair can be used. Either case gives $K_Q = 0$. For the rest of the proof, assume $C \geq 1$ and $r_0 \geq 1$.

Step 2: identify the only reusable pair in a star. Two distinct leaves have supports $\{a_i, b_i, w_t\}$ and $\{a_i, b_i, w_{t'}\}$. Their common pair is only $\{a_i, b_i\}$. Every other pair belongs to just one leaf. Pairs cannot be shared between stars, because the stars have disjoint vertices. The core of star i can therefore reduce at most $\min\{\ell_i, C\}$ terms. A private pair can reduce only one term.

Step 3: prove the upper bound. Consider any valid assignment with $r \leq r_0$ active pairs. Let I index the stars whose core pairs are active. Count one term for each active pair. An active core in star i can add at most $g_i = \min\{\ell_i, C\} - 1$ extra terms. Hence

$$|\mathcal{D}| \leq r + \sum_{i \in I} g_i.$$

There are at most $\rho_0 = \min\{r_0, s\}$ active cores. All gains are nonnegative. Replacing their sum by the largest ρ_0 gains and replacing r by r_0 can only increase this bound. Also, we cannot reduce more than all m terms. Therefore

$$|\mathcal{D}| \leq \min \left\{ m, r_0 + \sum_{j=1}^{\rho_0} g_{(j)} \right\}.$$

Step 4: construct an assignment that reaches the bound. Choose the ρ_0 cores with the largest gains. For each chosen core, assign $\min\{\ell_i, C\}$ leaves to it. This uses ρ_0 auxiliaries and reduces $\rho_0 + \sum_{j=1}^{\rho_0} g_{(j)}$ terms. Each such pair meets the penalty and degree budgets by Step 1.

If auxiliaries remain, choose an uncovered leaf and reduce it through a private pair. Repeat until the auxiliary budget is used or no leaf remains. These private pairs are distinct, and each holds one term. Each additional auxiliary therefore reduces exactly one additional term. The final number is $\min\{m, r_0 + \sum_{j=1}^{r_0} g_{(j)}\}$, as required. This construction also works when every gain is zero. $\square$

B.8 Proof of the infinite static mixed-only family

Proof of Corollary 2.6. The objective has L star terms and H isolated terms. All isolated terms use vertex sets disjoint from the star and from each other.

Step 1: check the proposed mixed design. Assign all L star terms to the core pair and retain all H isolated terms. The design uses one auxiliary. Its threshold is L , its penalty is $L + \varepsilon$, and its auxiliary degree is $L + 2$. It retains H cubic terms. Its resource vector is exactly

$$(1, L + \varepsilon, L + 2, H).$$

This equals the stated budget vector, so the design is feasible and exact.

Step 2: exclude native. Native retains all $L + H$ terms. Since $L \geq 1$, we have $L + H > H$. It fails the retained-term budget.

Step 3: exclude every full reduction. Reducing the star requires at least one pair. Reducing each isolated term requires a pair within that term. Disjoint vertex sets prevent reuse between these components. Every full reduction therefore needs at least $1 + H$ auxiliaries. Since $H \geq 1$, this exceeds the budget of one. This count applies to every pair assignment, regardless of the heuristic.

Step 4: obtain infinitely many instances. The argument holds for every integer $L, H \geq 1$. Increasing either parameter gives arbitrarily many terms and variables. Thus the construction is an infinite mixed-only family. $\square$

B.9 Proof of the compiled mixed-only family

Proof of Corollary 2.7. All counts use the fixed synthesis Π . Its ordering and layer rule are essential to the depth calculation.

Step 1: list the native star's Pauli supports. For one term,

$$x_a x_b x_{w_t} = \frac{1}{8}(I - Z_a)(I - Z_b)(I - Z_{w_t}).$$

Its weight-two supports are $\{a, b\}$, $\{a, w_t\}$ and $\{b, w_t\}$. Its weight-three support is $\{a, b, w_t\}$. When we sum the L terms, the common pair $\{a, b\}$ is collected once. The other pairs and triples are distinct. The coefficient of $Z_a Z_b$ is $L/8$. Each $Z_a Z_{w_t}$ and $Z_b Z_{w_t}$ coefficient is $1/8$. Each $Z_a Z_b Z_{w_t}$ coefficient is $-1/8$. All of these coefficients are nonzero, so collection removes none of them. The native star therefore has $2L + 1$ weight-two strings and L weight-three strings.

Step 2: list the reduced star's Pauli supports. Let y represent $x_a x_b$ and let $M = L + \varepsilon$. The reduced star is

$$y \sum_{t=1}^L x_{w_t} + M(x_a x_b - 2x_a y - 2x_b y + 3y).$$

Its weight-two supports are $\{a, b\}$, $\{a, y\}$, $\{b, y\}$ and $\{w_t, y\}$ for each leaf. Their Pauli coefficients are respectively $M/4$, $-M/2$, $-M/2$ and $1/4$. They are all nonzero. There are $L + 3$ weight-two strings and no weight-three strings.

Step 3: count CNOTs. A weight- k parity gadget uses $2(k - 1)$ CNOTs. The native star uses $2(2L + 1) + 4L = 8L + 2$. The reduced star uses $2(L + 3) = 2L + 6$. Each isolated native cubic term has three weight-two strings and one weight-three string, so it uses $3 \times 2 + 4 = 10$ CNOTs. Adding the H isolated terms gives

$$G_{2q}^{\Pi, \text{native}} = 8L + 2 + 10H, \quad G_{2q}^{\Pi, \text{mixed}} = 2L + 6 + 10H.$$

Step 4: schedule the native star's pair gadgets. The variable order is $a, b, w_1, \dots, w_L$. The pair gadgets are processed in this order:

$$(a, b), \quad (a, w_1), \dots, (a, w_L), \quad (b, w_1), \dots, (b, w_L).$$

Each pair gadget uses two CNOTs on its pair. After (a, b) , both core wires end at layer 2. After all (a, w_t) gadgets, wire a ends at layer $2L + 2$ and wire w_t ends at layer $2t + 2$. The first (b, w_1) gadget starts after layer 4 and ends at layer 6. The next ends at layer 8, and in general (b, w_t) ends at layer $2t + 4$. Thus after all pair gadgets,

$$d_a = 2L + 2, \quad d_b = 2L + 4, \quad d_{w_t} = 2t + 4.$$

Step 5: schedule the native star's triple gadgets. The first triple is (a, b, w_1) . Its CNOTs are $a \rightarrow w_1$, $b \rightarrow w_1$, $b \rightarrow w_1$, $a \rightarrow w_1$. Since $L \geq 2$, the four layers are

$$2L + 3, \quad 2L + 5, \quad 2L + 6, \quad 2L + 7.$$

The gap before the second CNOT is needed because wire b is busy through layer $2L + 4$. After this gadget, a ends at $2L + 7$ and b at $2L + 6$. Every remaining leaf wire is free by layer $2L + 4$. Each subsequent triple therefore adds four consecutive layers on the core wires. The final native star depth is

$$(2L + 7) + 4(L - 1) = 6L + 3.$$

Step 6: schedule the reduced star and isolated terms. The auxiliary is appended after the original variables. The reduced star's pair order is (a, b) , (a, y) , (b, y) , then $(w_1, y), \dots, (w_L, y)$. The first three gadgets finish at layers 2, 4 and 6. Each leaf gadget shares wire y and adds two layers. The reduced star depth is $6 + 2L$.

An isolated native triple has pair gadgets finishing at layers 2, 4 and 6. Its four triple CNOTs occupy layers 7, 8, 9 and 10. Its depth is 10. Different components use disjoint wires, so their layer counts combine by taking the maximum, not by adding them. Since $2L + 6 \geq 10$ for $L \geq 2$, we obtain

$$D_{2q}^{\Pi, \text{native}} = 6L + 3, \quad D_{2q}^{\Pi, \text{mixed}} = \max\{2L + 6, 10\} = 2L + 6.$$

Step 7: check gate and depth budgets. The mixed gate count equals $G_{\max}$ and its depth equals $D_{\max}$. Native exceeds the mixed gate count by $(8L + 2) - (2L + 6) = 6L - 4 > 0$. It exceeds mixed depth by $(6L + 3) - (2L + 6) = 4L - 3 > 0$. It therefore violates both budgets.

Step 8: check width, penalty and exactness. The mixed design uses one auxiliary, so its width is $n + 1 = Q_{\max}$. Its only penalty is $L + \varepsilon = M_{\max}$. Since $\varepsilon > 0$, Proposition 2.3 gives exactness and a unique correct auxiliary minimiser for every x .

Every full direct-pair reduction needs at least $H + 1$ auxiliaries by Step 3 of the static-family proof. Its width is at least $n + H + 1 > n + 1$, since $H \geq 1$. Thus all full reductions fail width, while the mixed design meets every budget. $\square$

C Compilation and Selection Details

This appendix preserves the historical reference compiler and search routines underlying the development evidence. The new primary resource-only formulation is in Section 4.1; historical fibre checks below are optional, and new target-specific compiler settings are reported with their study. We first fix how a complete design is compiled and scored, then describe candidate evaluation, beam search, and certification. The reference compiler is a mathematical cost model; the device-level protocol fixes the compiler used for a particular experiment.

C.1 Reference compilation

Use an all-to-all architecture with capacity $Q_{\max}$, directed CNOT gates, and arbitrary R_z rotations. Place original qubits in index order, followed by active auxiliaries in lexicographic pair order. A design with $N_{\sigma} > Q_{\max}$ is infeasible for this architecture.

Convert F_{σ} to the collected Pauli expansion of Eq. (34):

$$H_{F_{\sigma}} = \sum_T h_T^{(\sigma)} Z_T. \quad (54)$$

Combine equal strings before compilation and remove only exact zero coefficients. The parameter γ remains symbolic, so special numerical angles do not remove gates. The identity term is a global phase and is omitted. For a nonempty support $T = \{q_1 < \dots < q_k\}$, use

$$\text{CX}_{q_1, q_k} \cdots \text{CX}_{q_{k-1}, q_k} R_z^{(q_k)}(2\gamma h_T^{(\sigma)}) \text{CX}_{q_{k-1}, q_k} \cdots \text{CX}_{q_1, q_k}. \quad (55)$$

The CNOTs compute parity on q_k , the rotation supplies the required phase, and uncomputation restores the bits. The gadget therefore implements $e^{-i\gamma h_T^{(\sigma)} Z_T}$ using $2(k-1)$ CNOTs [6, 9]. For $k = 1$, it is just an R_z gate.

Process strings by increasing weight and then lexicographically, with no cross-gadget cancellation or reordering. Writing

$$p_{k, \sigma} := |\{T : |T| = k, h_T^{(\sigma)} \neq 0\}|, \quad (56)$$

the cubic design family has reference cost

$$G_{2q, \sigma}^{\text{ref}} = 2p_{2, \sigma} + 4p_{3, \sigma}, \quad S_{\sigma}^{\text{ref}} = 0. \quad (57)$$

These are costs of this synthesis, not lower bounds on arbitrary implementations. Phase-gadget optimisation can reduce gate count and depth [42]; commuting Pauli exponentials can also be synthesised jointly [43]. Such transformations define a different compilation oracle. The new compiler study evaluates two optimised gadget strategies directly on each target; the analytical reference counts remain statements about the synthesis above.

To compute two-qubit depth, scan the emitted CNOTs (a_j, b_j) in order. Initialise each qubit's last occupied layer to $d_q^{(0)} = 0$, and update

$$\lambda_j = 1 + \max\{d_{a_j}^{(j-1)}, d_{b_j}^{(j-1)}\}, \quad (58)$$

$$d_q^{(j)} = \begin{cases} \lambda_j, & q \in \{a_j, b_j\}, \\ d_q^{(j-1)}, & \text{otherwise.} \end{cases} \quad (59)$$

Then

$$D_{2q,\sigma}^{\text{ref}} = \max_q d_q^{(G_{2q,\sigma}^{\text{ref}})}, \quad (60)$$

with value zero if no CNOT is emitted. Each gate is placed in the first layer after both of its qubits are available. This preserves the gate order on each wire and is the earliest-layer schedule for that fixed CNOT stream; single-qubit durations are not counted. Routing is addressed separately below.

C.2 Fixed device-level protocol

For device-level costs, fix the complete protocol before comparing designs:

1. *Circuit construction*: coefficient aggregation, zero tests, parameter handling, qubit order, and synthesis templates.
2. *Device and compiler*: coupling map, basis gates, software versions, layout, routing, optimisation, scheduling, numerical tolerances, and tie-breaks.
3. *Evaluation*: a common finite seed bundle $\mathcal{S}$, an aggregation rule Agg , and the stage at which each resource is counted.

For a nonempty bundle, aggregate each metric over all its seeds; for an empty bundle, use one deterministic compiler call. These choices define the oracle

$$\mathcal{O}_{A,C}(\sigma) = (G_{2q,\sigma}^{A,C}, D_{2q,\sigma}^{A,C}, S_\sigma^{A,C}). \quad (61)$$

The certificate assumes that this oracle is deterministic and returns finite metrics for every design that passes the declared pre-compilation constraints. It is relative to this protocol, not to all possible compilations [10, 24]. A failed or interrupted compiler call is an unresolved evaluation, not evidence that a design is infeasible. Calibration-aware mapping can incorporate spatial and temporal error variations [44]; doing so would require a declared calibration target in addition to the coupling graph.

C.3 Computing the fibre diagnostics

Use the independent product law with probabilities (η, ξ) from Section 4.1. Raw selector moments and prepared-state probabilities can be substituted separately. For $p = \{i, j\}$, write $t_p = \eta_i \eta_j$. The mismatch probability and expected penalty are

$$\nu_p = \xi_p(1 - t_p) + (1 - \xi_p)t_p, \quad \ell_p = t_p + \xi_p(3 - 2\eta_i - 2\eta_j). \quad (62)$$

The first identity separates $Y_p = 1, X_i X_j = 0$ from $Y_p = 0, X_i X_j = 1$; the second follows by taking expectations in Eq. (4). Since A_p depends on neither X_i nor X_j , independence gives

$$E_{\text{fib}}(\sigma; \eta, \xi) = \sum_{p \in P_p} [(\xi_p - t_p)A_p(\eta) + M_p \ell_p], \quad A_p(\eta) = \sum_{A \in \mathcal{D}_p} c_A \prod_{k \in A \setminus p} \eta_k. \quad (63)$$

Indeed, subtract f from Eq. (9) and take expectations term by term. When $\xi_p = t_p$, the interaction correction is zero.

For a nonempty cubic support, a design-independent normalisation is

$$s_{\text{fib}} = 4 \sum_{p \in \partial_2 \mathcal{C}_3(f)} \left(\sum_{A \supset p, A \in \mathcal{C}_3(f)} |c_A| + \varepsilon_p \right).$$

If there are no cubic terms, set $s_{\text{fib}} = 1$; the unique design is native and its fibre excess is zero. Two optional diagnostics are

$$\Lambda_\sigma = \min \left\{ 1, \sum_{p \in P_\rho} \nu_p \right\}, \quad R_{\text{fib}}(\sigma) = \frac{E_{\text{fib}}(\sigma)}{4 \sum_{p \in P_\rho} M_p} \quad (P_\rho \neq \emptyset).$$

The sum of the individual mismatch probabilities bounds the probability that any auxiliary is wrong. Capping that sum at one gives Λ_σ . This union bound still holds when two mismatch events share original variables. To bound R_{fib} , consider one pair. Exactness makes its excess nonnegative. Its excess is at most $\|A_p\|_\infty + 3M_p \leq 4M_p$. Summing over pairs and dividing by the stated denominator gives $0 \leq R_{\text{fib}} \leq 1$. These diagnostics use the initial product distribution. They do not assume that the bits remain independent after QAOA evolution.

C.4 Evaluation and search routines

A tree of representation choices. Order the cubic supports as $\mathcal{C}_3(f) = \{A_1, \dots, A_m\}$. For each term A_t , choose one of four actions:

$$\alpha_t \in \mathcal{U}_t := \{\perp\} \cup \binom{A_t}{2}.$$

The action $\perp$ retains the term, while a pair p reduces it through y_p . A complete vector $\alpha = (\alpha_1, \dots, \alpha_m)$ therefore determines a design $\sigma(\alpha)$, with canonical penalties computed from all terms assigned to each pair. For example, if $A_1 = \{1, 2, 3\}$ and $A_2 = \{1, 2, 4\}$, the vector $(\{1, 2\}, \perp)$ reduces only the first term, whereas $(\{1, 2\}, \{1, 2\})$ reduces both using the same auxiliary. Thus the resource cost depends on the combined choices.

The search fixes one action at each level, giving a tree with 4^m leaves. A partial state $s = (k, \alpha_{1:k})$ fixes the first k actions; $\text{Desc}(s)$ denotes the complete designs extending that prefix. The root is $s_0 = (0, \emptyset)$. To evaluate a partial state, the beam uses its *native completion*

$$\bar{\sigma}(s) := \sigma(\alpha_{1:k}, \perp^{m-k}),$$

which leaves every undecided term native. This is one complete design in $\text{Desc}(s)$; its score need not bound the scores of the others.

Finding a feasible incumbent. The algorithm first evaluates the native design and a fixed fully quadratized reference. The width- B beam then expands the retained states, evaluates their native completions, and keeps at most B states at each level using resource-score and Pareto rankings. Each feasible evaluation can improve the *incumbent*, the best feasible design found so far. If $\mathcal{V}$ is the set of evaluated complete designs, its value is

$$U_J := \min_{\sigma \in \mathcal{V} \cap \mathcal{F}_{\text{sel}}} J_{\text{res}}(\sigma),$$

with $U_J = +\infty$ until a feasible design is found. A finite U_J is an upper bound on J_{res}^* , but beam truncation can miss better designs. It therefore provides no optimality certificate.

Certifying the remaining search. Certification instead needs a lower bound valid for *every* feasible completion below a state:

$$L_J(s) \leq \inf_{\sigma \in \text{Desc}(s) \cap \mathcal{F}_{\text{sel}}} J_{\text{res}}(\sigma), \quad \inf \emptyset := +\infty. \quad (64)$$

The bound used here accounts for auxiliaries and penalty strengths already forced by the prefix. These cannot decrease when further actions are fixed. For compiled gate count and depth, zero is a valid default lower bound; the native completion's compiled cost is not automatically a lower bound. Appendix C.4 gives the formula and deterministic search rules.

Branch and bound starts from the root, reusing previous evaluations and the incumbent. It maintains an unresolved frontier `Open`. A state is removed only if all its completions are proved infeasible, or if $L_J(s) > U_J$; otherwise, a leaf is evaluated and an internal state is replaced by its children. Thus every unevaluated design not excluded by a valid proof remains covered by the frontier. In particular, branches dropped by the beam are not discarded from the certification search.

At each completed frontier update, define

$$L_J^{\text{glob}} := \min \left\{ U_J, \min_{s \in \text{Open}} L_J(s) \right\}, \quad (65)$$

where the inner minimum is $+\infty$ for an empty frontier. This combines the best value already found with the lower bounds on unresolved designs; if the frontier is exhausted and an incumbent exists, it equals U_J .

Algorithm 1 Compiler-aware representation selector

Require: Ordered cubic supports; fixed exact family and protocol Π ; beam width B ; flag `Certify`.

Ensure: Incumbent and upper bound U_J ; optionally, a certified lower bound and stopping status.

```

1: Initialise the shared search state  $\mathbf{X}$  with empty evaluation records, no incumbent, and  $U_J = +\infty$ .
2: for all  $\sigma \in (\sigma_{\text{nat}}, \sigma_{\text{fq}}^{\text{ref}})$  do
3:    $(\mathbf{X}, \text{done}) \leftarrow \text{EVALUATECANDIDATE}(\sigma, \mathbf{X}, \Pi)$ .
4:   if  $\text{done} = \text{false}$  then
5:     break.
6:   end if
7: end for
8:  $\mathbf{X} \leftarrow \text{PARETOBEAM}(s_0, B, \mathbf{X}, \Pi)$ 
9: if Certify then
10:   $(\mathbf{X}, \text{Open}, L_J^{\text{glob}}, \text{status}) \leftarrow \text{BRANCHANDBOUND}(s_0, \mathbf{X}, \Pi)$ 
11:  return  $(\sigma_{\text{inc}}, U_J, L_J^{\text{glob}}, \text{status})$ 
12: else
13:  return  $(\sigma_{\text{inc}}, U_J)$ 
14: end if
```

For the action tree just defined, use a fixed action order: $\perp$ first, followed by pairs in lexicographic order. For $s = (k, \alpha_{1:k})$ and $k < m$,

$$\bar{\alpha}(s) = (\alpha_{1:k}, \perp^{m-k}), \quad \bar{\sigma}(s) = \sigma(\bar{\alpha}(s)), \quad \text{child}(s, u) = (k+1, \alpha_{1:k}, u). \quad (66)$$

Only complete designs are evaluated. A partial state's native completion is one such design, not a bound on its other completions.

Shared records and candidate evaluation. All routines share

$$\mathbf{X} = (\mathcal{K}, \mathcal{V}, \mathcal{P}_{\text{emp}}, \sigma_{\text{inc}}, U_J), \quad (67)$$

initialised to $(\emptyset, \emptyset, \emptyset, \text{NONE}, +\infty)$. The cache $\mathcal{K}$ stores completed records, $\mathcal{V}$ their designs, and $\mathcal{P}_{\text{emp}}$ the non-dominated feasible designs found so far. A completed record contains either compiled metrics or a proved pre-compilation rejection; an unfinished evaluation belongs to neither $\mathcal{K}$ nor $\mathcal{V}$.

Cache entries are local to one immutable context Θ : the instance, support order, canonical margins, relaxation data, normalisations, weights, budgets, and complete compiler protocol (including seed aggregation). Thus the cache key is

$$\text{key}(\sigma) = (\Theta, \alpha(\sigma)). \quad (68)$$

Changing any part of this context requires a new cache. This prevents a record from a different objective or compiler setting from being reused.

For compiled designs use the vector

$$r_{\text{sel}}(\sigma) = (a_\sigma/s_a, G_{2q,\sigma}/s_G, D_{2q,\sigma}/s_D, M_{\text{max},\sigma}/s_M, \hat{E}_{\text{fib}}(\sigma)). \quad (69)$$

A vector dominates another if it is no larger in every coordinate and is strictly smaller in at least one. Retain all non-dominated feasible designs, including distinct designs with equal vectors. Incumbent ties use

$$\sigma \prec_{\text{inc}} \tau \iff (J_{\text{res}}(\sigma), \hat{E}_{\text{fib}}(\sigma), \alpha(\sigma)) <_{\text{lex}} (J_{\text{res}}(\tau), \hat{E}_{\text{fib}}(\tau), \alpha(\tau)). \quad (70)$$

Algorithm 2 Evaluate one complete design

Require: Complete design σ ; state $\mathbf{X}$; fixed protocol Π .

Ensure: Updated $\mathbf{X}$ and a flag done.

```

1:  $q \leftarrow \text{key}(\sigma)$ .
2: if  $q \notin \mathcal{K}$  then
3:   Construct  $F_\sigma$ ; compute auxiliary count, penalties, and fibre excess.
4:   if a declared pre-compilation constraint is violated then
5:     Prepare a rejection record:  $\text{feas} = \text{false}$ ,  $\tilde{J} = +\infty$ , and no compiled resource vector.
6:   else
7:     if evaluation budget is unavailable then
8:       return  $(\mathbf{X}, \text{false})$ .
9:     end if
10:    Compile the complete common seed bundle and aggregate its metrics.
11:    if the evaluation fails or is interrupted then
12:      Discard its temporary record; return  $(\mathbf{X}, \text{false})$ .
13:    end if
14:    Prepare a record with  $r_{\text{sel}}(\sigma)$ ,  $\tilde{J} = J_{\text{res}}(\sigma)$ , and all feasibility flags.
15:  end if
16:  Commit the complete record to  $\mathcal{K}[q]$ .
17: end if
18: Load  $\mathcal{K}[q]$ ; add  $\sigma$  to  $\mathcal{V}$ .
19: if the record is feasible then
20:   Update  $\mathcal{P}_{\text{emp}}$  by dominance; update  $(\sigma_{\text{inc}}, U_J)$  using  $\prec_{\text{inc}}$ .
21: end if
22: return  $(\mathbf{X}, \text{true})$ .

```

When done is false, the routine also stores the reason for stopping. Later search stages report the last completed state and do not retry that call. The rejection score $\tilde{J} = +\infty$ is used only to sort records; it is not a physical resource value. Do not insert infinite resource values into Eq. (18): a zero weight could then give $0 \cdot \infty$. Only completed records can update feasibility or the incumbent.

Beam selection. At a level with child set $\mathcal{C}$, evaluate every native completion before ranking. Sort by the two keys

$$K_J(s) = (\mathbf{1}[\bar{\sigma}(s) \text{ infeasible}], \tilde{J}(\bar{\sigma}(s)), \hat{E}_{\text{fib}}(\bar{\sigma}(s)), \bar{\alpha}(s)), \quad (71)$$

$$K_P(s) = (\mathbf{1}[\bar{\sigma}(s) \text{ infeasible}], \text{rank}(s), -\text{crowd}(s), \bar{\alpha}(s)). \quad (72)$$

For K_P , first separate feasible and infeasible designs with finite compiled resource vectors. Compute the non-dominated ranks within each group. Within each rank, use the NSGA-II crowding rule [45]. Sort the values in each coordinate. Give the endpoints infinite crowding. For each interior point, add the normalised gap between its neighbours. Skip any coordinate with zero range. Use action order to break ties between equal coordinate values. If a design was rejected before compilation, give it rank $+\infty$ and crowding zero. Never calculate crowding from infinite resource coordinates.

Algorithm 3 Width- B Pareto-beam search

Require: Root s_0 ; $B \geq 1$; state X ; protocol Π .

Ensure: Updated X ; beam truncation carries no certificate.

```
1:  $\mathcal{B} \leftarrow \{s_0\}$ .
2: for  $k = 0, \dots, m - 1$  do
3:   if the beam stopping rule is met then
4:     return  $X$ .
5:   end if
6:    $\mathcal{C} \leftarrow \emptyset$ .
7:   for all  $s \in \mathcal{B}$ , then  $u \in \mathcal{U}_{k+1}$ , in fixed order do
8:     if the beam stopping rule is met then
9:       return  $X$ .
10:    end if
11:     $s' \leftarrow \text{child}(s, u)$ .
12:     $(X, \text{done}) \leftarrow \text{EVALUATECANDIDATE}(\bar{\sigma}(s'), X, \Pi)$ .
13:    if done = false then
14:      return  $X$ .
15:    end if
16:     $\mathcal{C} \leftarrow \mathcal{C} \cup \{s'\}$ .
17:  end for
18:   $b \leftarrow \min\{B, |\mathcal{C}|\}$ ;  $b_J \leftarrow \lceil b/2 \rceil$ .
19:   $\mathcal{S}_J \leftarrow$  first  $b_J$  states under  $K_J$ .
20:   $\mathcal{S}_P \leftarrow$  first  $b - b_J$  states of  $\mathcal{C} \setminus \mathcal{S}_J$  under  $K_P$ , computed on  $\mathcal{C}$ .
21:   $\mathcal{B} \leftarrow \mathcal{S}_J \cup \mathcal{S}_P$ .
22: end for
23: return  $X$ .
```

Every retained state has a completed representative record. An infeasible representative does not make its subtree infeasible, so it may still be retained. When $m = 0$, the loop is empty; the native design was already attempted by Algorithm 1 and can also be evaluated by certification.

Admissible bounds and pruning. For a prefix $s = (k, \alpha_{1:k})$, let P_s be its selected pairs and set

$$a_s = |P_s|, \quad C_{p,s}^+ = \sum_{t \leq k: \alpha_t = p, c_{A_t} > 0} c_{A_t}, \quad C_{p,s}^- = \sum_{t \leq k: \alpha_t = p, c_{A_t} < 0} |c_{A_t}|, \quad (73)$$

$$M_{p,s} = \max\{C_{p,s}^+, C_{p,s}^-\} + \varepsilon_p, \quad M_{\max,s} = \max(\{0\} \cup \{M_{p,s} : p \in P_s\}). \quad (74)$$

Then use

$$L_J(s) = w_a \frac{a_s}{s_a} + w_G \frac{G_{\text{lb}}(s)}{s_G} + w_D \frac{D_{\text{lb}}(s)}{s_D} + w_M \frac{M_{\max,s}}{s_M}, \quad (75)$$

where $G_{\text{lb}} = D_{\text{lb}} = 0$ unless stronger bounds have been proved for every completion. This is admissible: completing the prefix cannot remove an active pair or decrease either signed coefficient mass, and all weights are nonnegative. Summing these resource lower bounds proves Eq. (64).

A valid default $\text{CertInfeasible}(s)$ checks $n + a_s > Q_{\max}$ or $M_{\max,s} > B_M$. The same monotonicity justifies these tests. Other tests require their own subtree proof; in particular, the native completion's gate count or fibre excess cannot justify pruning. Prune by score only when $L_J(s) > U_J$. Keeping equality cases allows the search to certify the deterministic incumbent tie-break as well as its value.

Branch and bound. Start a new frontier at the root, retaining the evaluations from the beam. Every unevaluated leaf not excluded by a valid proof must stay below an open state. Replacing a parent by all its children preserves this property; evaluating a leaf or pruning a proved-unnecessary subtree also preserves it.

Algorithm 4 Anytime branch-and-bound certification

Require: Root s_0 ; state $\mathbf{X}$; protocol Π ; declared stopping rule.

Ensure: Updated $\mathbf{X}$, frontier, certified lower bound, and status.

```
1: Open  $\leftarrow \{s_0\}$ .
2: loop
3:    $L_J^{\text{glob}} \leftarrow \min(\{U_J\} \cup \{L_J(s) : s \in \text{Open}\})$ .
4:   if Open =  $\emptyset$  or a stopping rule is met then
5:     break.
6:   end if
7:   Choose  $s^*$  minimising  $(L_J(s), \alpha_{1:k(s)}(s))$ .
8:   Unpack  $s^* = (k, \alpha_{1:k})$ .
9:   if CertInfeasible( $s^*$ ) or  $L_J(s^*) > U_J$  then
10:    Remove  $s^*$  from Open.
11:   else if  $k = m$  then
12:     $(\mathbf{X}, \text{done}) \leftarrow \text{EVALUATECANDIDATE}(\sigma(\alpha_{1:m}), \mathbf{X}, \Pi)$ .
13:    if done = false then
14:      Keep  $s^*$  open; record the interruption reason; break.
15:    end if
16:    Remove  $s^*$  from Open.
17:   else
18:     $\mathcal{C} \leftarrow \{\text{child}(s^*, u) : u \in \mathcal{U}_{k+1}\}$ .
19:    Atomically replace  $s^*$  by all states in  $\mathcal{C}$ .
20:   end if
21: end loop
22: Recompute  $L_J^{\text{glob}}$  and assign status by Appendix C.5.
23: return  $(\mathbf{X}, \text{Open}, L_J^{\text{glob}}, \text{status})$ .
```

Children are tested when selected from the frontier. This keeps expansion atomic and avoids removing any branch before its test has been completed. In the absence of stopping or evaluation failure, the finite tree is exhausted: each iteration removes one state or expands it into fresh children.

C.5 Stopping, interruption, and returned status

Recompute the global bound from the current incumbent and frontier before testing a gap target or reporting a result. At every reported checkpoint, cache updates, incumbent updates, and frontier updates must be complete. If compilation is interrupted, retain the last completed record state and leave its leaf open. Pre-checking a seed budget alone does not guarantee that a wall-time deadline will permit a whole evaluation to finish.

Assign the first applicable status:

1. *Infeasible*: $U_J = +\infty$ and Open = $\emptyset$.
2. *Value and tie-break certified*: $U_J < +\infty$ and no open state has $L_J(s) \leq U_J$. No unresolved design can improve or tie the incumbent.
3. *Optimal value certified*: $U_J < +\infty$ and $L_J^{\text{glob}} = U_J$; unresolved equal-value designs may remain.
4. *Stopped*: report the budget, target-gap, or evaluation-failure reason. If $U_J < +\infty$, return the interval $[L_J^{\text{glob}}, U_J]$; otherwise explicitly report that no feasible incumbent has yet been found.

An additive target is met when $U_J < +\infty$ and $U_J - L_J^{\text{glob}} \leq \epsilon_{\text{abs}}$. A multiplicative target also requires $L_J^{\text{glob}} > 0$ and $U_J/L_J^{\text{glob}} \leq 1 + \epsilon_{\text{mult}}$. When $L_J^{\text{glob}} = 0 < U_J$, only the additive gap is available. The incumbent-normalised gap $(U_J - L_J^{\text{glob}})/U_J$ is defined only for $U_J > 0$ and is not an approximation ratio to the unknown optimum.

An exact certificate needs comparisons whose correctness is guaranteed. With floating-point arithmetic, round node lower bounds down and incumbent upper bounds up. These bounds must also include uncertainty in the constraint and score calculations. Ordinary rounding without error bounds, or an approximate model of compiler costs, cannot justify exact pruning or an exact equality certificate.

C.6 Why the certificate is valid

Proposition C.1 (Anytime selection certificate). *Every candidate evaluated by Algorithm 1 is pointwise exact. Suppose the frontier has the coverage property above and pruning uses only valid infeasibility tests or admissible bounds. If $U_J < +\infty$, then the bounds in Eq. (23) hold. In particular, $L_J^{\text{glob}} = U_J$ certifies global optimality relative to the declared design space, compiler, budgets, and objective.*

Proof of Proposition C.1. Step 1: check exactness of evaluated candidates. Every evaluated candidate belongs to $\Sigma_{\text{ex}}^\varepsilon$. Proposition 2.3 therefore makes it pointwise exact. This property does not depend on how well the search ranks candidates.

Step 2: prove the upper bound. A finite U_J is the score of an evaluated feasible design. The optimum over all feasible designs cannot be larger than that score. Hence $J_{\text{res}}^* \leq U_J$.

Step 3: prove the lower bound if an optimum has been evaluated. Let σ^* be an optimal feasible design. If it has been evaluated, the incumbent has score $U_J = J_{\text{res}}^*$. The global lower bound is the minimum of U_J and the open-state bounds. It is therefore at most $U_J = J_{\text{res}}^*$.

Step 4: prove the lower bound if the optimum is unresolved. The design σ^* cannot fail a valid infeasibility test, because it is feasible. It also cannot be pruned by $L_J(s) > U_J$. If it were, admissibility would give

$$J_{\text{res}}^* \geq L_J(s) > U_J,$$

contradicting Step 2. The frontier coverage assumption therefore places σ^* below some open state s . Admissibility gives $L_J(s) \leq J_{\text{res}}^*$. The minimum defining L_J^{glob} is no greater than this bound. Thus $L_J^{\text{glob}} \leq J_{\text{res}}^*$ in this case too.

Step 5: obtain the gap and equality certificate. Steps 2–4 give $L_J^{\text{glob}} \leq J_{\text{res}}^* \leq U_J$. Subtract the left inequality from U_J to obtain $U_J - J_{\text{res}}^* \leq U_J - L_J^{\text{glob}}$. If the bounds are equal, the optimum has that same value. This certifies the value only for the declared design space, compiler, budgets and objective. $\square$

The argument uses only completed evaluations and frontier coverage, so it also applies at an early stopping checkpoint. Exhausting the frontier without a feasible evaluation proves infeasibility. With a feasible incumbent, eliminating all states with bound at most U_J additionally certifies its tie-break, because no equal-value design remains unresolved.

D Warm-Start and Cost-Layer Circuit Details

For the matched single-qubit warm-start mixer [28, 30], let

$$\theta_\omega := 2 \arcsin \sqrt{\omega},$$

so that

$$e^{+i\beta h_M(\omega)} = R_y(\theta_\omega) R_z(2\beta) R_y(-\theta_\omega).$$

Hence the complete mixer factorises into parallel single-qubit gates,

$$U_{M,\sigma}^{\text{ws}}(\beta) = \prod_{i=1}^n e^{+i\beta h_M^{(i)}(\mu_i)} \prod_{p \in P_p} e^{+i\beta h_M^{(y_p)}(q_p)}.$$

For the cost layer, use F_σ from Eq. (9). After $u_j \mapsto N_j = (I - Z_j)/2$, all resulting operators are diagonal and commute [2, 6]. In particular,

$$e^{-i\gamma M_p P_R(N_i, N_j, N_{y_p})} = e^{-i\gamma M_p N_i N_j} e^{2i\gamma M_p N_i N_{y_p}} e^{2i\gamma M_p N_j N_{y_p}} e^{-3i\gamma M_p N_{y_p}}.$$

Thus quadratizing a cubic term replaces its three-local logical phase by an auxiliary qubit together with one- and two-local penalty phases [24, 25]. All representations are subsequently transpiled using the common protocol of Appendix C.2.

D.1 Preparation and diagnostic interpretation

Each term of $H_{M,\sigma}^{\text{ws}}$ has eigenvalues $\{-1, 1\}$ and acts on a different qubit. The product warm state puts every qubit in its local ground state. It is therefore the unique global ground state, with energy $-N_\sigma$. Exciting one qubit raises the energy by 2, so the spectral gap is 2. If the physical evolution is written as $\exp(-i\beta_{\text{phys}}H_M)$, we set $\beta_{\text{phys}} = -\beta$ at every objective call. This keeps the same optimiser coordinate as the cold start. For the null control, every qubit must have uniform probability, including the auxiliaries.

The prepared state gives an independent product distribution in the computational basis. Its expected encoded-energy excess is therefore $E_{\text{fib}}(\sigma; \mu, q)$, using the probabilities in Eq. (24). The historical selector instead uses relaxation moments before clipping. These values generally differ from the probabilities used to prepare the state. A threshold on the selector diagnostic therefore does not itself bound the prepared state’s excess. At the 0.02 threshold, the prepared-state diagnostic exceeded the selector guardrail for 5/60 oracle designs, 43/180 compilation designs and 8/72 QAOA-pool designs. These cases show the difference between the two diagnostics. They do not violate the selector constraint, which uses the raw moments. Even when $q_p = \mu_i \mu_j$, matching this marginal does not force $Y_p = X_i X_j$ in each measurement outcome. For $P_\rho = \emptyset$, omit the auxiliary tensor factor.

D.2 Original-objective and consistency diagnostics

For the final state $|\psi\rangle = |\psi_{r,\sigma}\rangle$, write $x \in \mathbb{B}^n$ and $y \in \mathbb{B}^{a_\sigma}$ for the measured original and auxiliary assignments, respectively, and let $p_\psi(x, y) := |\langle x, y | \psi \rangle|^2$, and $X^\star := \arg \min_{x \in \{0,1\}^n} f(x)$. We evaluate solution quality using the original-objective expectation and the probability of sampling an optimal original assignment:

$$\mathcal{E}_f(\psi) := \sum_{x,y} f(x) p_\psi(x, y), \quad P_{\text{opt}}(\psi) := \sum_{x \in X^\star} \sum_y p_\psi(x, y).$$

Both quantities depend only on the original variables and sum over all auxiliary outcomes. The classical optimiser minimises $\mathcal{E}_f$, using the common initialisation rule and evaluation budget of Section H.2.5.

We measure auxiliary consistency separately by

$$P_{\text{cons}}(\psi) := \mathbb{P}_\psi(Y = \phi_\rho(X)) = \sum_{x \in \{0,1\}^n} p_\psi(x, \phi_\rho(x)). \quad (76)$$

This is the probability that all auxiliary relations hold simultaneously, whether or not x is optimal. It equals one when there are no auxiliary variables. No independence assumption is made for the final distribution. The diagonal cost layer preserves P_{cons} , while the mixer can change it. The implementation records

$$R_{\text{inc}}(\psi) := \mathbb{P}_\psi(Y \neq \phi_\rho(X)) = 1 - P_{\text{cons}}(\psi) \quad (77)$$

under the field `auxiliary_inconsistency_rate`. Thus a sample is counted as inconsistent when at least one active relation $y_{ij} = x_i x_j$ is violated. Statevector runs sum the corresponding probability mass, while shot-based runs use the observed fraction. We set $R_{\text{inc}} = 0$ when the representation has no active auxiliary.

D.3 Alternative auxiliary initialisations

Two comparison initialisations may be used in ablation studies.

A rounded construction sets

$$\hat{x}_i = \mathbf{1}\{\mu_i \geq 1/2\}, \quad \hat{y}_p = \hat{x}_i \hat{x}_j,$$

giving an exactly consistent computational-basis state.

Alternatively, prepare the original warm-start superposition and then use a quantum circuit to compute each product into its auxiliary:

$$U_\rho|x\rangle|0\rangle^{\otimes|P_\rho|} = |x\rangle|\phi_\rho(x)\rangle.$$

This makes all auxiliary relations correct in the initial state. It needs extra entangling gates. A later local mixer can still break these relations.

E Max-3SAT Formulation Details

For a clause C , let P_C^+ and P_C^- be its positive- and negative-literal supports. The definition in Section H.1 expands in one formula for all sign patterns:

$$P_C(x) = \sum_{B \subseteq P_C^+} (-1)^{|B|} x_{B \cup P_C^-}.$$

Coefficients from all weighted clauses are aggregated before the cubic support, pair shadow, or exact penalties are computed. Cancellation can remove monomials or reduce the signed masses associated with a shared pair.

E.1 Worked instance: expansion and exactness

To illustrate the mixed-sign reduction of Proposition 2.3, consider the following weighted instance on five Boolean variables:

$$\begin{aligned} C_1 &= (\neg x_1 \vee \neg x_2 \vee \neg x_3), & w_1 &= 5, \\ C_2 &= (x_2 \vee x_3 \vee x_4), & w_2 &= 4, \\ C_3 &= (\neg x_1 \vee \neg x_4 \vee \neg x_5), & w_3 &= 2. \end{aligned} \tag{78}$$

The instance is constructed for illustration rather than as a difficult Max-3SAT benchmark. Its three clause-violation penalties are

$$P_{C_1}(\mathbf{x}) = x_1 x_2 x_3, \quad P_{C_2}(\mathbf{x}) = (1 - x_2)(1 - x_3)(1 - x_4), \quad P_{C_3}(\mathbf{x}) = x_1 x_4 x_5. \tag{79}$$

Hence the total violated weight of this example is

$$f_{3\text{SAT}}^{\text{ex}}(\mathbf{x}) = 5x_1 x_2 x_3 + 4(1 - x_2)(1 - x_3)(1 - x_4) + 2x_1 x_4 x_5. \tag{80}$$

After expansion and coefficient aggregation,

$$\begin{aligned} f_{3\text{SAT}}^{\text{ex}}(\mathbf{x}) &= 4 - 4x_2 - 4x_3 - 4x_4 + 4x_2 x_3 + 4x_2 x_4 + 4x_3 x_4 \\ &\quad + 5x_1 x_2 x_3 - 4x_2 x_3 x_4 + 2x_1 x_4 x_5. \end{aligned} \tag{81}$$

Here $g(\mathbf{x})$ is the quadratic part defined in Section H.1.1. The fully quadratized objective is

$$\begin{aligned} F_{\text{full}}(\mathbf{x}, y_{23}, y_{14}) &= g(\mathbf{x}) + 5x_1 y_{23} - 4y_{23} x_4 + 2y_{14} x_5 \\ &\quad + 6P_R(x_2, x_3, y_{23}) + 3P_R(x_1, x_4, y_{14}). \end{aligned} \tag{82}$$

Both reductions are pointwise exact. For the shared pair, the positive and negative residual masses are 5 and 4; for the remaining pair they are 2 and 0. Thus $M_{23} = 6 > 5$ and $M_{14} = 3 > 2$ satisfy the strict conditions of Proposition 2.3. Each consistent auxiliary assignment is the unique minimiser for fixed $\mathbf{x}$, and substitution recovers the original polynomial. Consequently,

$$\begin{aligned} f_{3\text{SAT}}^{\text{ex}}(\mathbf{x}) &= \min_{y_{23} \in \{0,1\}} F_{\text{sel}}(\mathbf{x}, y_{23}) \\ &= \min_{(y_{23}, y_{14}) \in \{0,1\}^2} F_{\text{full}}(\mathbf{x}, y_{23}, y_{14}) \quad (\mathbf{x} \in \{0,1\}^5). \end{aligned} \tag{83}$$

Enumeration over all 32 original assignments and all auxiliary assignments also verifies these equalities. All three representations have minimum zero. After expansion and aggregation, the numbers of distinct quadratic monomials are 3, 7, and 11, respectively, as reported in Table 11.

F Cubic Spin-Glass Formulation Details

For the higher-order spin-glass benchmark family [46–48], set $s_i = 1 - 2x_i$,

$$s_i s_j = 1 - 2x_i - 2x_j + 4x_i x_j, \quad (84)$$

$$s_i s_j s_k = 1 - 2(x_i + x_j + x_k) + 4(x_i x_j + x_i x_k + x_j x_k) - 8x_i x_j x_k. \quad (85)$$

Substitution into Equation (91) and coefficient aggregation produces the binary objective. In the main benchmark $c = 0$, $h_i = 0$, and $J_{ij} = 0$. The sampling and pair-reuse rules are those of Appendix G.3. The raw mathematical objective is not normalised as a function of n or m_e ; any later positive circuit rescaling is stored explicitly and is common to every representation of the same instance.

G Reproducibility and Additional Numerical Material

G.1 Archive identities and retained records

The original archive uses configuration hash 9d124ad72c0fe2b841fbfc927ec45c35cb413a6a0e01baada17f1025709d315c and result-package manifest hash 781132a5ac1a7f5d37ca128e24d52c15fe3e54e56bffc676b68d55d8966a749b. The E1–E3 and E4–E6 raw rows record commits 22e05ea2e1684256cfcefc889a4c658b5530de8a and 749d39f68d29d796da0060f6c180c3306254979e, respectively. The four-part addendum uses configuration hash dcae3585519be308040729bc8a4548c06f46493af50e454b389165349380c632 and code commit 5bdec984a409335ca07aa5bec867f3c5fee9cf14; its distributed ZIP has SHA-256 9909592333daa580bfc1d38f1c5ddd155ddac8506148be2312f573bdfcb47db3.

The correction archive is URSS_WARMSTART_CORRECTION_EXECUTED_v1.zip, with SHA-256 9e539a0aa387393e482e5ac0f186ace9695af39d05017fe3bc90eefefe5dd5e5. The correction’s execution-source snapshot and the later Git integration commit are distinguished in Appendix G.6. The repository contains the correction audit and summary records; The complete raw correction archive is publicly available in the [fixed-version repository archive](#).

The selector-v2 resource pack is URSS_E2_E5_LOCAL_RESULTS_20260913_020457_389.zip, with SHA-256 ebb8dc624edce8ee679d4e57f70134b95232d809784eab42b6c2e9de12df7429 and code commit fa5b8bb0644b007176fe5727d36f66b969e5c8b. The matching QAOA pack is URSS_QAOA_LOCAL_RESULTS_20260913_010958_585.zip, with SHA-256 cba9d6cd3182544dec12f0b9869575fd4034c25bcd2281108dd807042b0a8166 and code commit bcf47285d474a4bffb863984340978aa10816f80. Certification and selector-sensitivity figures retain their archived inputs; Resource, E3, E4 and E6 summaries use the selector-repaired merged rows. LP signal values retain the original diagnostics, with updated design-status markers in the original E4 signal plot. No experiment was rerun solely to produce a figure. An archival DOI and the final software licence will be added when available.

G.2 Joint-study freeze and Windows compatibility audit

The v3 protocol identifier is URSS_feedback_2_6_iteration_v3. Its original freeze has SHA-256 6884cab2f5c4fc54f0510d4189f75d41242b8ff7e5dd4005bf512e54755b1943. The Linux execution generated 72 unique objectives with no exact overlap against 462 supplied historical objectives. It executed 4,560 main compiler calls and 1,476,585 objective evaluations (309,960 equal-effort, 950,625 grid and 216,000 refinement calls). These are original execution counts, not new calls from the Windows verification.

On 18 September 2026, a Windows 11/Python 3.12.10 audit of the stored results completed with the pinned packages NumPy 2.5.3, SciPy 1.18.1, Qiskit 2.5.2, Matplotlib 3.11.2, PyYAML 6.0.3 and pypdf 6.10.0. Checks covered all 72 instances and 4,560 pointwise-exact representations, 9,120 product-law diagnostics, 1,086 saved parameter sets, 1,350 grid samples and 2,172 trace samples. The maximum independent numerical difference was 1.5632×10^{-13} . Nine representative circuits (native/full/mixed for the first instance of each family) were recompiled and checked against the independently simulated state. This does not recompile every candidate or re-optimize the full experiment. Six explanatory

secondary witnesses had additional independent compilation and full-endpoint exclusion checks in the original Linux evidence.

Table 8: All total-gate-count differences among the nine Windows circuit recompiles. In both rows, Q, G, D, M, J , logical width, auxiliary count and reduced-term count match exactly, and the sampled state check passes.

Instance	Representation	Archived total gates	Windows total gates
constructed_00	Full	273	272
constructed_00	Proper mixed	246	245

The original audit compared every field in the resource dictionary. It stopped at the first difference. A separate v3 portability amendment was declared after the experiment. It keeps all numerical checks, file-integrity checks and checks on resources used in decisions. It records differences in total gate count separately. The frozen budgets, score, selections and statistical analyses do not use total gate count, so the amendment changes none of those results or thresholds. The compatibility status is `PASS_WITH_COMPILER_METADATA_DRIFT`. The original strict replay status remains `FAIL`. All resource fields match for the other seven sampled circuits. These checks support compatibility for the sampled circuits and tested quantities. They do not establish strict compiler reproducibility across platforms or validation on hardware.

The original scientific sources, freeze and result files remain unchanged. The evidence includes the verifier, its two-line effective-audit diff, all compared resource dictionaries, full output log and separate strict/compatibility outcomes. The verifier SHA-256 is `1192912244d4145a8d8997e3138ae49ce921493facd5b525830c03c1fbcaf8a6`; the effective audit SHA-256 is `55a39265655b09b6af943ba01ed9db94929eafc05ae4676bd3170821028b29a4`. The v3 Nelder–Mead study does not resolve the earlier cross-platform COBYLA trajectory discrepancy; historical results and the new study remain separate.

G.3 Benchmark generation, screening, and provenance

The workload sizes and variable counts are specified in Section H.1.3. The following rules define the candidate pools and the separate warm-start screening study.

For weighted Max-3SAT, each clause uses three distinct variables. Each literal sign is sampled independently with probability $1/2$. We reject duplicate signed clauses and sample each positive integer weight uniformly from $\{1, 2, 3, 4, 5\}$. Candidate pools use target clause loads $m_C/n \simeq 2, 4, 6$. Both families use uniform proposals and anchor-pair proposals. This allows sparsity and reuse to vary separately during generation. In uniform Max-3SAT generation, we sample a variable triple and its literal signs. We accept it if that signed clause is new. The same unordered triple may therefore appear again with different signs. For spin glass, we sample unordered hyperedges without replacement. In anchor mode, we first sample an unordered pair. We combine it with distinct third variables and reject repeated signed clauses or spin-glass hyperedges. Literal signs, clause weights and spin-glass coefficient signs are sampled independently after the structural proposal. Each instance record stores the number of anchors and the requested uses of each anchor.

For spin glasses, each active hyperedge has an independent coefficient $K_{ijk} \in \{-1, +1\}$ with equal probability. Target hyperedge loads are $m_e/n \simeq 1, 2, 4$.

Candidate generation is stratified over the listed values of n , the three load targets, and the two proposal modes. The realised sparsity and reuse coordinates, rather than the requested cell, are used in later analysis. Clause and hyperedge targets are rounded to the nearest valid integer and capped by the number of distinct legal clauses or hyperedges. Duplicate raw instances, repeated hyperedges, invalid clauses, and instances with an unused variable are rejected. The realised generator mode and parameters are kept with each instance rather than inferred from the final structural bin.

For the targeted warm-start addendum, we scheduled 500 generation attempts for each family. We used $n \in \{6, 8, 10\}$ and the same uniform and anchor-pair proposal modes. The saved screening table contains 567 candidates: 93 spin-glass and 474 Max-3SAT. Thus the 1,000 scheduled attempts should not be read as successful generations. The historical generation-failure logs are unavailable, and we do not reconstruct them. Screening used only relaxation information. No QAOA result was inspected.

We kept candidates with at least one active auxiliary and at most 12 encoded qubits. We ranked them first by median original-variable bias $\text{median}_i |\hat{\mu}_i - 1/2|$, then by active-pair dependence, then by instance identifier. The eligible pools contained 93 spin-glass and 474 Max-3SAT instances. The ten highest-ranked instances from each family form the 20-instance addendum set. Seven spin-glass and all ten Max-3SAT cases exceed the frozen weak-signal threshold of 0.05. We call this the *strongest-screened* set because some selected instances still have weak single-variable bias. We use it only for the targeted warm-start analysis and do not pool it with E3 or E5.

The benchmark-generation protocol records generation seeds, raw data, canonical coefficients, structural metadata, and split assignments in versioned source manifests. These manifests and the corresponding generator scripts are included in the reproducibility repository identified in the Data and code availability statement. All representations of one raw instance share the same instance identifier and split. Instances are divided into training, validation, and test sets before representation or QAOA results are inspected; no test-labelled instance is removed because one representation performs unexpectedly.

G.4 Software and compiler configuration

The archived compilation and original numerical runs used Qiskit 2.4.2 [49], Qiskit Aer 0.17.2, and SciPy 1.18.1 [50]. The warm-start coordinate correction uses the separate environment recorded in Appendix G.6; compiled budgets remain those of the archived representations. We compiled every representation to the basis $\{\mathbf{rz}, \mathbf{sx}, \mathbf{x}, \mathbf{cx}\}$ with optimisation level 1, SABRE layout and routing [51], the translator pass, and no scheduling. The main study used all-to-all coupling and a bidirectional 3×4 grid. The connectivity addendum retained these targets and added a bidirectional 12-qubit line, a bidirectional 12-qubit ring, and the 19-qubit graph returned by `CouplingMap.from_heavy_hex(3)`. The undirected grid edges were

$$(0, 1), (1, 2), (2, 3), (4, 5), (5, 6), (6, 7), (8, 9), (9, 10), (10, 11), \\ (0, 4), (1, 5), (2, 6), (3, 7), (4, 8), (5, 9), (6, 10), (7, 11).$$

Both directions of every listed grid, line, and ring edge were supplied to the compiler. The complete directed coupling maps, including all heavy-hex edges, are stored in the hashed addendum configuration.

Equal polynomial terms were combined before circuit construction, and only exact zero coefficients were removed. Original variables were placed first, followed by auxiliary variables in lexicographic pair order. Pauli strings were ordered first by increasing weight and then lexicographically. Each parity gadget used the largest qubit index in that string as its target. We did not cancel gates across different gadgets, reorder commuting gadgets, or apply approximate coefficient simplification. Compilation used transpiler seeds 2026082721–2026082725, and the per-instance resource value was the median over these five seeds. Keeping these choices fixed is important because compiled resources can otherwise change with the software stack and routing seed [10, 24, 25].

G.5 Optimisation hyperparameters

The equal-layer study used QAOA depths $p \in \{1, 2\}$. The equal-gate study used compiled two-qubit-gate budgets $B \in \{128, 256\}$ and converted each budget to a representation-specific depth using Equation (95). COBYLA was run for at most 60 objective evaluations in each of three restarts, with tolerance 10^{-3} . The optimiser seeds were 2026082731–2026082733, with initialisation intervals $\gamma \in [0, 2\pi]$ and $\beta \in [0, \pi]$; these are not imposed optimisation bounds. Each representation was optimised independently. The noiseless experiment used exact statevector expectations, whereas every E6 evaluation, including its sampled noiseless reference, used 4096 shots.

Numerical MaxCut studies illustrate the role of parameter search in observed QAOA performance [4, 52]. Annealing-based parameter initialisation [53] offers another comparison for future work; the present experiments retain their archived starts.

The targeted warm-start run design is specified in Appendix H.6 and uses the same COBYLA settings.

The five matched-random constructions used seeds 2026082711–2026082715. The noisy circuits used seeds 2026082741–2026082743 and measurement seeds 2026082751–2026082753. For benchmark generation, the master, validation, and split-assignment seeds were 2026082701, 2026082702, and 2026082703, respectively. All scheduled restarts, random constructions, and compiler seeds were retained in the aggregation; no best-restart or best-seed selection was made. The relaxation was solved with SciPy’s HiGHS interface [54] and the fixed variable ordering and tie-breaking rule stated in Appendix H.4.

G.6 Warm-start coordinate correction and execution provenance

The correction follows frozen plan `warmstart_coordinate_correction_plan_v1` and execution `exec_20260912_v2`. It retains the original 20 targeted and 14 E4 instances, all representations and matched draws, archived probabilities, initial parameters, seeds, compiled budgets, and optimisation loss. The correction execution audit records that the 132 materialised designs and all 4,800 historical saved-parameter replays passed validation before the corrected comparisons were interpreted. It also records that the full-register null-control tests passed at the fixed tolerances (phase-aligned statevector and probability errors at most 10^{-12} on the small regression cases). The audit records the observed errors.

The execution used Linux, Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0, whereas the archived numerical environment was Windows with Python 3.12.10, NumPy 2.5.2, and SciPy 1.18.1. Because optimiser-environment equivalence was not established, all 1,221 positive-depth original E4 cold tasks were rerun, in addition to the 2,631 required positive-depth warm tasks and all 480 targeted tasks. The 468 zero-layer original E4 records were re-evaluated and validated for reuse; their historical count of 60 evaluations is bookkeeping, with one current initial-state evaluation and zero effective optimisations. There were 4,332 fresh optimisations and no unresolved failed or missing tasks. No new LP, instance screening, or selector change was introduced by this correction.

The actual execution-source snapshot is `1301a48c88d27ddc47c4fe10c547745d6fe28d9b`. It is distinct from the subsequent repository integration commit `d9f5de9b7bef80a428b973274a44758c4909f98a`; the latter records code and audit integration, not a new experimental run. The binding, source manifests, raw task rows, attempts, reuse audit, instance-level paired data, all five prespecified policy contrasts, and separate depth-stratum summaries are retained with the correction. Historical results were available when the plan was fixed, so this is an exploratory correction study rather than an outcome-blind confirmatory preregistration. Historical and corrected studies are never pooled.

The subsequent selector-v2 rerun retained the frozen LP moments, instances, budgets and screening. It reoptimised 420 positive-depth jobs affected by changed representations, with 492 new observations. All new encodings passed the exactness checks and all null controls passed. The final merged files contain 1,344 E3 rows, 4,176 original E4 rows, 480 targeted E4 rows and 1,296 E6 rows; unchanged rows retain their original lineage.

G.7 Noise model, files, and provenance

Analyses of local noise relate QAOA quality to noise rate, circuit size and depth [55]. Other results identify noise-induced loss of trainability [56] and conditions limiting optimisation advantage [57]. These motivate reporting noise separately; our limited experiment does not establish which mechanism controls performance at larger scales.

The limited noise study used local depolarising noise after each `sx`, `x`, and `cx` gate, together with symmetric readout error. The low-noise setting used one-qubit, two-qubit, and readout error rates (0.001, 0.01, 0.01), while the high-noise setting used (0.003, 0.03, 0.03). It covered 18 structurally selected instances, nine from each benchmark family, on the bidirectional 3×4 grid. Parameters were optimised once in the corresponding noiseless circuit and then reused without re-optimisation at both noise levels. The subset was frozen from the QAOA candidate pool by sparsity and pair-reuse coverage before noise outcomes were examined. Its 18 exact instance identifiers are retained in the noise-run file and can be joined to the frozen QAOA manifest for the structural labels. For each design, the

E6 circuit uses the smallest transpiler seed among 2026082721–2026082725 that attains the median one-layer two-qubit cost.

For E6, the one-layer estimate supplies an initial depth. The complete measured circuit is then transpiled, and depth is decreased until its actual two-qubit gate count fits the declared budget. The recorded realised depth is used for the noisy and sampled-noiseless comparisons. Initial and realised depths are retained in `D08_e6_initial_and_realised_depth.csv`; the E3 and E6 instance-level Student-*t* summaries are generated by the selector-rerun post-processing entrypoint.

The machine-readable result archive contains the raw per-run CSV and JSON records, summary files, experiment configuration, package manifests, and completion audit. Each result row carries its configuration hash, the relevant source or tier manifest hash, and the run code commit. The accompanying GitHub repository contains the source manifests, executable scripts, environment information, and commands used to rebuild the summaries. The outer package manifest and completion-audit commit are reported in the Data and code availability statement. The tables in the main text are therefore summaries rather than replacements for the underlying instance-level records. The E3 and E6 intervals and tables use the selector-repaired instance-level aggregations. Original and targeted E4 performance uses the selector-repaired merge of the coordinate-corrected rows, with the separate provenance in Appendix G.6. The four-part addendum separately stores the 60 certification traces, three added-topology compilation outputs, the systematic selector ablation, and all 480 targeted warm-start rows. Its manifest was verified against every archived file before the new summaries were produced.

Table 9: Origin of the numerical results used in this paper. Historical files remain immutable; selector-v2 packs merge only the observations requiring recomputation and preserve source lineage for every reused row.

Analysis in the paper	Source group	Main archived record
Exactness and penalty checks	Original E1	<code>e1_exactness.csv</code> , <code>e1_penalty_trials.json</code>
Base resource comparison	Selector-v2 resource merge	<code>results/E2_merged_rows.csv</code>
Noiseless QAOA and secondary metrics	Selector-v2 QAOA merge	<code>results/E3_merged_runs.csv</code>
Original E4 warm-start comparison	Selector-v2 QAOA merge	<code>results/E4_corrected_original_merged_runs.csv</code>
Structural regime analysis	Selector-v2 resource pack	<code>analysis/E5_instance_level.csv</code>
Limited noise study	Selector-v2 QAOA merge	<code>results/E6_merged_runs.csv</code>
Branch-and-bound certificates and traces	Four-part addendum	<code>certification/</code>
Line, ring, and heavy-hex compilation	Selector-v2 resource merge	<code>results/E2_added_topologies_merged_rows.csv</code>
Selector sensitivity study	Four-part addendum	<code>selector_ablation/selector_ablation_raw.csv</code>
Signal-enriched warm-start study	Selector-v2 QAOA merge	<code>results/E4_corrected_targeted_merged_runs.csv</code>

The main-text tables and figures are derived from these instance-level records using the aggregation rules in Section H.2.6. Because the same benchmark pool was available during protocol refinement, a future confirmatory study would still require newly generated instances.

G.8 Higher-degree checks and limits of certification

An additional correctness audit used 24 degree-four/five polynomials (12 of each), each with four overlapping high-order monomials and affine terms. Across 91 sequential shared substitutions, exhaustive checks covered 42,752 assignments of the current variables. At each step, $M = \|A\|_\infty$ preserved values, $M = \|A\|_\infty + 1/4$ gave a unique consistent lift, and $M = \|A\|_\infty - 1/4$ produced a counterexample. The

final composed representations were checked over 45,056 original-plus-auxiliary assignments; all point-wise and unique-lift checks passed. These are small truth-table checks, not an efficient arbitrary-degree implementation. The existing implementation and an independent rational-polynomial compiler also agreed in all 744 representation compilations from 248 pair-star-family instances.

The sharp threshold can itself require exponential Boolean optimisation. For an aggregate residual $A(v) = \sum_B a_B v_B$, the readily computed bound $M \geq \sum_B |a_B|$ is sufficient but need not be sharp. For instance, $A = uv - uw$ has $\|A\|_\infty = 1$ although its coefficient absolute sum is 2; $A = uvw - uvz$ gives the same gap in a degree-five substitution. Sharpness here is restricted to the specified Rosenberg construction.

We also tested the existing branch-and-bound code with early stopping. For each support size below, we used three synthetic eight-variable instances. The settings were the stated reference compiler, resource weights (0.05, 0.45, 0.45, 0.05), scales (6, 128, 96, 64) in auxiliary/gate/depth/penalty order, penalty margin one, and budgets $Q_{\max} = 28$, $G_{\max} = 512$, $D_{\max} = 512$ and $M_{\max} = 128$. Nonzero cubic coefficients were sampled from integers of magnitude one to three. Moments from the all-zero assignment made the fibre constraint inactive. These tests therefore make no claim about relaxation quality or guardrail benefit. Each run had limits of 10,000 nodes, 2,000 candidate evaluations and 2 seconds. Limits were checked between complete search operations.

m	4^m	Complete	Median time (s)	Median gap	Termination
3	64	3/3	0.009	0.0000	optimal
5	1,024	3/3	0.191	0.0000	optimal
7	16,384	0/3	0.549	0.2370	evaluation_limit
9	262,144	0/3	0.771	0.3135	evaluation_limit
11	4,194,304	0/3	0.889	0.4305	evaluation_limit

Separate wall-time-only runs on the first eleven-term instance stopped at 0.1, 0.5 and 2 seconds with remaining score gaps of 0.4688, 0.4315, 0.4266, respectively. For all six cases with three or five cubic terms, a separate exhaustive enumeration verified that the reported interval contained the optimum. Four explicit node-limit interruptions on a three-term case also retained a valid interval, including the zero-node case with no incumbent and infinite upper bound. Detailed node, evaluation, frontier and bound traces accompany the reproducibility bundle. Positive remaining gaps at interruption are not certificates of near-optimality. These results demonstrate correct bound accounting and rapid growth of the search burden, not scalability. The bound sets gate and depth contributions to zero, which leaves loose intervals even when the incumbent improves; stronger bounds would be needed for larger proofs. Certificates remain relative to the declared representation grammar, score, compiler and implemented numerical feasibility rule; time limits may be exceeded by the duration of one atomic operation.

G.9 Additional new-study resource controls

Table 10 reports the same score for every retained comparator on grid-12. These are medians of 30 instance values per row, not independent compiler-run observations. The score is Eq. (16); smaller is better. The distance heuristic is explicitly not a reproduction of a named external implementation. Unconstrained native preference does not determine feasibility under hard budgets.

The grid-12 mean paired strict-minus-resource-full score differences, with descriptive 95% percentile bootstrap intervals from 10,000 instance resamples, are:

Max-3SAT: canonical -0.139 $[-0.152, -0.126]$; reuse -0.189 $[-0.202, -0.176]$.

Cubic spin: canonical -0.210 $[-0.223, -0.196]$; reuse -0.239 $[-0.253, -0.226]$.

Constructed: canonical -0.095 $[-0.103, -0.086]$; reuse -0.127 $[-0.136, -0.119]$.

These secondary intervals are unadjusted and compare the best designs found within each search budget. They do not certify the best possible full assignment or imply any difference in solution quality. Rank comparisons across topologies use candidates shared by both settings. Pareto-set overlap also depends on the candidate sets. Those sets can differ because the distance-aware rule proposes different designs. The overlaps therefore measure stability within the sampled sets; they do not isolate the effect of connectivity alone.

Table 10: Median compiled resource scores on grid-12. MinA: minimum-auxiliary full; reuse: maximum-frequency pair full; distance: distance-aware full; resFull: resource-optimised full within its candidate budget; strict: best proper mixed; random: control matched on auxiliary and reduced-term counts, with penalty matching when possible.

Family	Synthesis	Native	MinA	Reuse	Distance	resFull	Strict	Random
Max-3SAT	canonical	0.176	0.434	0.455	0.509	0.415	0.271	0.292
Max-3SAT	reuse	0.140	0.446	0.471	0.525	0.431	0.238	0.259
Cubic spin	canonical	0.107	0.448	0.485	0.520	0.445	0.237	0.248
Cubic spin	reuse	0.096	0.472	0.513	0.546	0.470	0.225	0.235
Constructed	canonical	0.206	0.395	0.395	0.396	0.387	0.296	0.305
Constructed	reuse	0.160	0.393	0.393	0.395	0.389	0.264	0.268

Certified strict-mixed-only feasibility across 30 constructed instances

Fixed slice: CX depth ≤ 96 , penalty ≤ 4.4 ; full endpoint excluded by exhaustive logical enumeration.

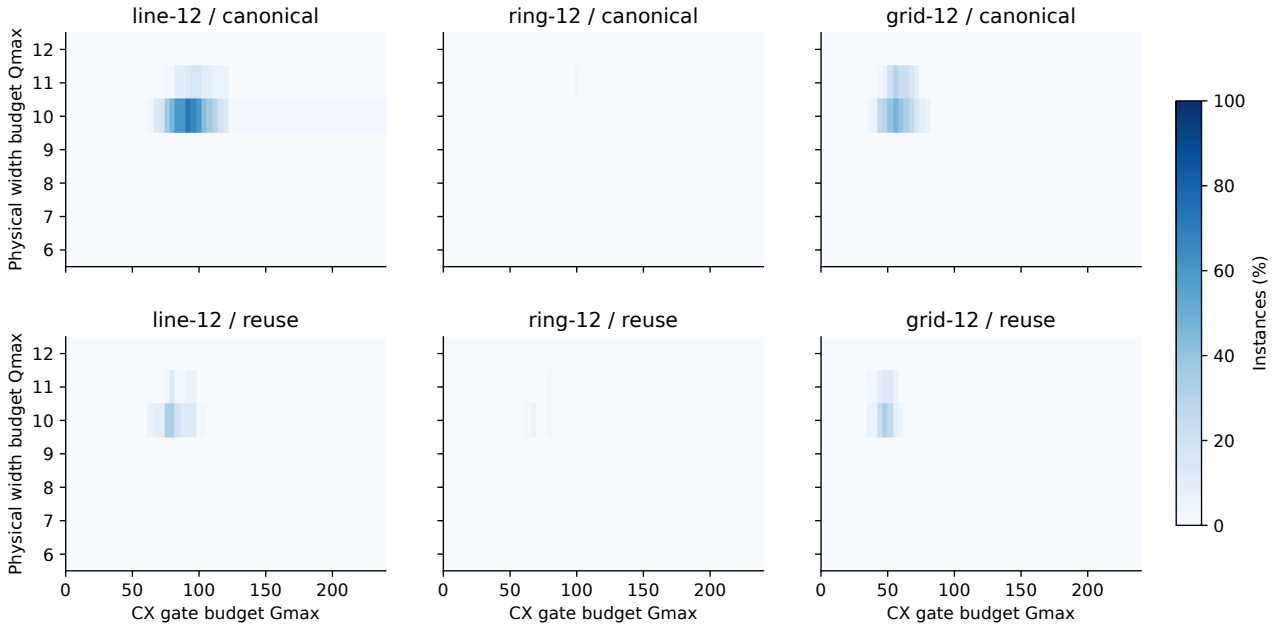


Figure 5: Certified mixed-only incidence across the 30 constructed mechanism instances in the fixed $(D_{\max}, M_{\max}) = (96, 4.4)$ slice. Each plotted cell counts independent instances, with the native and full endpoints excluded as specified in Section 2.5.1. Budget cells and compiler settings are not independent samples.

G.10 Five-seed delivery and independent checks

The additional data come from `URSS_ALL_PROJECT_CODE_20260918T190812Z.zip`, supplied inside `delivery.zip`. The inner archive has SHA-256 `54e5629261b6bceb1b8922f7a33aa9f75f87fae5af80f7ccb4ba116349a32014`. The analysed run is `results/multiseed_20260918T163717Z/`. Earlier incomplete run folders are retained as provenance and are not pooled with the completed run.

The saved run contains 101,460 completed compiler rows. We independently recompute all 1,800 budget-classification arrays from these rows. Every array matches, covering 37,661,400 classified points. The 5,662-file package manifest and 3,742-file run manifest also match. Independent enumeration checks one native, one full and one strictly mixed representation per instance: all 180 pass exactness and unique-lift checks, with maximum numerical error 1.78×10^{-15} . Six classification tests and ten small regression tests pass. These checks do not rerun all compilation tasks. Candidate checks, circuit digests and source hashes provide separate records of what was run.

The delivered audit records three unresolved output problems. Its baseline-winner feasibility field was not computed; its YAML file omitted settings present in the authoritative JSON file; and its region-presence figure reused one value across seeds. A further check finds that its “lost or changed” field counts points lost in every other seed, rather than points lost in at least one seed. The two counts differ. They cannot support the same stability label.

We retain the supplied files unchanged and provide a separate correction script, derived tables and a new figure. The manuscript uses these checked outputs. The retained-point total, 79,069 out of 149,439, is unchanged. The original audit failure remains part of the record; the corrected analysis does not turn the original archive into an unqualified PASS.

The delivery also records two inaccessible source directories and omits the environment used for the formal Qiskit 2.5.2 run. The copied frozen inputs and saved results allow the data checks reported here. They do not establish complete reconstruction of that environment or coverage of the inaccessible directories. No additional QAOA optimisation was run for this follow-up.

H Additional Experimental Protocol Details

H.1 Benchmark definitions and structural coordinates

We use weighted Max-3SAT and cubic spin glasses to study how selective reduction trades interaction locality against auxiliary variables and compiled resources. Max-3SAT supplies clause-derived signs and overlap; spin glasses allow sparsity and pair reuse to be varied directly.

H.1.1 Max-3SAT

Consider a clause $C = (\ell_i \vee \ell_j \vee \ell_k)$. Define the false-literal indicator

$$u_{\ell_r}(x_r) := \begin{cases} 1 - x_r, & \ell_r = x_r, \\ x_r, & \ell_r = \neg x_r. \end{cases} \quad (86)$$

The clause-violation penalty is

$$P_C(\mathbf{x}) = u_{\ell_i}(x_i)u_{\ell_j}(x_j)u_{\ell_k}(x_k), \quad (87)$$

which equals one exactly when the clause is violated. A weighted Max-3SAT instance with $w_C > 0$ therefore yields the cubic PUBO [58, 59]

$$f_{3\text{SAT}}(\mathbf{x}) = \sum_C w_C P_C(\mathbf{x}). \quad (88)$$

After expansion and coefficient aggregation, cubic terms that share a variable pair can use the same auxiliary. Cancellation must be accounted for before selecting substitutions and setting penalties; the expansion for arbitrary literal signs is given in Appendix E.

Table 11: Logical comparison for the worked Max-3SAT instance. Quadratic couplings count distinct degree-two monomials after aggregation.

Encoding	Total variables	Auxiliary variables	Retained cubic terms	Quadratic couplings	Penalties
All-native	5	0	3	3	–
Selective	6	1	1	7	$M_{23} = 6$
Fully quadratized	7	2	0	11	$M_{23} = 6, M_{14} = 3$

A small worked example Consider the five-variable weighted instance

$$f_{3\text{SAT}}^{\text{ex}}(\mathbf{x}) = g(\mathbf{x}) + 5x_1x_2x_3 - 4x_2x_3x_4 + 2x_1x_4x_5, \quad (89)$$

where $g(\mathbf{x}) = 4 - 4(x_2 + x_3 + x_4) + 4(x_2x_3 + x_2x_4 + x_3x_4)$. The clauses and full expansion are given in Appendix E.1. The first two cubic terms share $\{2, 3\}$. Substituting $y_{23} = x_2x_3$ gives the aggregate residual $A_{23} = 5x_1 - 4x_4$, so Proposition 2.3 gives the sharp threshold $M_{23}^* = \max\{5, 4\} = 5$. Taking $M_{23} = 6$ yields

$$F_{\text{sel}}(\mathbf{x}, y_{23}) = g(\mathbf{x}) + 5x_1y_{23} - 4y_{23}x_4 + 2x_1x_4x_5 + 6P_{\text{R}}(x_2, x_3, y_{23}). \quad (90)$$

The native representation retains all three cubic terms. Full quadratization also substitutes $y_{14} = x_1x_4$ in the remaining term, with threshold $M_{14}^* = 2$ and penalty $M_{14} = 3$. All three representations preserve the original objective under auxiliary minimisation, as verified in Appendix E.1. Table 11 shows the trade-off: the selective choice removes two cubic terms using one auxiliary, while saving one auxiliary and four quadratic couplings relative to full reduction. These logical counts do not establish a compiled-resource or QAOA advantage; those are tested separately.

H.1.2 Cubic spin glasses

Higher-order spin-glass models provide a complementary benchmark [46–48]. For $s_i \in \{-1, +1\}$, write

$$E(\mathbf{s}) = c + \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \sum_{i < j < k} K_{ijk} s_i s_j s_k. \quad (91)$$

The main study uses the pure cubic case $c = h_i = J_{ij} = 0$. Substituting $s_i = 1 - 2x_i$ gives a PUBO of degree at most three. Appendix F specifies the conversion and common rescaling convention. Varying the cubic supports provides different opportunities for pair reuse without deriving the interactions from clauses. The two families are contrasting tests, not a controlled separation of application effects from hypergraph effects.

H.1.3 Structural coordinates and benchmark scope

Analytical QAOA guarantees for regular MaxCut graphs [60] and high-depth analyses on large-girth graphs and hypergraphs [61] concern specified graph families and limiting regimes. The finite benchmark ensembles below serve a different purpose: measuring paired resource and optimisation changes under exact representation choices.

The main structural axes are cubic sparsity, potential auxiliary reuse, and hardware connectivity. Coefficient signs affect the penalty through Equation (10), but sign balance is recorded as a diagnostic rather than varied as an additional main axis.

All clauses or hyperedges are expanded and equal monomials are aggregated before structural features are calculated. Let $\mathcal{C}_3(f)$ denote the remaining nonzero cubic supports and $m_3 = |\mathcal{C}_3(f)|$. We record the canonical cubic density $m_3/\binom{n}{3}$ and the load m_3/n . The pair shadow $\partial_2 \mathcal{C}_3$ is the set of all pairs contained in at least one cubic support. Potential pair sharing is summarised by the pair-incidence score

$$r_{\text{pair}} = 1 - \frac{|\partial_2 \mathcal{C}_3|}{3m_3}, \quad m_3 > 0, \quad (92)$$

together with $d(p) = |\{S \in \mathcal{C}_3(f) : p \subseteq S\}|$ and $\Delta_2 = \max_{p \in \partial_2 \mathcal{C}_3} d(p)$. This score is zero when every cubic term has three private pairs and increases when pair incidences are repeated. It measures potential sharing, not the number of auxiliaries actually saved by a chosen representation. When the exact pair cover is available, the corresponding full-reduction saving is

$$r_{\text{aux}} = 1 - \frac{\tau_2(\mathcal{C}_3(f))}{m_3}, \quad m_3 > 0, \quad (93)$$

where τ_2 is the set-cover number defined in Equation (53).

Connectivity is assessed by compiling representations of the same instance to common target graphs. The main targets are all-to-all coupling and a 12-qubit 3×4 grid; additional line, ring, and heavy-hex targets extend the routing comparison. Capacity failures and common-fit comparisons are handled by the protocol in Section 4.

The three workload tiers have different purposes. The oracle tier targets 30 instances per family with $n \in \{5, 6, 7, 8\}$; the QAOA tier targets 36 per family with $n \in \{6, 8, 10\}$ and a 12-qubit width limit; the compilation tier targets 90 per family with $n \in \{8, 12, 16, 20\}$. These generation targets are distinct from the evaluated subsets reported with each result. The limited noise subset contains nine instances per family. A separate warm-start study uses ten instances per family selected using relaxation information only, without inspecting QAOA outcomes. This strongest-screened set is not pooled with the original representation comparison. Complete sampling, filtering, screening, and provenance rules are given in Appendix G.3.

H.2 Detailed evaluation protocol

We test exactness, compiled resources, and QAOA performance separately. All comparisons use representations of the same raw instances and a common execution protocol; full implementation details are given in Appendices G and H.

H.2.1 Benchmark scope and comparison rules

The oracle tier contains 60 instances for exhaustive exactness and selector checks, and the compilation tier contains 180 instances for resource comparisons. The main noiseless QAOA comparison uses 14 test-labelled instances; the noise and targeted warm-start studies use separate subsets of 18 and 20 instances, respectively. Each set is balanced between the two families. These evaluated sets are distinct from the generation targets in Section H.1.3 and are analysed separately by study.

We compare the native objective, a fully quadratized reference, the selective design from Section 4.1, and a matched-random selective control. Full quadratization uses a minimum-cardinality pair cover with deterministic tie-breaking. The random control matches only the number of active pair auxiliaries, not gate count, depth, or penalty size. Five seeded controls are used; they need not be distinct when the feasible choices are limited. The exact construction and fallback rules are in Appendix H.3.

Within each family and tier, the recorded training-validation-test split is 60/20/20. Settings were chosen using training and validation data, with no direct selector retuning on test-labelled rows. However, the pool was available during earlier protocol refinement, so this is an exploratory study, not a fresh untouched-test evaluation. The broad resource comparison pools all 180 compilation instances across split labels.

H.2.2 Exactness and penalty checks

On each oracle instance we enumerate the original assignments and minimise over the auxiliary variables, checking

$$f(\mathbf{x}) = \min_{\mathbf{y}} F_\rho(\mathbf{x}, \mathbf{y}) \quad \text{for every } \mathbf{x}. \quad (94)$$

The main experiments use $M_p = T_p + 1$ and check that the minimising auxiliary assignment is unique and consistent. Tightness checks use $T_p - 1$, T_p , and $T_p + 1$, recording mismatches, ties, inconsistent minimisers, and explicit failure witnesses. Penalty-margin sensitivity runs are reported separately.

Table 12: Main selector settings. The resource order is auxiliary count, two-qubit gates, two-qubit depth, and maximum penalty. The broader compilation tier does not impose the 12-qubit QAOA width limit.

Setting	Value
Resource weights	(0.05, 0.45, 0.45, 0.05)
Normalisation scales	(6, 128, 96, 64)
QAOA-scope limits	12 qubits; 256 two-qubit gates per cost layer; two-qubit depth 192; maximum penalty 128
Selector fibre-diagnostic threshold	0.02
Beam width; main penalty margin	64; 1
Warm-start clipping	$\delta = 0.05$

H.2.3 Logical and compiled resources

We record register width, auxiliary count, retained cubic terms, aggregated quadratic couplings, maximum penalty, and coefficient dynamic range. Every representation uses the same synthesis and compiler settings and the same five transpiler seeds on all-to-all coupling and the 12-qubit 3×4 grid. The compiled metrics are two-qubit gate count, two-qubit depth, and routing overhead, including inserted SWAPs.

A width failure is recorded separately, not assigned an artificial gate count. Direct comparisons use instances on which all representations in that comparison fit the same device. The connectivity extension compares only selective and matched-random designs on a 12-qubit line, a 12-qubit ring, and a 19-qubit heavy-hex graph, without rerunning QAOA. Software, coupling maps, and seeds are specified in Appendix G.

H.2.4 Relaxation, selection, and warm starts

We solve a sparse RLT-style, strengthened recursive-McCormick LP for the original cubic objective. It uses single, pair, and supported triple pseudo-moments in $[0, 1]$; only the single and pair coordinates are exported for selection and warm starts. Cubic terms are represented directly in the LP, without Rosenberg penalties. The complete constraints and solver rules are given in Appendix H.4.

The selector uses the settings in Table 12. In these experiments, selection and oracle certification use the exact all-to-all reference synthesis of Appendix C.1. Device-transpiled resources are evaluated subsequently for the selected representations. Complete enumeration and branch-and-bound certification are run on the 60 oracle instances. A separate sensitivity study varies selector settings on that tier, without rerunning QAOA; details are in Appendix H.5.

Warm-start comparisons use cold start, original-only warming, and original-plus-auxiliary warming with either pair moments or the independence approximation. The corresponding probabilities and matched mixers follow Section 4.2; the experimental policies are specified in Appendix H.6.

The original warm-start runs (E4) use the first LP optimum returned by HiGHS. The selector and targeted warm-start study use a second, deterministic optimisation to choose one of the LP’s optimal solutions. The two rules can return different pseudo-moments for the same instance. The 20 targeted instances were screened using relaxation information, before inspecting QAOA outcomes. They have the strongest screening scores in the candidate pool and do not form a representative random sample. We analyse them separately from original E4. The coordinate correction restores the saved probability vectors and fixed designs in both studies. It does not solve the LP again or screen new instances. Appendix G.6 describes the execution and checks.

H.2.5 QAOA budgets and optimisation objective

Each representation is optimised independently using COBYLA, with three restarts and at most 60 objective evaluations per restart. The loss is the original-objective expectation $\mathcal{E}_f$, obtained by ignoring the auxiliary register. Encoded energy and auxiliary consistency are diagnostics, not terms in this classical loss.

Equal-layer comparisons use $p \in \{1, 2\}$. For a two-qubit-gate budget $B \in \{128, 256\}$, let g_σ be the median one-layer two-qubit gate count on the grid over the five transpiler seeds. For $g_\sigma > 0$, the assigned depth is

$$p_\sigma(B) = \left\lfloor \frac{B}{g_\sigma} \right\rfloor, \quad p_\sigma(B)g_\sigma \leq B. \quad (95)$$

If $g_\sigma > B$, the stored $p = 0$ row is an initial-state baseline with no trainable parameters. In particular, the 128-gate comparison includes such rows; their counts and the subset on which every design fits at least one layer are reported separately. The targeted warm-start study uses only $p = 1, 2$ and contains no zero-layer rows.

H.2.6 Aggregation, uncertainty, and supplementary analyses

The raw instance is the statistical unit. For resources, we take the median over transpiler seeds for each design, then average the five random-control design values within each instance. For QAOA, we average the three restart scores per design, then the five random-control design means. No best-seed, best-restart, or best-random-design selection is used. Paired differences are formed only after these within-instance aggregations.

Resource comparisons report medians of paired instance differences. QAOA, warm-start, and noise comparisons report means and two-sided 95% Student- t intervals across instances. The corrected warm-start intervals use $n - 1$ degrees of freedom, are unadjusted for multiple comparisons, and are exploratory; an interval containing zero does not establish equivalence or absence of an effect. Repeated seeds and restarts are not additional independent samples. The exploratory regime labels use a separate within-endpoint stability rule, not population-level confidence intervals; its thresholds are in Appendix H.7.

Noiseless statevector results are the primary QAOA comparison. The limited noise study uses 18 structurally selected instances on the grid, with representative depolarising and readout errors rather than a calibrated real device. Parameters are optimised noiselessly and reused at both noise levels without re-optimisation. Full noise, shot, and seed settings remain in Appendix G. The structural study uses the sparsity, pair-reuse, and connectivity axes of Section H.1.3. All these analyses are exploratory; intervals describe uncertainty, and unadjusted multiple-setting p -values are not used to claim significance.

This appendix contains the implementation details supporting Section 4. Appendix G retains software versions, compiler configuration, generation records, optimiser seeds, and noise settings.

H.3 Fully reduced and matched-random references

The historical full-quadrization baseline uses a minimum-cardinality pair cover of all cubic supports. Ties are resolved by the lexicographically smallest active pair set and then by a lexicographic term-to-pair assignment. This fixes the auxiliary count and penalty profile rather than leaving “full” as an unspecified quadrization.

The random control matches only the number of active pair auxiliaries. We generate five feasible controls with seeds 2026082711–2026082715. First sample a candidate pair set. Then use a deterministic augmenting-path procedure to assign reduced terms to those pairs. If 256 pair-set attempts fail, reuse the selected pair set and choose a random valid assignment. If no distinct control exists, keep the selected design itself. A native selected design therefore has a native matched control. Every representation keeps the raw instance identifier, which lets us compare designs within the same instance.

H.4 Relaxation and optimum-face selection

The implementation records its relaxation as `SA_RLT_level_2`. Because conventions for hierarchy levels differ, we define the exact model used here. It is a sparse RLT-style [62], strengthened recursive-McCormick [63] relaxation of the canonical cubic objective

$$f(\mathbf{x}) = c_\emptyset + \sum_i c_i x_i + \sum_{i < j} c_{ij} x_i x_j + \sum_{i < j < k} c_{ijk} x_i x_j x_k. \quad (96)$$

Let $\mathcal{T} := \{(i, j, k) : i < j < k, c_{ijk} \neq 0\}$. The LP has one single-variable pseudo-moment $\hat{\mu}_i$ for every original variable, one pair pseudo-moment $\hat{q}_{ij}$ for every $i < j$, and one triple pseudo-moment $\hat{t}_{ijk}$ for every $(i, j, k) \in \mathcal{T}$. All are bounded in $[0, 1]$, and the primary objective is

$$\min_{\hat{\mu}, \hat{q}, \hat{t}} \left(\sum_i c_i \hat{\mu}_i + \sum_{i < j} c_{ij} \hat{q}_{ij} + \sum_{(i, j, k) \in \mathcal{T}} c_{ijk} \hat{t}_{ijk} \right). \quad (97)$$

The constant $c_\emptyset$ is added back only when the objective value is reported. Cubic terms are represented directly by $\hat{t}_{ijk}$; no Rosenberg penalty is used inside this relaxation.

For every $i < j$, the pair envelope is

$$\hat{q}_{ij} \leq \hat{\mu}_i, \quad \hat{q}_{ij} \leq \hat{\mu}_j, \quad \hat{q}_{ij} \geq \hat{\mu}_i + \hat{\mu}_j - 1. \quad (98)$$

For each $(i, j, k) \in \mathcal{T}$, the triple variable obeys

$$\hat{t}_{ijk} \leq \hat{\mu}_i, \hat{\mu}_j, \hat{\mu}_k, \quad \hat{t}_{ijk} \geq \hat{\mu}_i + \hat{\mu}_j + \hat{\mu}_k - 2, \quad (99)$$

where the first shorthand denotes three separate upper bounds. For each pair-remaining-index decomposition $(ab, c) \in \{(ij, k), (ik, j), (jk, i)\}$, we additionally impose

$$\hat{t}_{ijk} \leq \hat{q}_{ab}, \quad \hat{t}_{ijk} \geq \hat{q}_{ab} + \hat{\mu}_c - 1. \quad (100)$$

These constraints recover the intended products at Boolean points. At a fractional solution they are relaxation pseudo-moments and need not be the marginals of one joint probability distribution.

The LP is solved with SciPy 1.18.1 using `scipy.optimize.linprog`, the HiGHS method, and presolve. The v2 selector uses a second solve after obtaining the primary value z^* . It keeps the primary objective within

$$|\mathbf{c}^\top \mathbf{v} - z^*| \leq 10^{-9} \quad (101)$$

and minimises a positive secondary objective whose weights are derived deterministically from SHA-256 labels. The variable order is all singles, lexicographic pairs, and then sorted supported triples. Under the pinned solver stack, this supplies the selector with a reproducible representative of the primary optimum face. The coordinates $\hat{\mu}_i$ and $\hat{q}_{ij}$ are read directly from the returned vector and clipped to $[0, 1]$ only to remove small numerical violations; the triple coordinates are not exported as selector moments.

E4 uses the same feasible LP and primary objective, but its frozen runs use the first optimal vector returned by HiGHS rather than the selector's secondary representative. Therefore, when the primary optimum face is not a single point, the selector and E4 can use different pseudo-moment vectors. Before interpreting the E4 results, we report the distribution of $|\hat{\mu}_i - 1/2|$; instances whose values are almost uniform carry little warm-start information and are identified as such.

H.5 Certification and selector sensitivity

The main selector settings are listed in Table 12; final score ties are broken lexicographically.

For the 60 oracle-tier instances, we also ran the best-bound branch-and-bound procedure of Algorithm 4. At each step, the unresolved state with the smallest valid lower bound is expanded first, and one trace row is stored after every 256 explored nodes. A run is declared complete only when the frontier is resolved, the final lower and upper bounds are equal, and the selected value matches the previously frozen complete-enumeration optimum. The compiler contribution in a partial state uses the valid but conservative lower bound of zero. These runs therefore check the certificate implementation on small finite search spaces; they are not a claim about large-instance scaling.

We performed a one-factor-at-a-time sensitivity study on the same 60 oracle instances. Starting from weights (0.05, 0.45, 0.45, 0.05), beam width 64, fibre threshold 0.02, and penalty margin 1.0, we varied the resource weights, tested gate-only, depth-only, and joint gate-depth scores, used beam widths 16, 32, and 128, changed the fibre threshold to 0, 0.01, 0.05, and 0.10, and changed the penalty margin to 0.5 and 2.0. We also included a deterministic greedy baseline and a deterministic maximum-pair-reuse baseline. This study compares representation choices and resource outputs only; it does not rerun QAOA.

H.6 Warm-start policies

The warm-start comparison uses clipping value $\delta = 0.05$ and compares four policies including cold start. From the relevant LP vector we set

$$p_i = \text{clip}_\delta(\hat{\mu}_i), \quad p_{ij}^{\text{pair}} = \text{clip}_\delta(\hat{q}_{ij}), \quad p_{ij}^{\text{ind}} = p_i p_j. \quad (102)$$

Cold start puts every qubit in $|+\rangle$ and uses the usual X mixer. Original-only warm start uses p_i and its matched mixer on the original qubits, while each auxiliary remains at probability $1/2$ with an X mixer. The original-plus-auxiliary policy also applies a matched mixer to the auxiliaries, using p_{ij}^{pair} as the main rule. The product of the individually clipped probabilities, p_{ij}^{ind} , is the single auxiliary ablation, without further clipping. The LP supplies pair coordinates in these experiments, so this ablation deliberately omits available pair information.

Null control at fixed optimiser coordinates. The cold single-qubit unitary is $e^{-i\beta X}$. With $B(\omega) = 2\sqrt{\omega(1-\omega)}X + (1-2\omega)Z = -h_M(\omega)$, the implemented warm gate is $e^{-i\beta B(\omega)}$ on every warmed qubit. Setting *all* original and auxiliary probabilities to $1/2$ therefore gives the same statevector, probabilities, and metrics as cold start at each parameter vector. The regression covers $p = 1, 2$, native and auxiliary designs, and all four policy paths. Setting only the original probabilities to $1/2$ is insufficient: the independence rule would set an auxiliary probability to $1/4$. The control explicitly overrides the full register after policy construction.

Targeted warm-start addendum. The 20 strongest-screened instances are evaluated at equal layer depths $p = 1$ and $p = 2$. Every selected representation has at least one active auxiliary, and no zero-layer bookkeeping row is included. We compare cold start, original-variable-only warm start, original-and-auxiliary warm start using the pair pseudo-moments, and the independence closure $p_{ij} = p_i p_j$. Each setting uses three COBYLA restarts and 60 objective evaluations per restart, giving 480 run rows in total. All runs use noiseless statevector evaluation. The archived addendum probabilities were obtained with the deterministic secondary optimum-face selection used by the v2 selector, whereas the original E4 probabilities came from the first primary optimum. The correction restores both archived probability sets and reruns all 480 targeted tasks; it does not invoke either LP selection procedure.

H.7 Secondary statistics and regime labels

For paired instance differences d_i , let $\bar{d}$ denote their mean. A paired standardised effect is reported as $d_z = \bar{d}/s_d$, where s_d is the sample standard deviation of the paired instance differences. Win/tie/loss counts use the same raw instances and the declared equivalence band. The E5 endpoint classification retains its frozen paired normal interval over repeated observations within each endpoint and is used only for the exploratory regime counts. The regime study uses frozen equivalence bands of 5% for gate count, depth, and routing overhead, and 0.01 objective units for the QAOA score. A result is called helpful or detrimental only when its paired interval lies beyond the corresponding band; otherwise it is assigned to the residual (inconclusive) category. This category includes intervals overlapping the band and does not establish practical equivalence.

These endpoint intervals describe algorithmic stability, not across-instance uncertainty. Sign balance and coefficient heterogeneity remain diagnostics, not additional factorial axes.

Supplementary Information

S1 Historical development studies

The earlier studies use 60 oracle instances, 180 compilation instances and 14 noiseless QAOA instances. Each set is balanced between the two families. They compare native, full, selective and random representations. The old random controls match only auxiliary count.

QAOA is scored on the original objective after ignoring auxiliary bits. We form resource differences within each instance. Instances, not restarts, are the independent units for QAOA intervals. We analyse the families separately because their objective scales differ. These are exploratory studies. Methods (Section 4) and Appendix S4 give their protocols.

S1.1 Development evidence: exactness and selector validation

We check every original assignment and minimise over every auxiliary assignment in the small oracle cases. Above the penalty threshold, the reduced minimum equals the original value. The minimising auxiliaries are unique and correct. At the threshold, an incorrect assignment can tie, but the minimum value remains correct. Below the threshold, we find explicit failures. These checks illustrate the distinction between value exactness and unique correct auxiliaries in Proposition 2.3. Appendix S4.1 gives the counts.

The old v2 Pareto beam obeys its raw-moment guardrail on all 60 instances. It finds the enumerated optimum in 58 cases: all 30 spin cases and 28 of 30 Max-3SAT cases. Its largest score regret is 0.010156 (Table S1 and Figure S4). Both misses have training labels. All 12 test-labelled cases match. The greedy baseline matches the 30 spin optima, which are native. It matches no Max-3SAT optimum.

This validates a small historical search space with a diagnostic constraint. It does not certify the new resource-only selector or the new device-specific searches.

Supplementary Table S1: Oracle-tier validation on 60 instances. Regret is the selector score minus the enumerated optimum. Certification is a separate exact run; work and timing details are in Supplementary Section S4.2.

Method	Zero-regret cases	Mean regret	Maximum regret
Complete enumeration	60/60	0	0
Pareto beam	58/60	2.3003×10^{-4}	0.010156
Greedy native-completion	30/60	0.019371	0.107292
Branch-and-bound certification	60/60	0	0

Separate branch-and-bound runs find the enumerated optimum on all 60 cases. Their final gaps are zero. The lower bounds never decrease, and the upper bounds never increase. Capacity pruning is used in 26 runs. The conservative resource bound is used in none. This validates the completed small-instance certificates. It does not show that branch and bound is faster than enumeration or scales to large cases (Table S10 and Figure S5).

These certificates use the declared score, direct-pair search space and fixed all-to-all compiler. They do not certify optimality on every device. The implemented guardrail uses frozen floating-point moments and the rule $\hat{E}_{\text{fib}}^{\text{computed}} \leq \tau_{\text{fib}} + 10^{-9}$. The certificate applies to this numerical feasible set. It does not apply to exact LP moments with zero tolerance.

Changing beam width or penalty margin moderately changes fewer selections than changing resource weights or the fibre threshold. In the 48-case calibration ablation, the guardrail removes all 12 recorded threshold violations. This checks compliance with the guardrail, not its ability to predict QAOA outcomes. Supplementary Section S4.2 gives the sweeps. We do not rerun QAOA for these changes.

S1.2 Development evidence: resource trade-offs

Repairing the old selector changes 285 representation/draw groups on 65 fixed compilation instances. We rebuild 7,125 affected resource rows with the same compiler seeds and topologies. Of these, 2,430

compile and 4,695 are excluded by width as expected. There is no unexpected compiler failure. The other 23,475 rows keep their archived values and provenance.

All scheduled all-to-all circuits compile. Each family has 90 instances. For Max-3SAT, selective reduction saves a median of 25 gates and 60.5 depth units relative to native. For spin glass, it saves 18 depth units but adds six gates (Table S2). The differences are paired within instances. This is a gate–depth trade-off, and the outcome depends on the family. We pool the split labels for this descriptive comparison. Appendix S4.3 gives resource levels and diagnostics.

Supplementary Table S2: Median within-instance resource difference from the native representation on the all-to-all topology. Differences are calculated after transpiler-seed aggregation, so a negative value means a smaller circuit. The table pools all 90 instances in each family.

Family	Representation	Δ two-qubit gates	Δ two-qubit depth
Max-3SAT	Fully reduced	−37	−72
	Selective	−25	−60.5
	Matched random	+4	−15.5
Cubic spin glass	Fully reduced	+42	−29
	Selective	+6	−18
	Matched random	+8.2	0

Device width limits change which instances can be compared. On the 12-qubit grid, 108 native designs fit, but only 57 selective and 21 full designs fit (Table S3). Only 21 instances fit all four representations: 12 Max-3SAT and nine spin cases. We use these same instances for direct grid comparisons. This avoids comparing different success-only subsets.

Supplementary Table S3: Capacity results on the fixed 12-qubit sparse topology. The counts refer to unique benchmark instances rather than compiler-seed rows.

Representation	Feasible instances	Fraction
Native	108/180	60.0%
Fully reduced	21/180	11.7%
Selective	57/180	31.7%
Matched random	57/180	31.7%

Supplementary Table S4: Median within-instance difference from native on the 21 instances that fit the sparse topology in all four representations. Differences are calculated after seed aggregation; negative values mean lower resource use.

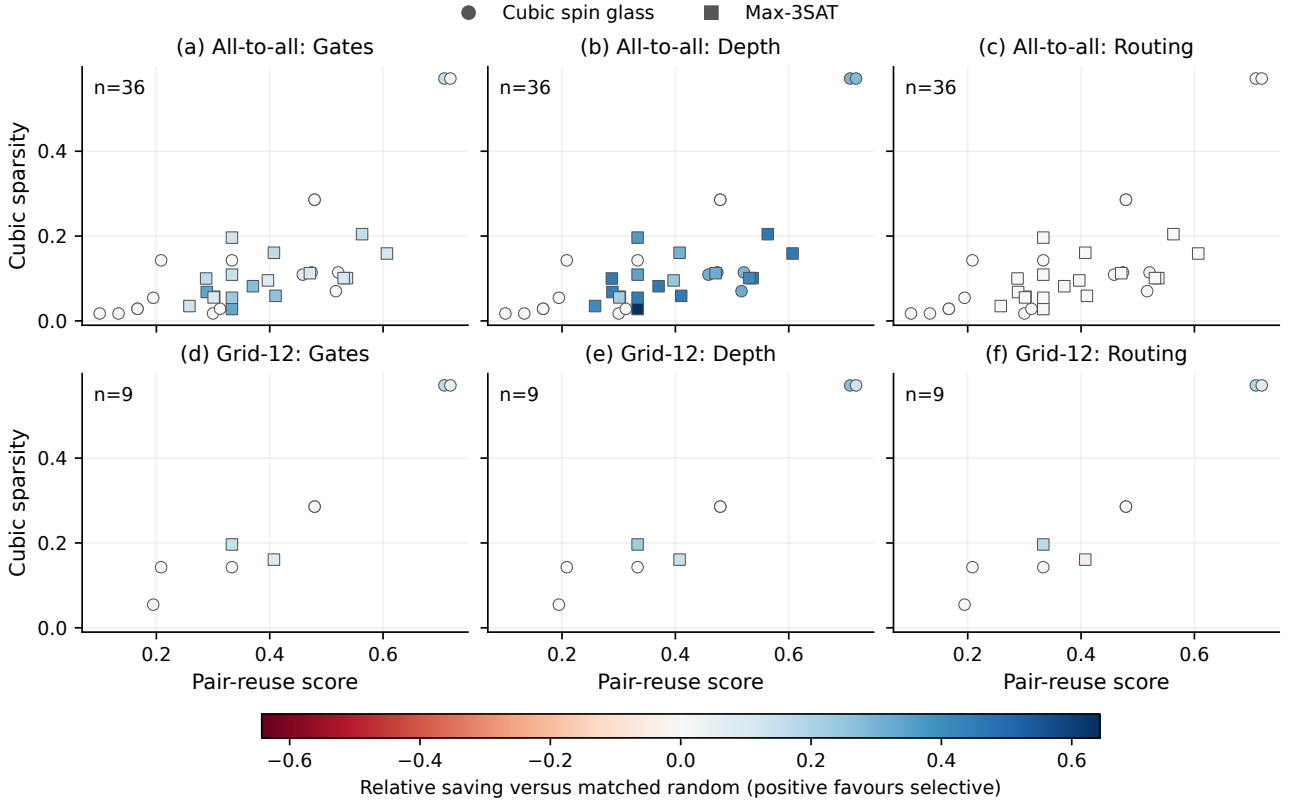
Family	Representation	Δ gates	Δ depth	Δ SWAP	Δ routing
Max-3SAT	Fully reduced	−16	−29	−3	−5.5
	Selective	−10	−21	−1.5	−2
	Matched random	+6.5	−3.9	+1.8	+4.8
Cubic spin glass	Fully reduced	+45	+13	+7	+22
	Selective	0	0	0	0
	Matched random	0	0	0	0

On the common Max-3SAT subset, selective reduction saves a median of 10 gates and 21 depth units relative to native. Random reduction increases gate count. The old control fixes auxiliary count, but not reduced-term count, penalty or pair assignment. The difference therefore does not isolate the effect of partial reduction.

Eight of nine selective spin designs are native. This explains the zero selective and random medians. Full reduction increases every listed resource. Only three common-fit instances have test labels, so these results are descriptive. The new controls strengthen the comparison by matching both auxiliary count and reduced cubic count.

The additional line, ring and heavy-hex tests show the same Max-3SAT pattern against random reduction. Selective depth is lower on every common-fit instance. Gate count is lower in every line and ring case and in 54 of 55 heavy-hex cases. Spin medians remain zero. Heavy-hex has more qubits and fits more instances. These comparisons therefore mix connectivity and capacity effects (Appendix S4.4).

Figure S1 groups the savings by cubic sparsity and pair reuse. The supplementary stability screen favours selective over random in 20/20 estimable Max-3SAT depth endpoints and 19/20 gate endpoints. These endpoints are descriptive summaries, not independent samples. Some bins are sparse, and width exclusions change the available cases. We cannot conclude that more pair reuse always gives larger savings. Appendix S4.7 gives all counts.



Supplementary Figure S1: Two-dimensional representation-regime map from the selector-repaired compilation results. Pair reuse and canonical cubic sparsity form the two axes. Panels (a)–(c) show all-to-all gates, depth, and routing; (d)–(f) show the same metrics on the fixed 12-qubit grid. Colour is the relative saving of selective reduction over its matched-random control, so positive values favour the selective design. Marker shape distinguishes the two benchmark families.

S1.3 Boundary result: native quality under the historical budget

Native has the lowest mean original objective on the 14 development instances, historically labelled test. This holds in each family at $p = 1, 2$ and gate budgets $B = 128, 256$ (Table S5). In this optimisation protocol, saving circuit resources does not give a general QAOA-quality improvement.

Four of the eight selective-minus-native 95% intervals include zero. They are the spin cases at $p = 1$, $p = 2$ and $B = 128$, and Max-3SAT at $B = 128$. For Max-3SAT, the difference at $p = 2$ is 1.935 [0.597, 3.273], favouring native. At $B = 128$ it is 0.040 [−1.013, 1.093]. The spin difference at $B = 256$ is 0.821 [0.007, 1.635]. An interval that includes zero does not prove equal performance.

At the 128-gate budget, 35 of 112 design/draw groups have $p = 0$. These groups account for 105 restart rows. No full cost layer fits, so these are initial-state baselines with no QAOA optimisation.

If every representation and random draw must fit at least one layer, only five spin and three Max-3SAT instances remain. Their selective-minus-native differences are −0.072 [−0.544, 0.400] and 0.982 [−0.890, 2.854], respectively. Both intervals include zero. Table S16 and Figure S8 give the

Supplementary Table S5: Mean paired objective difference relative to the native representation in the noiseless QAOA study. Lower is better, so positive entries favour the native representation. Results from the two problem families are kept separate because their objective scales differ.

Family	Representation	$p = 1$	$p = 2$	$B_{2q} = 128$	$B_{2q} = 256$
Cubic spin glass	Fully reduced	1.837	2.041	2.044	2.458
	Selective	0.379	0.275	0.431	0.821
	Matched random	0.394	0.271	0.448	0.543
Max-3SAT	Fully reduced	1.774	2.112	0.245	1.855
	Selective	1.899	1.935	0.040	1.511
	Matched random	1.390	1.192	0.385	1.516

breakdown. The optimised loss is always the original expectation. Encoded energy and auxiliary inconsistency are separate diagnostics.

S1.4 Exploratory warm-start mechanisms and their limits

The corrected warm-start runs keep the same frozen instances, LP probabilities and budgets. After selector repair, the audit accounts for 4,176 original E4 rows and 480 targeted rows. It records which zero-layer rows are reused and which redundant warm policies are removed when a design has no auxiliaries. Appendix G.6 gives the records.

The corrected warm mixer uses the cold mixer’s angle convention. This removes a coordinate mismatch in the optimisation comparison. Warming the original bits and warming the auxiliaries are still separate changes.

In the original E4 set, 11/14 instances have weak median original-bit bias: $\text{median}_i |\hat{\mu}_i - 1/2| \leq 0.05$. This median does not bound every bit’s bias or the whole state’s distance from uniformity.

The common selective subset with auxiliaries has only one spin instance and seven Max-3SAT instances. A spin interval cannot be estimated from one instance. All four primary Max-3SAT pair-minus-cold intervals include zero. We keep zero-layer labels at $B = 128$ (Table S17 and Figure S10).

The targeted study uses 20 strongest-screened instances, ten per family. Screening uses relaxation information, not QAOA outcomes. All 480 runs at $p = 1, 2$ have positive depth. We analyse this signal-enriched set separately. It does not represent ordinary random instances. Its rule for choosing an LP optimum also differs from original E4. Three selected spin cases still fall below the single-bit signal threshold (Figure S11).

Supplementary Table S6: Corrected warm-start results on the 20 strongest-screened instances. We average three restarts, then form paired differences within each instance. Each family has ten instances. Intervals are exploratory 95% Student- t intervals, without adjustment for multiple comparisons. Negative differences favour the first policy.

Family	Depth	Pair minus cold	Pair minus original-only	Pair minus independence
Spin glass	$p = 1$	−0.010 [−0.281, 0.260]	−0.116 [−0.344, 0.112]	−0.105 [−0.320, 0.109]
Spin glass	$p = 2$	0.152 [−0.026, 0.330]	−0.025 [−0.228, 0.179]	0.008 [−0.289, 0.304]
Max-3SAT	$p = 1$	−6.761 [−9.263, −4.260]	−0.149 [−0.219, −0.080]	0.092 [−0.017, 0.202]
Max-3SAT	$p = 2$	−6.722 [−9.143, −4.300]	−0.500 [−0.875, −0.124]	−0.153 [−0.656, 0.350]

For targeted Max-3SAT, all three warm policies improve on cold start (Figure S12). Pair-minus-cold differences are −6.761 at $p = 1$ and −6.722 at $p = 2$. Both intervals are below zero. Under pair warming, mean optimum probability rises from 0.0360 to 0.5987 at $p = 1$, and from 0.0419 to 0.6262 at $p = 2$. These probabilities are secondary diagnostics. Spin differences remain uncertain.

For Max-3SAT, pair-minus-original-only differences are −0.149 [−0.219, −0.080] at $p = 1$ and −0.500 [−0.875, −0.124] at $p = 2$. These exploratory intervals suggest a benefit over warming only the original bits. They are not adjusted for multiple comparisons. Both pair-minus-independence intervals include zero. The study therefore does not establish a stable extra benefit from pair moments over independence warming. The limited noise study is separate (Appendix S4.8).

S2 Optional fibre diagnostics: results

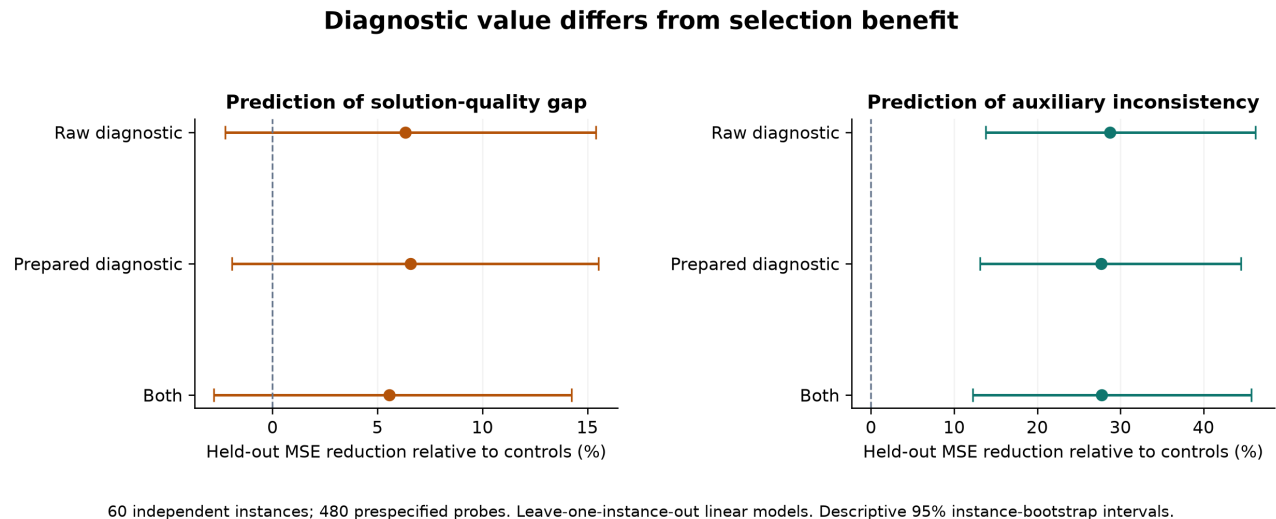
We compare no guardrail, raw-moment filtering and prepared-state filtering. All use the same mixed candidate bank. The methods, threshold and prediction model are in main Section 4.4.

All three rules select the same design on all 72 instances. This is not because the rules are identical. Of 1,920 random-cohort mixed candidates, the raw rule rejects 1,243 and the prepared rule rejects 1,329. They disagree on 86 candidates. All 324 mechanism candidates pass both. In every instance, the lowest-score candidate passes both rules, so filtering changes no action. This gives no observed benefit from this intervention. It does not prove that guardrails always help or never help. Resampling identical zero differences cannot establish a population-zero effect.

The separate prediction test uses 480 probes chosen in advance from 60 random instances. To predict an instance, it fits the model using only other instances. Adding the prepared-state diagnostic lowers inconsistency MSE from 0.02862839 to 0.02070590. The improvement is 0.00792249, with descriptive 95% interval [0.00376080, 0.01273606]. This is a 27.7% relative reduction in held-out MSE. For objective gap, MSE changes from 0.00148790 to 0.00139012. The improvement interval $[-0.00002862, 0.00023108]$ includes zero. Raw diagnostics give a similar pattern. These secondary intervals are unadjusted. They resample held-out errors by instance without refitting the models. The results give limited evidence that this diagnostic helps predict consistency for the tested model and instance distribution. They do not establish reliable quality prediction or show that changing selection causes a gain. The optional threshold rule still has no demonstrated algorithmic benefit.

Supplementary Table S7: Held-out prediction on 480 preselected candidates from 60 random instances. Positive MSE improvement is favourable; intervals are secondary, descriptive and unadjusted.

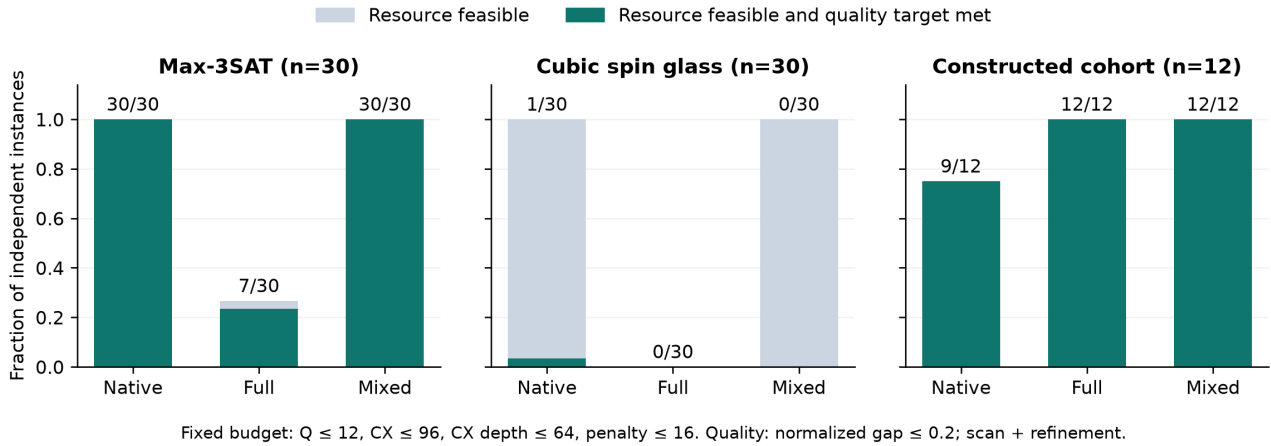
Outcome	Controls MSE	+ Prepared MSE	Improvement [95% interval]
Quality gap	0.00148790	0.00139012	0.00009778 $[-0.00002862, 0.00023108]$
Inconsistency	0.02862839	0.02070590	0.00792249 $[0.00376080, 0.01273606]$



Supplementary Figure S2: Relative held-out MSE improvement after adding diagnostics to the control model. Bars are descriptive instance-bootstrap intervals. Prediction gains do not demonstrate selection gains.

S3 Additional joint-study display

Primary joint resource-quality test: no mixed-only success



Supplementary Figure S3: Primary joint-feasibility outcomes. Grey bars show resource availability and green bars joint resource-and-quality success. The three cohorts are displayed separately. The main text reports the zero primary mixed-only result.

S4 Supplementary Numerical Results

These results supplement the four comparisons in Section 2. Protocols and within-instance aggregation follow Section 4 and Appendices G–H.

S4.1 Complete exactness counts

We first checked the reduction by exhaustive enumeration on 60 small instances, with 30 Max-3SAT instances and 30 cubic spin-glass instances. The selected designs were not all the same: 30 were all-native, 23 were strictly mixed, and seven were fully reduced. These instances contain 5,696 distinct instance–original-assignment pairs. Across 1,384 representation, design, penalty, and target-pair trials, the verifier repeatedly enumerated the same assignments where required, giving 124,672 original-assignment evaluations and 585,408 joint original–auxiliary evaluations in total. These totals measure the full verification workload rather than the number of distinct original assignments.

Section S1.1 summarises the threshold behaviour. Table S8 reports the counts: 5,351 at-threshold inconsistent ties, 5,639 below-threshold mismatches, and 452 saved failure witnesses, all with non-zero error.

Supplementary Table S8: Summary of the exhaustive exactness checks. The above-threshold setting produced no mismatch or inconsistent minimiser in any tested case.

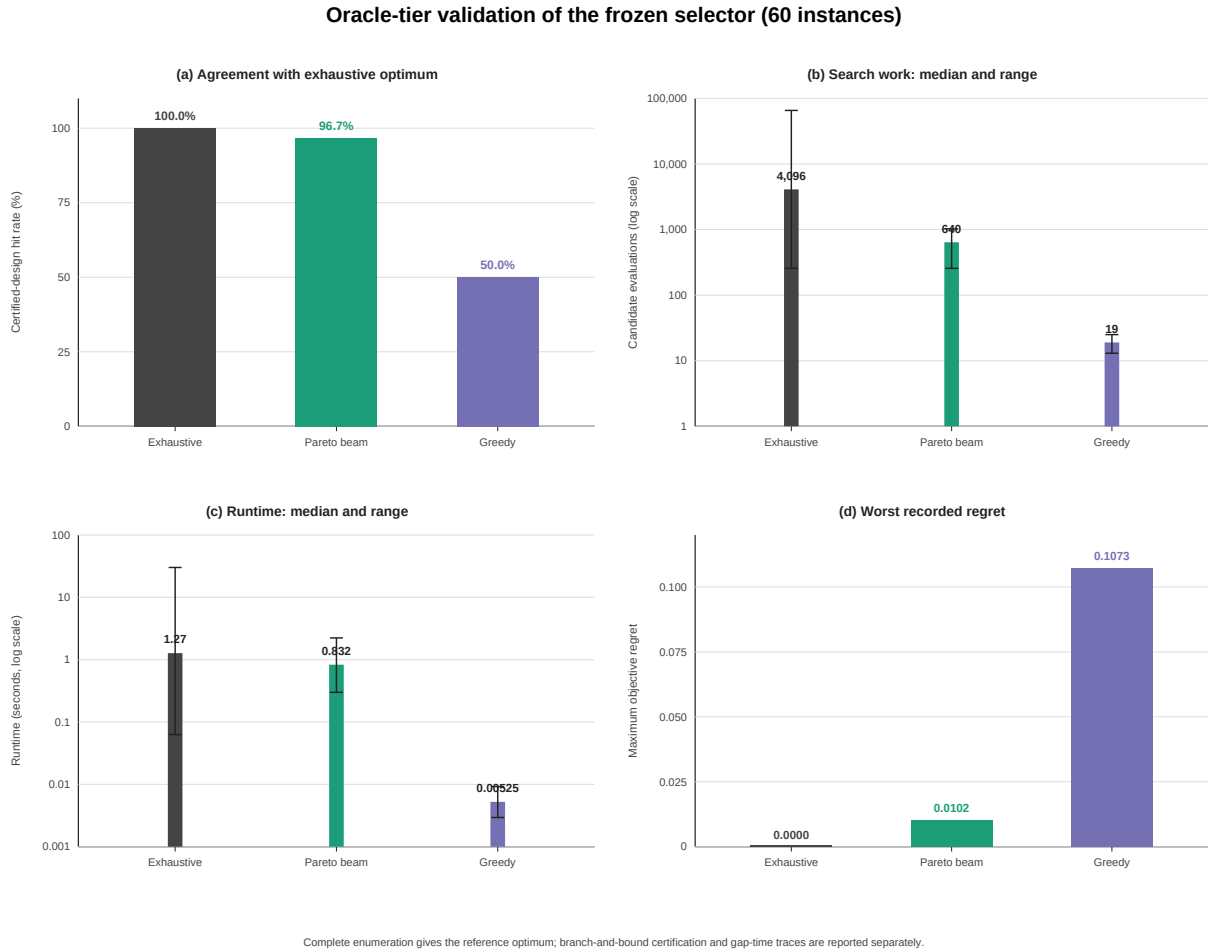
Family and representation	At-threshold ties	Below-threshold mismatch	Saved below-threshold witnesses
Cubic spin glass, fully reduced	970	970	78
Max-3SAT, fully reduced	268	268	62
Max-3SAT, selective	311	315	52
Max-3SAT, matched random	3,802	4,086	260
Total	5,351	5,639	452

Supplementary Table S9: Work and runtime for the oracle-tier methods in Table S1. Entries are medians [interquartile ranges]; timings describe the machine used in this study.

Method	Candidate evaluations	Runtime (s)
Complete enumeration	4096 [1024, 16384]	1.274 [0.370, 6.112]
Pareto beam	640 [448, 832]	0.832 [0.603, 1.158]
Greedy native-completion	19 [16, 22]	0.00525 [0.00414, 0.00667]

S4.2 Selector validation, certification, and sensitivity

Figure S4 gives the graphical counterpart of Table S1. Certification timings and stored gap traces are given in Table S10 and Figure S5.



Supplementary Figure S4: Numerical oracle-tier selector check using the frozen results. Complete enumeration supplies the reference optimum, and the practical methods are compared with it using hits, work, runtime, and regret. Branch-and-bound certification and gap traces are reported separately in Table S10 and Figure S5.

Supplementary Table S10: Branch-and-bound certification on the oracle tier. Runtime is descriptive for the machine used in this study.

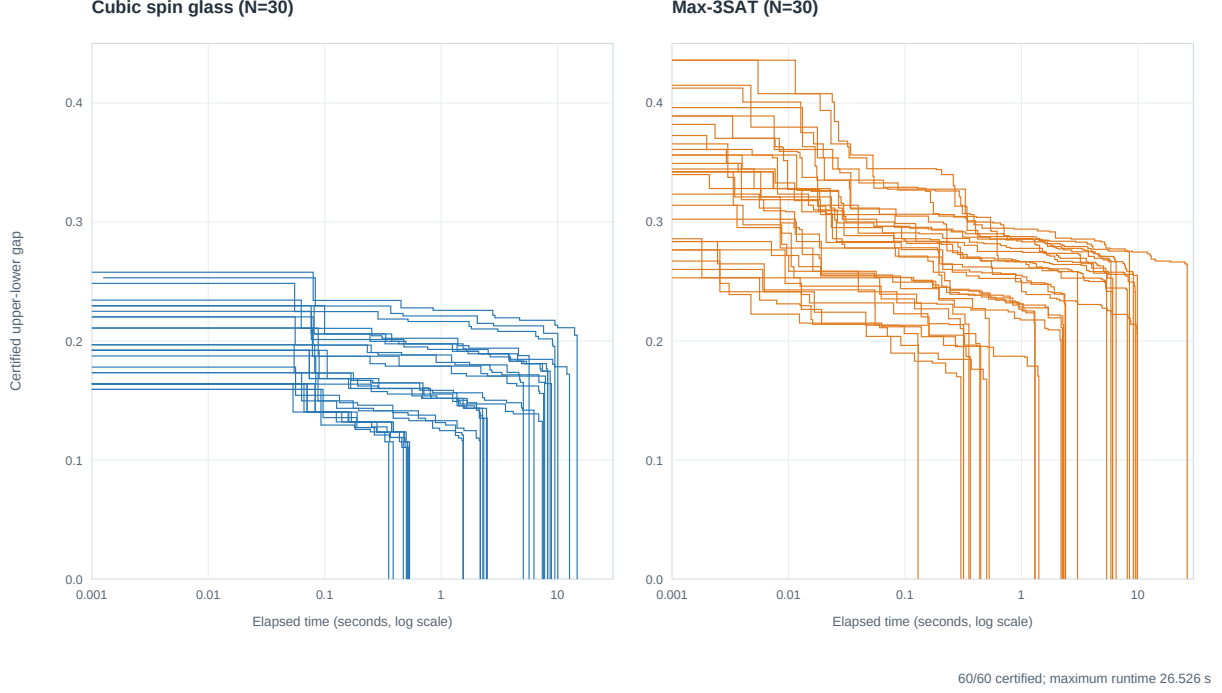
Family	Certified	Median nodes	Max. nodes	Median time (s)	Max. time (s)	Max. final gap
Cubic spin glass	30/30	5,461	33,462	2.469	14.715	0
Max-3SAT	30/30	5,461	55,453	2.336	26.526	0

The repaired certifier can start before any feasible design has been found. If it stops early, it returns bounds from the remaining search frontier. The experiments keep the saved floating-point

raw moments fixed. A design passes the fibre check when $\hat{E}_{\text{fib}}^{\text{computed}} \leq \tau_{\text{fib}} + 10^{-9}$. It must also meet the stated resource constraints. Scores and reference resource bounds use rational arithmetic. The certificate therefore applies to the feasible set defined by these fixed values and this tolerance. It does not certify the feasible set based on exact LP moments or the zero-tolerance definition in Section 4.1. All 60 completed historical traces remain valid under the repaired interface.

Branch-and-bound certificate trajectories

Gap against elapsed time for 60 oracle instances; every trace terminates at zero.



Supplementary Figure S5: Stored certificate gap against elapsed time for the 60 oracle instances: (a) spin glass and (b) Max-3SAT. Each trace terminates at zero. The figure checks the small certification cases and is not used as a scaling claim.

Supplementary Table S11: Guardrail on/off ablation on the 48 training and validation calibration instances. Fibre-risk medians and score changes use only the instances whose selected design changed. Here $\Delta S = S_{\text{guarded}} - S_{\text{unguarded}}$.

Split	n	Changed	Violations off $\rightarrow$ on	Median fibre risk off $\rightarrow$ on	Median ΔS	Max. ΔS
Train	36	10	10 $\rightarrow$ 0	0.03142 $\rightarrow$ 0.01820	0.01654	0.03802
Validation	12	2	2 $\rightarrow$ 0	0.03301 $\rightarrow$ 0.01753	0.03451	0.04870
Combined	48	12	12 $\rightarrow$ 0	0.03142 $\rightarrow$ 0.01820	0.01849	0.04870

Without the guardrail, 12 of 48 calibration choices exceeded the threshold. Enabling it changed exactly these choices and removed all recorded violations, at a median resource-score increase of 0.01849 among changed cases. All changes were in Max-3SAT. This is a training/validation calibration ablation. The separate one-factor-at-a-time study covers all 60 oracle instances.

The deterministic greedy baseline was feasible in all 60 cases but differed from the main selector in 30. The maximum-reuse rule was feasible on only ten Max-3SAT instances and on none of the spin-glass instances, so it is not a generally usable replacement for the constrained selector.

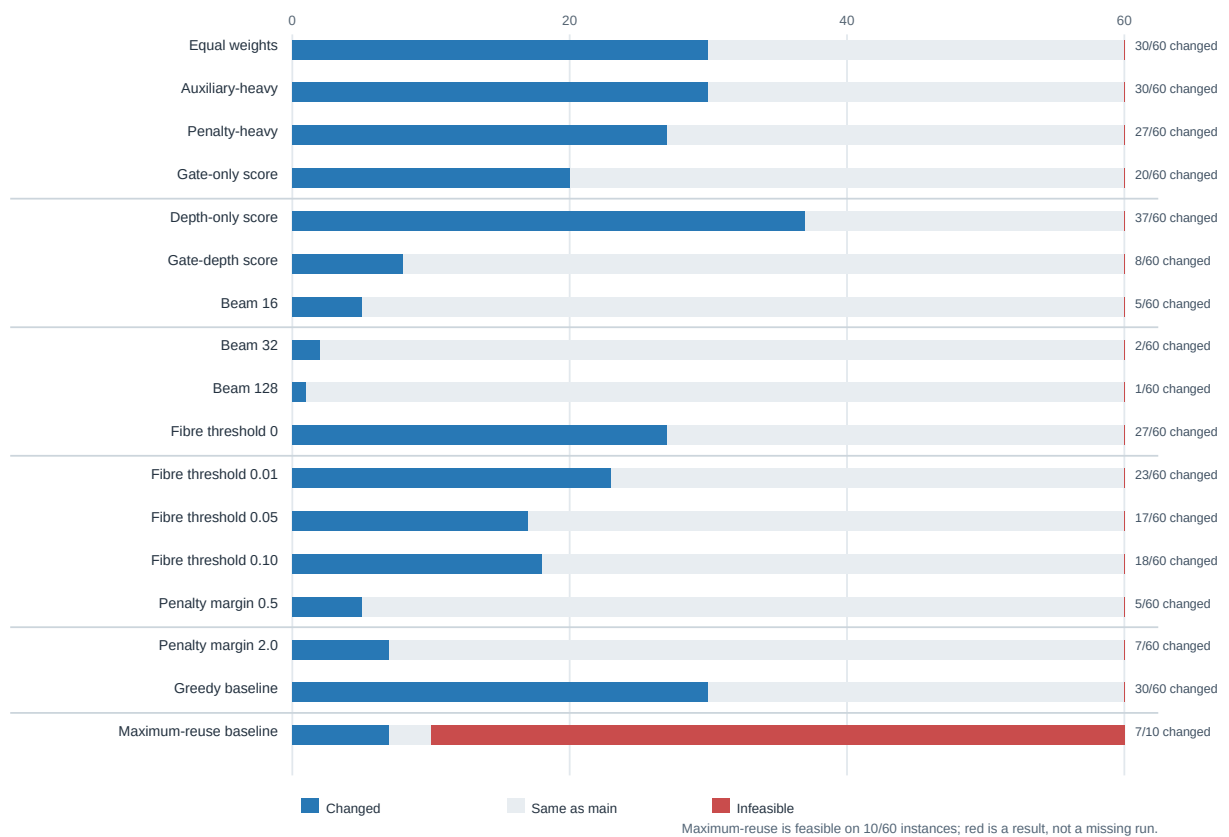
Changing the weights changes what the resource score measures. We therefore do not compare score values directly between weight settings. We use this test to see whether the selected design changes and to compare its gate count, depth, auxiliary count and penalty. It does not identify one

Supplementary Table S12: One-factor-at-a-time selector sensitivity on 60 oracle instances. “Changed” counts representations different from the main setting. Every row uses the same raw instances.

Block	Alternative values	Changed choices	Feasible runs
Resource weights	equal, auxiliary-heavy, penalty-heavy	30, 30, 27	60/60 each
Resource mode	gate-only, depth-only, gate-depth	20, 37, 8	60/60 each
Beam width	16, 32, 128	5, 2, 1	60/60 each
Fibre threshold	0, 0.01, 0.05, 0.10	27, 23, 17, 18	60/60 each
Penalty margin	0.5, 2.0	5, 7	60/60 each
Greedy baseline	deterministic	30	60/60
Maximum-reuse baseline	deterministic	—	10/60

Selector sensitivity on the oracle tier

Changed, unchanged, and infeasible cases relative to the frozen main setting (N=60).



Supplementary Figure S6: Choice stability and feasibility in the selector sensitivity study. A changed choice differs from the frozen main setting. Red bars identify infeasible cases rather than missing runs.

weight choice that is best for every problem.

S4.3 Marginal resource levels and selector diagnostics

The all-to-all tier contains 108 training, 36 validation, and 36 test-labelled instances. Marginal medians describe resource scale; their differences are not paired effects. For example, selective and native Max-3SAT gate medians are 145 and 178, while the median paired difference is -25 , not -33 . On the grid, 4,845 of 7,200 scheduled rows are expected width failures, with no unexpected compiler failures. Main common-fit results are in [Table S4](#).

Supplementary Table S13: Marginal median logical and compiled resource levels on the all-to-all reference topology. These levels show scale, but the difference between two entries is not the within-instance paired effect.

Family	Representation	Qubits	Auxiliaries	Two-qubit gates	Two-qubit depth
Max-3SAT	Native	12	0	178	119.5
	Fully reduced	18	7	137	46
	Selective	18	4	145	56
	Matched random	18	4	180	94
Cubic spin glass	Native	12	0	96	79
	Fully reduced	18	6	148	47
	Selective	14	1	116	63.5
	Matched random	14	1	117	82

Supplementary Table S14: Median diagnostics for the 90 compilation-tier selective designs in each family. Fibre risk is the normalised raw-moment excess used by the v2 selector guardrail; the threshold is 0.02. Time is selector runtime in seconds.

Family	Retained cubic	$M_{\max}$	Dynamic range	Fibre risk	Score	Compiler calls	Time
Cubic spin glass	19	9	5.5	0.00115	0.7233	4096	2.19
Max-3SAT	6.5	12	59.5	0.01925	0.8156	4288	1.94

All 180 compilation-tier selected designs satisfied the raw-moment 0.02 selector guardrail; the largest stored normalised excess was below 0.02. The different median retained cubic counts also show that the repaired v2 selector did not apply the same amount of reduction to the two families.

We do not report a compilation-tier certificate gap because these rows are beam-search incumbents and no common certified lower bound was produced.

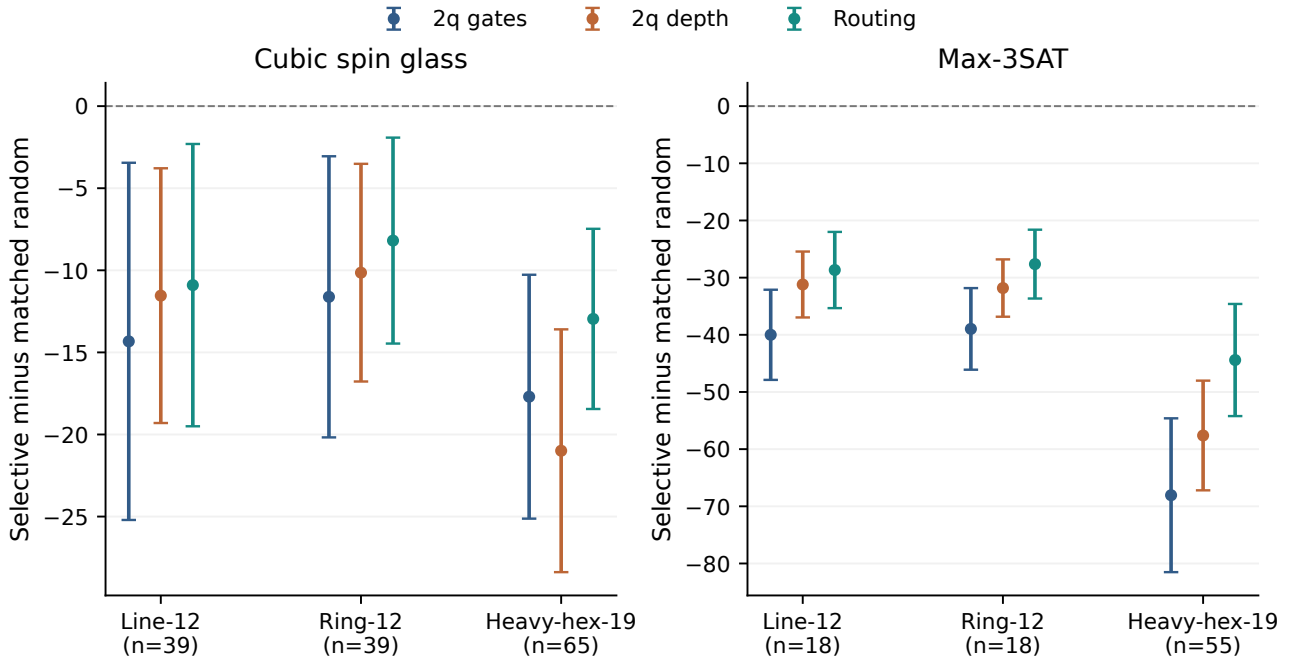
S4.4 Additional coupling maps

Of 16,200 added compiler rows, 7,020 succeeded and 9,180 were expected width failures; no unexpected compiler failure was recorded. The table reports paired medians, while the figure reports means and instance-level intervals.

Capacity remains part of the result. The 12-qubit line and ring each had 39 common-fit spin-glass and 18 common-fit Max-3SAT instances, while the 19-qubit heavy-hex graph had 65 and 55. The corresponding test-labelled counts were only 7 and 2 on line and ring, and 12 and 9 on heavy-hex. Since heavy-hex also has greater capacity, its numerical differences cannot be attributed to connectivity alone. The pooled results broaden the connectivity check, but they remain exploratory rather than a universal hardware rule.

Supplementary Table S15: Selective-minus-matched-random resource differences on the three added coupling maps. Values are paired medians after taking the median over transpiler seeds and the mean over matched-random designs within each instance. Negative values favour the selective choice. Parentheses give the number of instances on which selective is strictly lower.

Topology	Family	Common N	Δ two-qubit gates	Δ two-qubit depth	Δ routing
Line-12	Spin glass	39	0.00 (8)	0.00 (9)	0.00 (8)
	Max-3SAT	18	−38.4 (18)	−33.3 (18)	−26.0 (18)
Ring-12	Spin glass	39	0.00 (8)	0.00 (9)	0.00 (8)
	Max-3SAT	18	−36.3 (18)	−31.8 (18)	−26.6 (18)
Heavy-hex-19	Spin glass	65	0.00 (24)	0.00 (26)	0.00 (24)
	Max-3SAT	55	−55.8 (53)	−52.4 (55)	−35.6 (52)



Supplementary Figure S7: Mean paired selective-minus-matched-random resource changes on the additional coupling maps: (a) spin glass and (b) Max-3SAT, with 95% Student-*t* intervals across raw instances. The comparison uses only instances that fit in both representations. Negative values favour selective reduction; the primary medians are reported in Table S15.

S4.5 Zero-layer accounting and secondary QAOA diagnostics

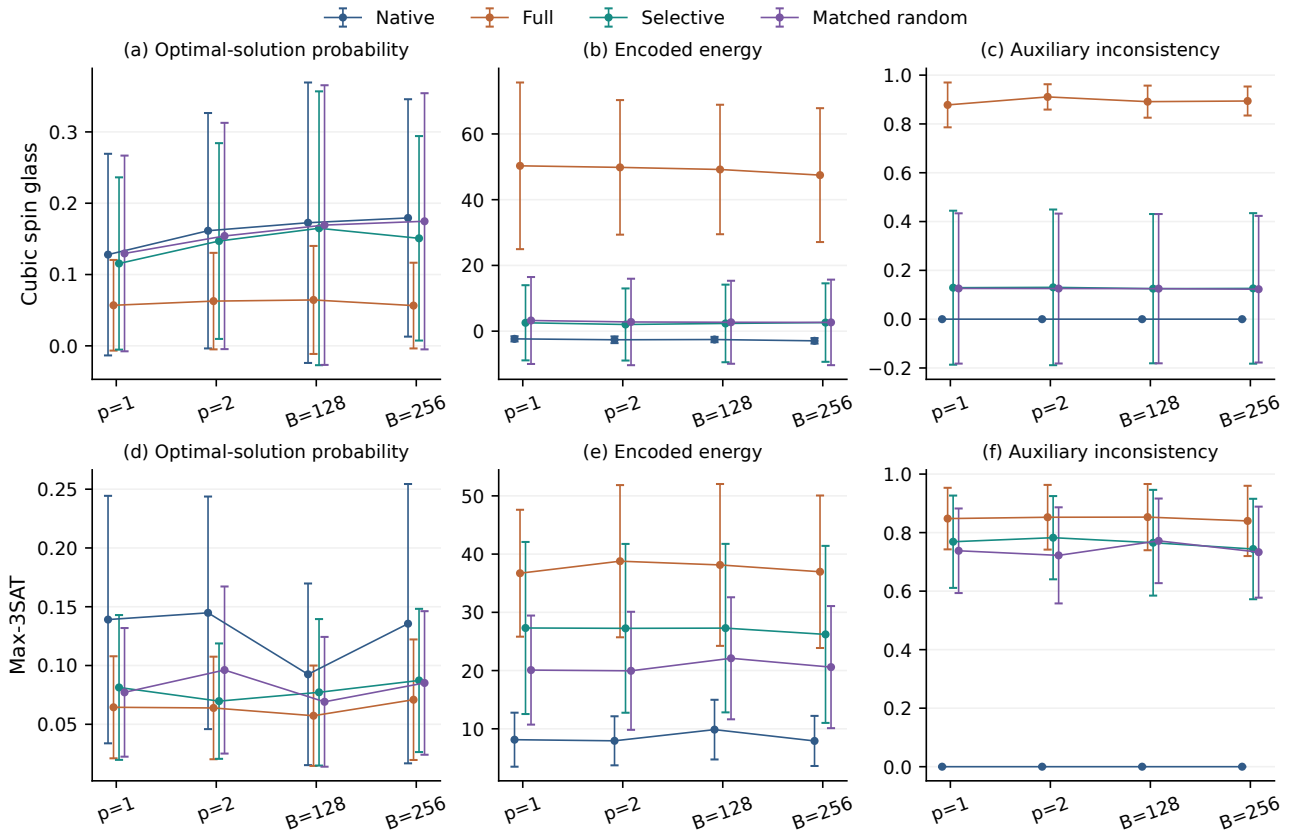
Table S16 separates the initial-state baselines from designs that fit at least one layer. Matched-random groups count five draws per raw instance, and each group contains three restart rows.

Supplementary Table S16: Separation of the stored zero-layer baselines at $B_{2q} = 128$. A design/draw group contains three restart rows. Matched random has five frozen draws per raw instance. “Fully normal” means that all draws for the representation have $p \geq 1$.

Family	Representation	$p = 0$ groups	$p = 0$ rows	Instances with any $p = 0$	Fully normal instances
Spin glass	Native	0/7	0	0/7	7/7
	Full	2/7	6	2/7	5/7
	Selective	1/7	3	1/7	6/7
	Matched random	5/35	15	1/7	6/7
Max-3SAT	Native	4/7	12	4/7	3/7
	Full	2/7	6	2/7	5/7
	Selective	2/7	6	2/7	5/7
	Matched random	19/35	57	4/7	3/7

Requiring every representation and all five matched-random draws to fit at least one layer leaves only five spin-glass and three Max-3SAT instances. On this strict common subset, the spin-glass selective and matched-random paired objective differences from native are -0.072 $[-0.544, 0.400]$ and -0.183 $[-0.981, 0.615]$, respectively. Both intervals include zero. For Max-3SAT, the corresponding means are 0.982 $[-0.890, 2.854]$ and 0.941 $[0.603, 1.280]$. This small common-fit analysis is exploratory. Overall, the experiment does not show a general QAOA performance benefit from selective reduction; it shows that a resource saving does not automatically lead to a better finite-depth objective value.

The encoded energies should not be compared as a performance ranking because the four representations use different encoded Hamiltonians. The stored field `auxiliary_inconsistency_rate` implements the event in Equation (77): it is the probability mass, or sampled fraction, for which at least one active auxiliary relation is violated. It is therefore exactly $1 - P_{\text{cons}}$ for the active auxiliary set.

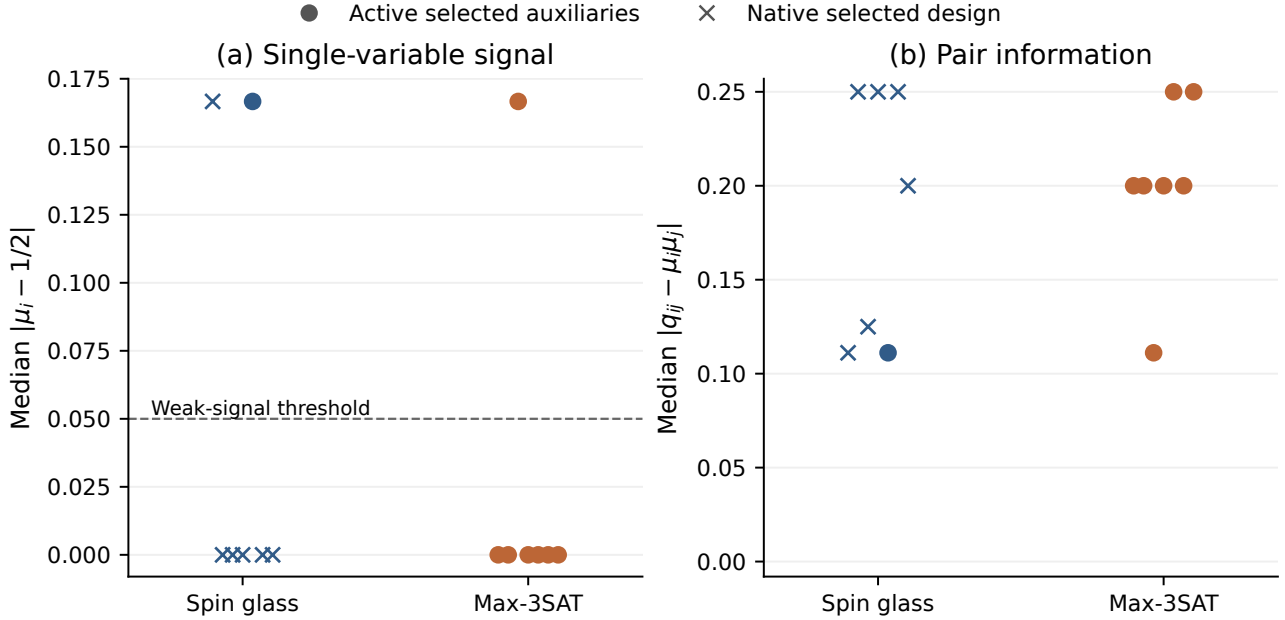


Supplementary Figure S8: Additional noiseless-QAOA diagnostics recalculated from the frozen run rows. Panels (a)–(c) show spin-glass optimal-solution probability, encoded energy, and auxiliary inconsistency R_{inc} ; (d)–(f) show the same metrics for Max-3SAT. Restarts and matched-random draws are averaged before the raw instance is used as the statistical unit. Encoded energy is representation-specific and is shown only as a diagnostic.

S4.6 Original and targeted warm-start diagnostics

The first sparse-RLT solve reported an optimum for all 14 QAOA instances labelled as test. The saved vectors contain 96 original-variable pseudo-moments and 292 pair pseudo-moments, for 388 diagnostic values in total. After the selector repair, 62 distinct instance–representation/draw groups have active auxiliaries. Their auxiliary warm-start probabilities are the clipped pair pseudo-moments.

Figure S9 compares original-variable bias with pair information. The pair pseudo-moments can differ from the independence closure; this diagnostic alone does not establish a performance benefit.



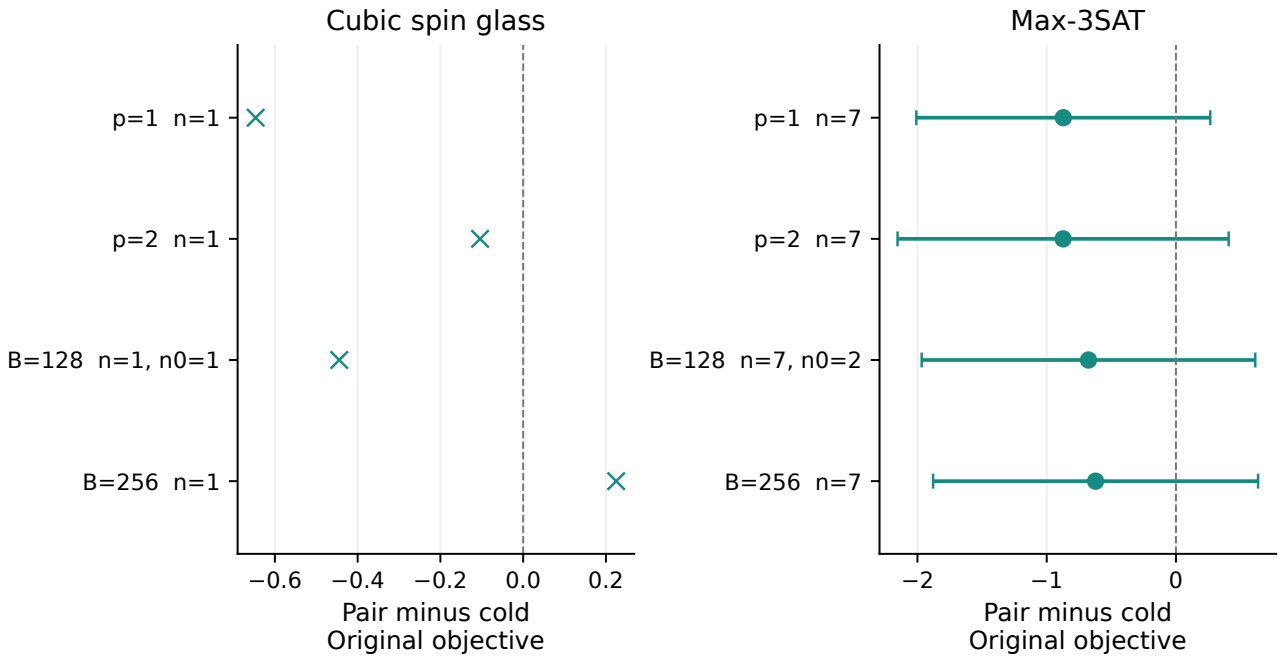
Supplementary Figure S9: Signal available to the E4 warm start. Each point summarises one raw QAOA instance. Circles identify cases where the repaired selected representation has active auxiliaries; crosses identify a native selected endpoint. The LP values are unchanged; design-status markers are updated. Eleven of fourteen instances have weak single-variable signal. Two instances combine stronger signal with an active selected auxiliary design.

Only three instances exceed the weak-signal threshold, and two belong to the eight-case common selective active-auxiliary subset. Table S17 reports the corrected outcomes on this fixed subset, after averaging three restarts per instance and keeping each budget separate. Before the selector repair, the coordinate-corrected original E4 schedule contained 4,320 rows. Positive-depth, zero-depth, and all-scheduled summaries for the full design set are separately available in the correction archive. The selector-repaired merged schedule retains 4,176 rows after 144 redundant warm-policy rows for newly zero-auxiliary designs are removed.

Spin-glass intervals are not estimable at $n = 1$; all four Max-3SAT primary pair-minus-cold intervals include zero. In the direct pair-versus-original-only comparison, the Max-3SAT $p = 2$ contrast is $-0.311 [-0.556, -0.065]$; the other such intervals and every pair-versus-independence interval include zero. This secondary exploratory contrast does not establish the primary pair-versus-cold claim.

Supplementary Table S17: Coordinate-corrected original E4 results on the frozen common selective active-auxiliary subset. Cold is the mean original objective; other columns are paired differences from cold with exploratory, unadjusted 95% Student- t intervals across n row instances. n_0 counts zero-layer instances retained in the scheduled comparison. Negative differences favour warm starts. Pair uses archived clipped pair moments; Ind. multiplies clipped original probabilities without further clipping.

Budget	n	n_0	Cold	Original-only minus cold	Pair minus cold	Ind. minus cold
Spin glass						
$p = 1$	1	0	-1.093	-0.715	-0.647	-0.681
$p = 2$	1	0	-1.722	0.735	-0.104	0.164
$B = 128$	1	1	0.000	-0.444	-0.444	-0.444
$B = 256$	1	0	-1.215	0.290	0.225	0.307
Max-3SAT						
$p = 1$	7	0	10.035	-0.562 [-1.763, 0.640]	-0.872 [-2.009, 0.265]	-0.552 [-1.872, 0.768]
$p = 2$	7	0	9.872	-0.562 [-1.741, 0.616]	-0.873 [-2.154, 0.408]	-0.824 [-2.036, 0.387]
$B = 128$	7	2	9.884	-0.647 [-1.935, 0.641]	-0.677 [-1.967, 0.614]	-0.808 [-2.083, 0.467]
$B = 256$	7	0	9.437	-0.582 [-1.795, 0.632]	-0.622 [-1.879, 0.635]	-0.457 [-2.034, 1.121]

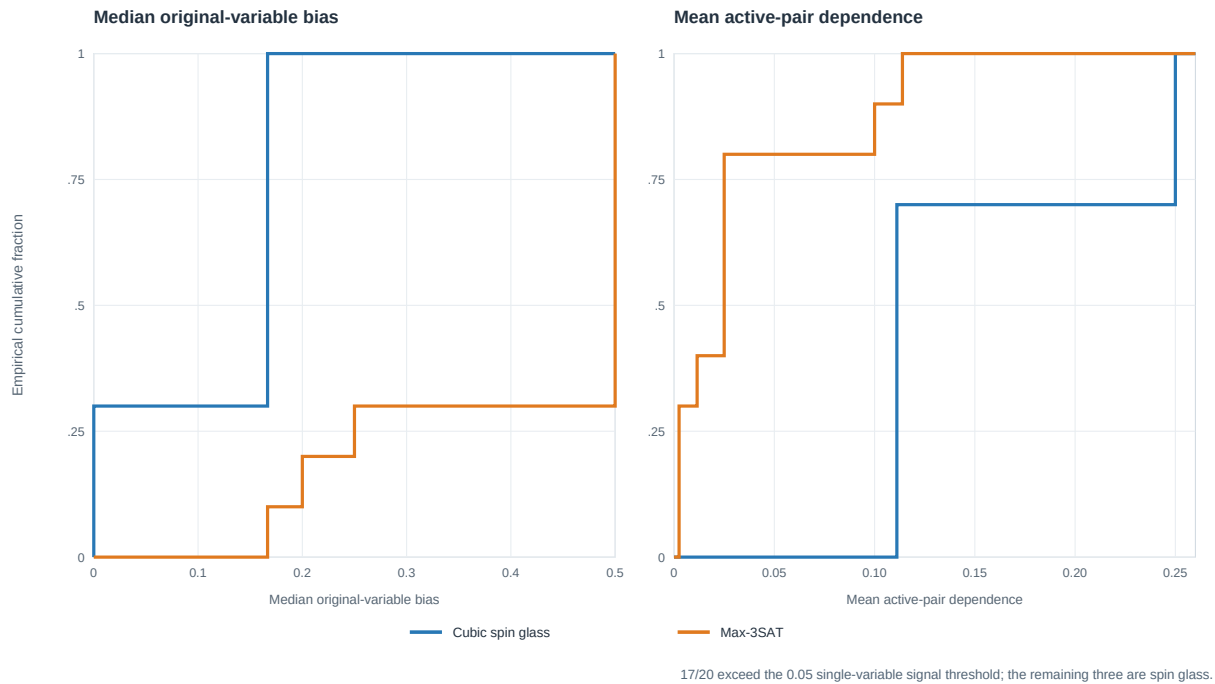


Supplementary Figure S10: Corrected original E4 pair-minus-cold differences on the selected active-auxiliary subset. Error bars are exploratory instance-level 95% Student- t intervals where estimable. The n_0 annotations identify retained zero-layer instances; spin-glass crosses have $n = 1$ and therefore no interval.

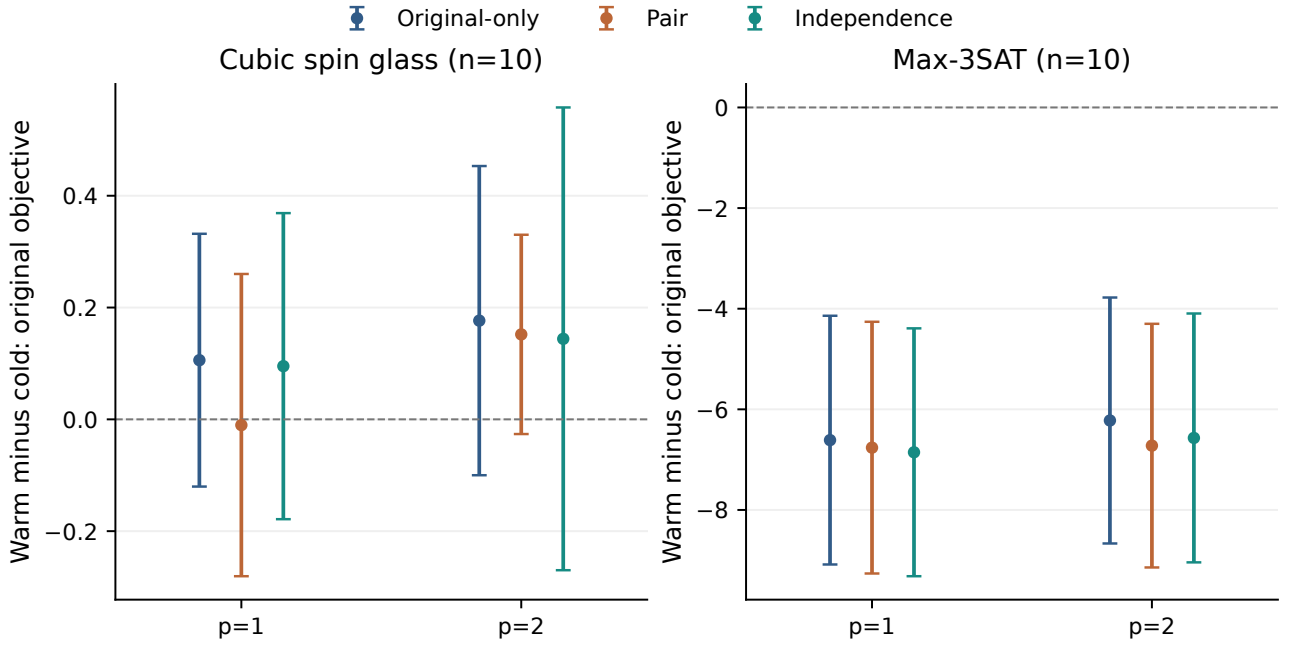
The targeted diagnostics and corrected warm-minus-cold plots below complement [Table S6](#). The original screening preceded QAOA evaluation; seven spin-glass and all ten Max-3SAT instances exceed the signal threshold. Corrected mean optimal-solution probabilities for targeted Max-3SAT under independence are 0.6094 and 0.6224 at $p = 1, 2$, compared with 0.5987 and 0.6262 under pair warming.

Relaxation signal in the targeted warm-start set

Empirical distributions for the 20 strongest-screened instances; screening used no QAOA outcomes.



Supplementary Figure S11: Relaxation-signal diagnostics for the 20 strongest-screened instances: (a) original-variable bias and (b) pair dependence. Screening used only relaxation quantities and was completed before any QAOA outcome was inspected. Seven spin-glass and all ten Max-3SAT instances exceed the frozen single-variable signal threshold.



Supplementary Figure S12: Coordinate-corrected mean paired warm-minus-cold original-objective differences on the fixed strongest-screened set: (a) spin glass and (b) Max-3SAT. The three policies are shown separately. Error bars are exploratory, unadjusted 95% Student- t intervals over ten raw instances per family after averaging three restarts. Negative differences favour warm starts.

S4.7 Complete descriptive regime counts

We next compared selective reduction with its auxiliary-count-matched random control. The planned endpoints were two-qubit depth, two-qubit gate count, routing overhead, and QAOA objective. Outcomes not meeting either directional interval criterion were assigned to the residual (inconclusive) category; this does not establish equivalence.

Supplementary Table S18: Help, residual (inconclusive), hurt and not-estimable counts for selective reduction relative to the matched-random control. The residual category includes uncertain intervals and does not establish equivalence.

Family	Endpoint	Help	Residual	Hurt	Not estimable
Cubic spin glass	Two-qubit depth	8	17	0	11
Cubic spin glass	Two-qubit gates	3	22	0	11
Cubic spin glass	Routing overhead	1	24	0	11
Cubic spin glass	QAOA objective	1	25	2	0
Max-3SAT	Two-qubit depth	20	0	0	16
Max-3SAT	Two-qubit gates	19	1	0	16
Max-3SAT	Routing overhead	1	19	0	16
Max-3SAT	QAOA objective	1	13	14	0

Of 272 scheduled instance-metric endpoints, 191 were estimable: 54 helped, 121 were classified as residual (inconclusive) and 16 hurt; 81 were not estimable. In the

QAOA rows, each family contributes seven instances at four budgets. Thirty-nine residual intervals extend beyond the tolerance band. Classification follows Appendix H.7; these are endpoint counts, not independent-instance confidence statements.

The 81 not-estimable endpoints were expected sparse-device infeasibilities. All-to-all data were estimable for all 18 instances in each family, while the sparse-topology data were estimable for only seven of 18 spin-glass instances and two of 18 Max-3SAT instances. Some sparsity and pair-reuse bins also contain only one or two observations. The regime counts therefore show useful patterns, but they are too small for a universal rule about problem structure.

S4.8 Limited study with a representative noise model

The 18-instance noise study follows the fixed simulator configuration in Appendix G.7. The tables below report absolute objectives, degradation from each representation’s own sampled noiseless baseline, and paired selective-minus-native differences.

Supplementary Table S19: Descriptive means in the limited noise study. Lower objective values are better. Degradation is the high-noise objective minus the noiseless objective. The final column is the high-noise auxiliary inconsistency rate R_{inc} from Equation (77).

Family	Representation	Noiseless	Low noise	High noise	Degradation	Inconsistency
Cubic spin glass	Native	-3.052	-0.699	-0.087	2.965	0.000
	Fully reduced	-0.620	-0.175	-0.032	0.588	0.872
	Selective	-2.923	-0.729	-0.123	2.800	0.083
	Matched random	-2.953	-0.769	-0.124	2.829	0.082
Max-3SAT	Native	7.953	9.305	9.700	1.748	0.000
	Fully reduced	9.024	9.396	9.633	0.609	0.821
	Selective	8.701	9.455	9.692	0.990	0.775
	Matched random	8.402	9.302	9.642	1.239	0.771

Noise worsened every representation’s mean original objective. Full reduction changed least from its noiseless value, but started worse than native in both families and retained high auxiliary inconsistency. Selective reduction had lower inconsistency than full reduction for spin glass, but remained highly inconsistent for Max-3SAT. Smaller degradation is not better absolute performance.

This table is descriptive. The raw file contains three repeated rows per instance for the native, fully reduced, and selective representations, and 15 rows for the five matched-random designs. These repeated rows were therefore averaged inside each raw instance before any uncertainty calculation.

Supplementary Table S20: Instance-level selective-minus-native objective differences in the noise study. Lower objective values are better, so a negative mean favours selective reduction. The 95% intervals are Student- t intervals across nine raw instances per family. W/T/L gives the numbers of instance wins, ties, and losses using the predeclared 0.01 objective equivalence band.

Family	Noise	Mean difference	95% interval	d_z	W/T/L
Cubic spin glass	Noiseless	0.129	[−0.868, 1.126]	0.099	5/0/4
	Low	−0.030	[−0.312, 0.251]	−0.083	6/0/3
	High	−0.036	[−0.089, 0.016]	−0.532	4/3/2
Max-3SAT	Noiseless	0.749	[0.140, 1.358]	0.945	3/0/6
	Low	0.149	[−0.046, 0.344]	0.589	1/0/8
	High	−0.009	[−0.133, 0.116]	−0.053	5/0/4

After within-instance aggregation, neither non-zero noise level showed a clear selective advantage over native. All spin-glass intervals included zero; Max-3SAT favoured native without noise and was inconclusive under noise. The smaller Max-3SAT degradation starts from a worse noiseless baseline and does not establish better absolute noisy performance. With nine instances per family, this is an exploratory robustness check.

S5 Five-seed compilation results

These results use the same 30 Max-3SAT and 30 constructed instances as the earlier cost-layer study. They are not new independent samples. Main Section 4.5 gives the methods and resource definitions. Main Appendix G.10 describes the saved records and the independent checks.

S5.1 Budget points retained across seeds

Table S21 compares the new seed-1729 baseline with the intersection of all five seed-specific TRUE sets. Each entry pools the two synthesis strategies and 30 instances in its family. TRUE means mixed-only within the saved candidate library. It does not exclude full representations outside that library. The same budget point must pass in each seed, but a different mixed candidate may make it pass. Points share instances, so the large counts are not large independent sample sizes.

Supplementary Table S21: Mixed-only budget points within the saved library. The last column divides the all-five-seed count by the new seed-1729 count.

Family	Topology	Seed 1729	All five seeds	Retained
Max-3SAT	Line-12	23,734	13,294	56.01%
Max-3SAT	Ring-12	24,083	14,677	60.94%
Max-3SAT	Grid-12	30,024	13,228	44.06%
Constructed	Line-12	22,110	16,412	74.23%
Constructed	Ring-12	7,946	6,683	84.11%
Constructed	Grid-12	41,542	14,775	35.57%
Pooled		149,439	79,069	52.91%

Supplementary Table S22: Seed-specific results over 360 instance–topology–synthesis settings. A setting has a region if at least one budget point is TRUE. The settings share 60 instances. No point is UNKNOWN.

Seed	TRUE budget points	Settings with a region	Fraction
1729	149,439	255/360	70.83%
2718	156,824	255/360	70.83%
31415	155,289	262/360	72.78%
57721	165,626	254/360	70.56%
65537	155,638	256/360	71.11%

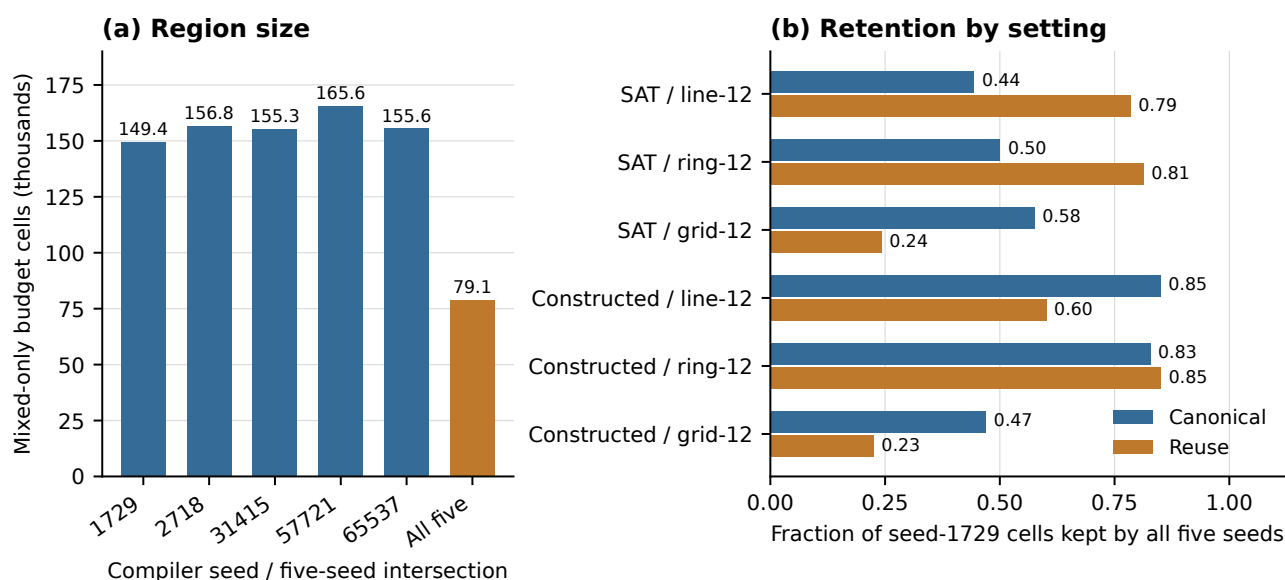
The total number of TRUE points can rise in another seed even when many baseline points are lost: new points may appear elsewhere in the grid. Thus similar totals do not show that the regions contain the same points. There are $149,439 - 79,069 = 70,370$ baseline points that fail the TRUE condition in at least one other seed.

S5.2 Changes in regions, frontiers and selected candidates

There are 360 instance–topology–synthesis settings. At least one common mixed-only budget point survives all five seeds in 220 settings. Having some common points does not mean that the whole region is unchanged. Table S23 separates these questions. The earlier analysis labelled 158 changed regions as stable because it counted points lost in every other seed rather than points lost in any other seed. The table uses direct comparisons of the five complete sets. A separate baseline-retention classification asks only whether the seed-1729 TRUE points survive: 49 settings retain all baseline points, 171 retain some, 35 retain none of a nonempty baseline, and 105 have an empty baseline. These four categories sum to 360. An unchanged baseline can gain points in another seed, so the 49 baseline-stable settings are not the same as the 15 settings with identical nonempty TRUE sets below.

We also compare each of the four other seeds with seed 1729 in each setting. This gives 1,440 paired comparisons, all based on the same 60 instances. The median Spearman correlation of candidate resource scores is 0.970; the minimum is 0.765. The median Jaccard similarity of the Pareto candidate sets is 0.667; the minimum is 0.125. Thus score rankings often agree even when the Pareto sets change. These summaries compare seeds within a synthesis strategy. They are different from the comparisons between synthesis strategies in the main text.

The lowest-score candidate changes in 8 of the 1,440 comparisons. This selector includes native and does not apply the four hard budgets. At seed 1729, it chooses native in 353 of 360 settings



Supplementary Figure S13: Five-seed compilation check within the saved candidate libraries. (a) Number of mixed-only budget points for each seed, followed by their intersection. (b) Fraction of new seed-1729 points retained by all five seeds, shown separately for each family, topology and synthesis strategy. SAT denotes weighted Max-3SAT. Each bar pools 30 instances. Points and compiler settings share instances and are not independent samples.

Supplementary Table S23: Region changes across the five seeds. The four rows partition the 360 settings. The third row requires a nonempty seed-1729 region; the last row includes any remaining change, including newly appearing regions.

Relation between the five TRUE sets	Settings
Identical and nonempty	15
Identical and empty	81
Baseline nonempty, but no point common to all five	35
All other changes	229

and strictly mixed in seven. No full representation is selected. The low winner-change rate therefore mainly describes native choices; it is not evidence that mixed choices are generally stable.

To complete the missing feasibility check, we keep the seed-1729 winner fixed. We first record every point in the saved budget grid where its seed-1729 circuit fits. We then use that candidate's saved circuits from the other seeds to test those same points. In 320 of 1,440 comparisons it loses at least one previously feasible point. This result explains why keeping the same selected representation does not guarantee keeping its feasible budget region. It is a grid-based diagnostic, not a result at a newly chosen primary budget.

The corrected figure and tables were computed from the saved compiler rows. They require no new compiler or QAOA calls. The correction script, full setting-level tables and audit results accompany this revision.

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